Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

SIDE KICK

Physics

Thirteen chapters written fresh for the 2027 syllabus: laws, derivations, ray diagrams, numericals and a practical lab companion.



EVERY TOPPER HAS A SIDEKICK

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Hey. I'm your Sidekick.

I live inside this book. Wherever things get tricky, I pop up with a tip, a trap to dodge, or a question the board loves to ask. Read the chapter, then try my HOT questions. Deal?



Aligned to the official ICSE 2027 syllabus. If it's in the syllabus, it's in here.

SIDEKICK TIP
Shortcuts, mnemonics and exam hacks.

FUN FACT
Things that make the concept stick.

REAL LIFE
Where this shows up outside the exam hall.

KEY POINT
Definitions and laws you must write exactly.

HOT QUESTION
Higher-order thinking. Try before you peek.

✦ NEW
Added in this edition to match the 2027 syllabus.

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01

CHAPTER · PHYSICS

Force, Moments and Circular Motion

Why does a long spanner loosen a stuck nut? Why does a see-saw balance with unequal children? And why do you slide outward when a car takes a turn? This chapter explains the turning effect of forces, balance, and motion in a circle.



IN THIS CHAPTER

- Moment of a force
- Equilibrium
- Principle of moments
- Centre of gravity
- Uniform circular motion
- Centripetal force

Two Kinds of Motion

A force can make an object move in two basic ways. In **translational motion** every part of the body moves the same distance in the same direction, like a box sliding across a table. In **rotational motion** the body turns about a fixed line called the **axis of rotation**, like a fan blade or a door on its hinges.

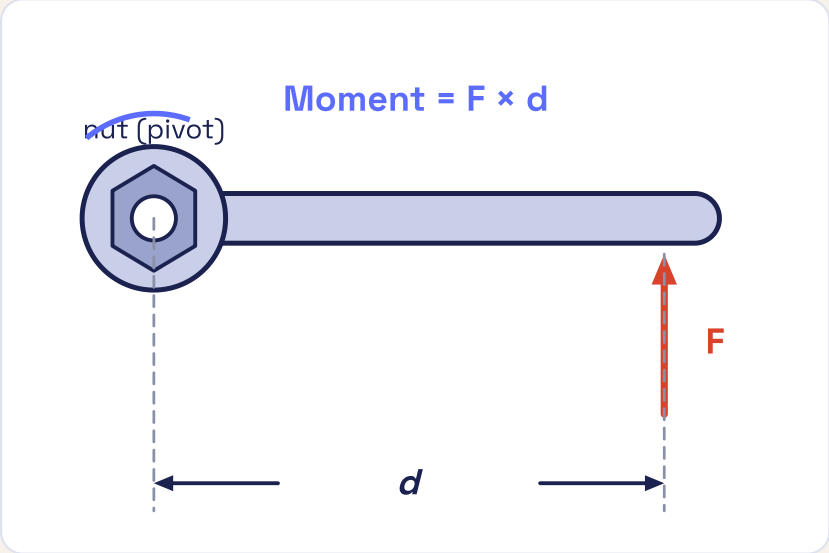
A push through the middle of a book slides it. The same push at one edge also spins it. The difference is where the force acts, and that brings us to the turning effect of a force.

Moment of a Force (Torque)

KEY POINT

Moment of a force about a point is the turning effect of the force. It equals the force multiplied by the **perpendicular distance** of the line of action of the force from the pivot. The moment of a force is also called **torque**.

$$\text{Moment of force} = \text{Force} \times \text{perpendicular distance from the pivot} \quad \mathbf{M = F \times d}$$



A spanner turning a nut. The turning effect depends on both the force and its perpendicular distance from the pivot.

- **SI unit:** newton metre (N m). **cgs unit:** dyne centimetre (dyn cm).
- **Conversion:** 1 N m = 10^5 dyn × 10^2 cm = 10^7 dyn cm.
- Moment is a vector: it is **anticlockwise** if it tends to turn the body anticlockwise and **clockwise** otherwise. By convention anticlockwise moments are taken as positive.
- A force whose line of action passes through the pivot has d = 0, so its moment is **zero**.

What Decides the Turning Effect?

- A **larger force** gives a larger moment.
- A **larger distance** from the pivot gives a larger moment, which is why door handles sit far from the hinges and spanners have long handles.

- The force must act **perpendicular** to the arm to give the full effect. Pushing along the arm gives no turning at all.

EVERYDAY EXAMPLE	PIVOT	HOW THE MOMENT HELPS
Opening a door	Hinges	Pushing at the handle needs far less force than pushing near the hinge
Steering wheel	Steering shaft	A big wheel lets two hands turn it with small forces
Bicycle pedal	Crank axle	Pressing the pedal at the end of the crank turns the sprocket
Using a spanner	Nut	A longer spanner loosens a tight nut with less force
Opening a tap	Tap spindle	A wide handle makes turning easy



SIDEKICK TIP

Whenever a question says 'calculate the moment', first **mark the pivot**, then measure the perpendicular distance from it to the line of the force. Most mistakes are wrong distances.

Worked Example 1: Moment of a Force

A mechanic pulls at the end of a spanner 0.30 m long with a force of 45 N perpendicular to the spanner. Find the moment about the nut. How much force is needed if the spanner is changed to 0.45 m?

```
M = F x d = 45 x 0.30 = 13.5 N m
Same moment with d = 0.45 m:
F = M / d = 13.5 / 0.45 = 30 N
```

Conditions of Equilibrium

A body is in **equilibrium** when it does not change its state of motion: it stays at rest or keeps moving in a straight line at constant speed, and it does not start to rotate. Two conditions must hold together.

1. Translational equilibrium

The resultant force on the body is zero. Forces up equal forces down, forces left equal forces right, so the body does not accelerate.

2. Rotational equilibrium

The sum of clockwise moments about any point equals the sum of anticlockwise moments about that point, so the body does not start turning.



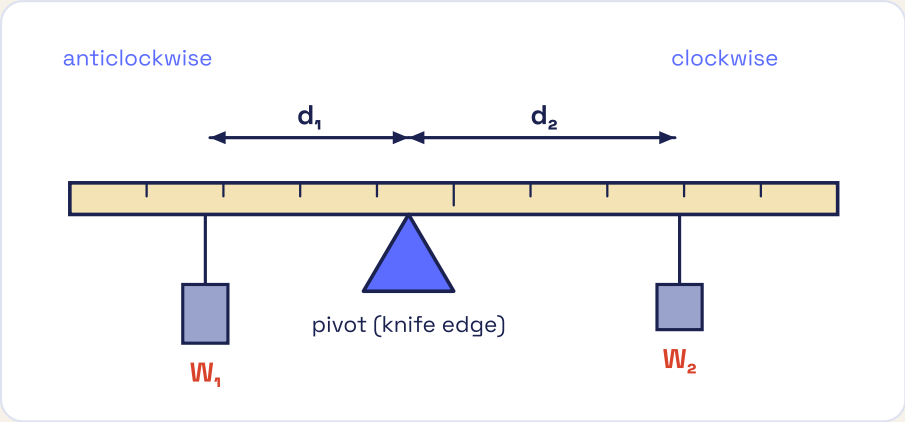
BOARD ALERT

A body can have zero resultant force and still spin. Two equal and opposite forces applied at opposite sides of a wheel give no net force yet turn the wheel. Both conditions are needed.

Principle of Moments

KEY POINT

Principle of moments: When a body is in rotational equilibrium, the sum of the clockwise moments about the pivot equals the sum of the anticlockwise moments about the same pivot.



Balanced rule: W_1 × d_1 (anticlockwise) = W_2 × d_2 (clockwise).

$$W_1 \times d_1 = W_2 \times d_2$$

A see-saw is the classic case: a heavy child sits close to the pivot, a light child sits far, and the plank balances because both moments are equal.

Verifying the Principle of Moments (Metre Rule Method)

Aim: to verify that, for a body in rotational equilibrium, the sum of clockwise moments equals the sum of anticlockwise moments about the pivot.

1. Suspend
Hang a metre rule horizontally from two spring balances, one near each end, using loops of thread. Note both readings.

2. Hang weights
Hang slotted weights from the rule at chosen points, for example one on each side of the middle.

3. Balance
Adjust the positions until the rule stays horizontal. Read the force on each spring balance.

4. Measure
Measure each load's distance from the pivot (the line of action of the force) in metres.

5. Calculate
Clockwise moment = F x d on one side. Anticlockwise moment = F x d on the other. Compare them.

SIDEKICK TIP

Your two sides will not match exactly. That is normal. Name **friction at the thread and reading error** as the sources of the small difference.

Worked Examples: Moments

Example 2. A 20 N weight hangs 0.30 m to the left of a pivot on a light rod. Where must a 15 N weight hang on the right to balance it?

```
Anticlockwise moment = 20 x 0.30 = 6.0 N m
Clockwise moment = 15 x d
For balance: 15 d = 6.0   so  d = 0.40 m
```

Example 3. A uniform metre rule is pivoted at the 30 cm mark. A 5.0 N weight hangs from the 5 cm mark and the rule balances horizontally. Find the weight of the rule. (The weight of a uniform rule acts at its 50 cm mark.)

```
Distance of 5 N weight from pivot = 30 - 5 = 25 cm = 0.25 m (anticlockwise)
Distance of rule's weight from pivot = 50 - 30 = 20 cm = 0.20 m (clockwise)
W x 0.20 = 5.0 x 0.25
W = 1.25 / 0.20 = 6.25 N   (mass = 0.625 kg, taking g = 10 m s-2)
```

Example 4. A uniform plank 4.0 m long weighing 200 N rests on supports at its two ends A and B. A 100 N load sits 1.0 m from A. Find the reaction at each support.

```
Take moments about A:
R_B x 4.0 = 200 x 2.0 + 100 x 1.0 = 500
R_B = 125 N
Translational balance: R_A + R_B = 200 + 100 = 300
R_A = 300 - 125 = 175 N
```

SIDEKICK TIP

To get a reaction or tension quickly, **take moments about the point where the unknown force acts**. Its own moment is zero, so it drops out of the equation.

Centre of Gravity

KEY POINT

Centre of gravity (G) of a body is the point through which the whole weight of the body appears to act, whatever the orientation of the body.

- A body hung at its centre of gravity balances perfectly, because the net moment of its weight about G is zero.
- G need not lie inside the material. A ring or a bent bar has its G in empty space.
- For a regular, uniform body G is at the geometric centre.

BODY	POSITION OF G
Uniform rod or metre rule	Midpoint of its length
Rectangular or square lamina	Point where the diagonals cross
Circular disc, ring, solid sphere	Centre of the circle or sphere
Triangular lamina	Point where the three medians meet
Solid cylinder	Midpoint of its axis
Solid cone	On the axis, one quarter of the height from the base

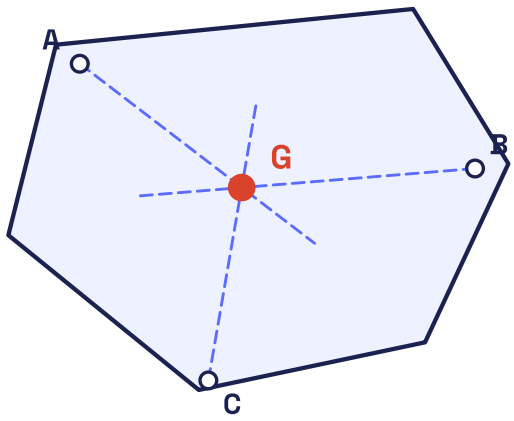
Finding G of an Irregular Lamina

1. Make holes
Punch three small holes A, B, C near the edge of a cardboard sheet cut to any shape.

2. Hang
Hang the card freely from a pin through A, with a plumb line (thread with a weight) hanging from the same pin.

3. Mark
When everything is still, mark where the thread passes over the card and draw the line.

4. Repeat and read
Repeat for B and C. All lines meet at one point. That point is G.



plumb lines from A, B and C all cross at the centre of gravity G

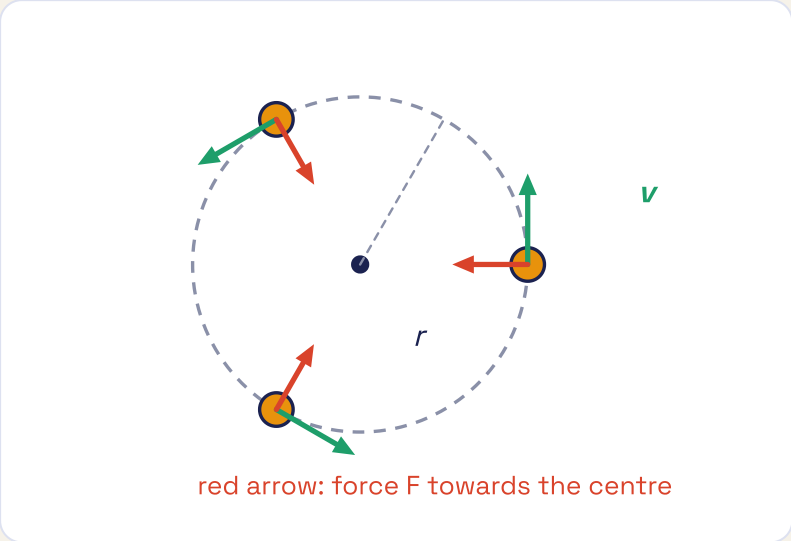
Finding G of an irregular lamina: hang it from different holes, draw the plumb line each time. They meet at G.

REAL LIFE

A cyclist leans into a bend, a tightrope walker holds a long pole, and a waiter carries a tray with an extended arm underneath it. All of them are keeping the centre of gravity above their support.

Uniform Circular Motion

When a body moves along a circle at **constant speed** the motion is called uniform circular motion. Its speed is steady, yet its **velocity changes continuously** because the direction keeps changing. A change in velocity is an acceleration, so a net force must act.



Uniform circular motion: speed is constant but the velocity (green) keeps turning, so a force (red) towards the centre is always needed.

- The velocity at every instant is along the **tangent** to the circle.
- The force, and hence the acceleration, always points **towards the centre** and is perpendicular to the velocity, so it changes direction but not speed.
- Because the force is perpendicular to the displacement along the path, it does **no work**.

Centripetal and Centrifugal Force

Centripetal force is the force that keeps a body moving in a circle. It acts **towards the centre**. Its magnitude for mass m, speed v and radius r is:

$$F = m v^2 / r$$

SITUATION	WHAT PROVIDES THE CENTRIPETAL FORCE
Moon around the Earth	Gravitational pull of the Earth
Stone whirled on a string	Tension in the string
Car on a level bend	Friction between tyres and the road
Electron around the nucleus	Electrical attraction of the nucleus
Clothes in a spinning dryer	The inward push of the drum wall

	CENTRIPETAL FORCE	CENTRIFUGAL FORCE
Direction	Towards the centre	Away from the centre
Nature	A real force acting on the body, due to an agent such as tension, friction or gravity	A pseudo (apparent) force felt in a rotating frame. It is the body's own inertia resisting the turn
Needed for the motion?	Yes, without it the body leaves the circle	No, it only seems to exist to a rotating observer
Example	The string pulling a whirled stone inward	You feeling pushed outward in a turning car



BOARD ALERT

If the string breaks, the stone does **not** fly radially outward. It moves along the tangent, in the direction of its velocity at that instant.

Example 5. A car of mass 1200 kg takes a bend of radius 40 m at 12 m s^{-1}. Find the centripetal force that friction must supply.

```
F = m v^2 / r = 1200 x 12^2 / 40 = 1200 x 144 / 40 = 4320 N
```



FUN FACT

Cream separators, washing machine spin cycles and centrifuges all use circular motion: the liquid is thrown to the outer wall while the heavier part is pushed outward fastest.



HOT QUESTION

A bucket of water is whirled in a vertical circle and no water spills at the top. Which force provides the centripetal force at the top, and why does the water stay in?

Mind Map Recap

IDEA	KEY FACTS
Moment	M = F × d, units N m, cgs dyn cm, 1 N m = 10^7 dyn cm
Equilibrium	Resultant force = 0 and total clockwise moment = total anticlockwise moment
Principle of moments	$W_1d_1 = W_2d_2$ about the pivot
Centre of gravity	Point where weight acts. Qualitative only; plumb-line method for irregular lamina
Circular motion	Constant speed, changing velocity, force towards the centre, $F = mv^2/r$
Centripetal vs centrifugal	Real inward force vs apparent outward effect due to inertia

Practice Questions

#	QUESTION	ANSWER
1	Define the moment of a force and state its SI and cgs units.	Turning effect of a force = force × perpendicular distance from the pivot. SI: N m; cgs: dyn cm.
2	Convert a moment of 3.5 N m into dyn cm.	$3.5 \times 10^7 = 3.5 \times 10^7$ dyn cm.
3	A force of 25 N acts at the end of a 0.8 m wrench. Find the moment.	25 × 0.8 = 20 N m.
4	State the two conditions for a body to be in equilibrium.	Net force zero, and total clockwise moment equals total anticlockwise moment about any point.
5	A 2 m uniform beam of weight 60 N is pivoted at its centre. A 40 N load is hung 0.5 m on one side. Where must a 25 N load hang on the other side?	40 × 0.5 = 25 × d, d = 0.8 m.
6	Why is it easier to open a door by pushing near the handle?	Larger distance from the hinges gives a larger moment for the same force.
7	Where is the centre of gravity of (a) a square sheet (b) a triangular lamina?	(a) Where the diagonals meet. (b) Where the medians meet.
8	Describe how to find G of an irregular lamina.	Hang it from three points in turn with a plumb line, mark each line. They meet at G.
9	Why is the motion of a satellite in a circular orbit accelerated though its speed is constant?	The direction of velocity keeps changing, and a change in velocity is acceleration.
10	A 0.5 kg stone is whirled at 4 m s^{-1} on a 1.0 m string. Find the tension.	$T = mv^2/r = 0.5 \times 16 / 1.0 = 8$ N.
11	Give two differences between centripetal and centrifugal force.	Centripetal: real, towards the centre, causes circular motion. Centrifugal: apparent, away from the centre, due to inertia.
12	A uniform metre rule of weight 1.0 N is pivoted at the 40 cm mark. Where must a 2.0 N weight hang to balance the rule?	Rule's weight acts 10 cm from pivot: 1.0 × 10 = 2.0 × d, d = 5 cm, on the side opposite to the 50 cm mark, i.e. at the 35 cm mark.

02

CHAPTER · PHYSICS

Work, Power and Energy

In physics, holding a heavy bag does no work, but lifting it does. This chapter gives work, energy and power exact meanings, derives the formulas for potential and kinetic energy, and shows that energy is never lost, only changed.



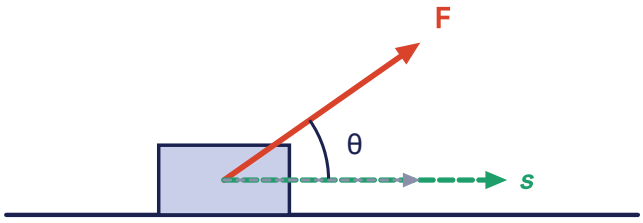
IN THIS CHAPTER

- Work and its units
- Energy and its forms
- Potential and kinetic energy
- Power
- Conservation of energy

Work

KEY POINT

Work is done when a force moves its point of application. Work done = force × displacement in the direction of the force.



F cosθ does the work

Only the component of force along the displacement does work: W = F s cosθ.

$$W = F\,s\cos\theta \quad (\theta \text{ is the angle between the force and the displacement})$$

ANGLE θ	COS θ	WORK DONE	EXAMPLE
0°	1	Maximum, W = F s	Pushing a trolley straight ahead
90°	0	Zero	Carrying a bag while walking on level ground; the earth's pull on a satellite in a circular orbit
180°	−1	Negative, W = −F s	Friction on a sliding block; a brake slowing a car

When a body of mass m is raised through a height h at steady speed, the force needed equals its weight mg and the displacement is vertical, so:

$$W = m\,g\,h$$

The work done **by** gravity as the body falls through h is the same, +mgh. Going up, gravity does negative work.

BOARD ALERT

No displacement means no work in physics. Pushing a wall with all your strength does zero work on the wall, even though you get tired.

Energy and Its Units

KEY POINT

Energy is the capacity to do work. The work a body can do is a measure of its energy, so energy has the same units as work.

UNIT	WHERE IT IS USED	VALUE IN JOULE
joule (J)	SI unit of work and energy: work done when a force of 1 N moves a body 1 m in its own direction	1 J
erg	cgs unit	1 J = 10^7 erg
calorie (cal)	Heat energy	1 cal = 4.2 J
kilowatt hour (kW h)	Electrical energy bills, the commercial unit	1 kW h = 3.6 × 10^6 J
electron volt (eV)	Atomic and nuclear energies	1 eV = 1.6 × 10^{-19} J

Forms of Energy

FORM	MEANING	EVERYDAY SOURCE
Mechanical	Energy due to motion (kinetic) or position (potential)	Moving car, stretched spring, water in a dam
Heat	Energy of the random motion of molecules	Fire, hot water
Chemical	Energy stored in bonds between atoms	Food, petrol, a battery
Electrical	Energy of moving charges	Current in a wire
Light	Energy carried by electromagnetic waves	Sun, a lamp
Sound	Energy of vibrating particles of a medium	Speaker, thunder
Nuclear	Energy released when nuclei change	Uranium in a reactor, the Sun

Energy Conversions

DEVICE	ENERGY CHANGE
Electric heater	Electrical → heat
Loudspeaker	Electrical → sound
Microphone	Sound → electrical
Electric motor	Electrical → mechanical
Dynamo or generator	Mechanical → electrical
Solar cell	Light → electrical
Battery	Chemical → electrical
Nuclear reactor	Nuclear → heat → electrical
Photosynthesis	Light → chemical

Mechanical Energy: Potential Energy



KEY POINT

Gravitational potential energy (U) of a body is the energy it has because of its height above the ground. It equals the work done in lifting it to that height.

Derivation. Lift a body of mass m slowly through a height h. The upward force needed equals the weight, F = mg, and the displacement is h in the direction of the force.

```
Work done against gravity = F x s = (m g) x h
This work is stored as potential energy:
    U = m g h
```

- Height is measured from a chosen reference level (usually the ground), so U is a relative quantity.
- A compressed spring and a stretched rubber band also store potential energy, called elastic potential energy.
- Water behind a dam, a raised hammer and a drawn bow are all examples.

Mechanical Energy: Kinetic Energy



KEY POINT

Kinetic energy (K) is the energy a body has because of its motion. It equals the work needed to bring the body from rest to its speed.

Derivation. A constant force F acts on a body of mass m, starting from rest, and moves it through a distance s, giving a final speed v.

```
From the equation of motion  v^2 = u^2 + 2 a s  with u = 0:   s = v^2 / (2a)
Force:  F = m a
Work done = F x s = m a x v^2 / (2a) = 1/2 m v^2
This work is stored as kinetic energy:
    K = 1/2 m v^2
```

- K depends on the **square** of speed. Doubling the speed makes K four times larger, which is why high speed crashes are so damaging.
- K is never negative.
- **Forms of kinetic energy:** translational (a rolling ball's motion along the ground), rotational (a spinning top or a fan) and vibrational (a vibrating guitar string or atoms in a solid).

Power



KEY POINT

Power is the rate of doing work, or the rate at which energy is transferred.

$$P = W / t \quad \text{and also} \quad P = F v$$

UNIT	VALUE
watt (W), SI	1 W = 1 J s^{-1}
erg per second, cgs	1 W = 10^7 erg s^{-1}
kilowatt (kW)	1 kW = 10^3 W
megawatt (MW)	1 MW = 10^6 W
gigawatt (GW)	1 GW = 10^9 W
horsepower (hp)	1 hp = 746 W



SIDEKICK TIP

kW h is a unit of **energy** (power × time) and not of power. 1 kW h = 1000 W × 3600 s = 3.6 × 10^6 J.

Worked Examples

Example 1. A force of 50 N pulls a sledge through 8.0 m along a rope making 60° with the ground. Find the work done.

```
W = F s cos(theta) = 50 x 8.0 x cos 60 = 50 x 8.0 x 0.5 = 200 J
```

Example 2. How much work is done in lifting a 12 kg bucket of water through 2.5 m? (g = 10 m s^{-2})

```
W = m g h = 12 x 10 x 2.5 = 300 J
```

Example 3. A car of mass 800 kg moves at 20 m s^{-1}. Find its kinetic energy. What is it at 40 m s^{-1}?

```
K = 1/2 m v^2 = 0.5 x 800 x 20^2 = 160 000 J = 160 kJ
At 40 m/s: K = 0.5 x 800 x 1600 = 640 000 J  (4 times larger)
```

Example 4. A student of mass 40 kg climbs a staircase 3.0 m high in 6.0 s. Find the power. (g = 10 m s^{-2})

```
Work = m g h = 40 x 10 x 3.0 = 1200 J
P = W / t = 1200 / 6.0 = 200 W
```

Example 5. A pump lifts 600 kg of water per minute through 20 m. Find its power in kW and horsepower.

```
W = m g h = 600 x 10 x 20 = 120 000 J in 60 s
P = 120 000 / 60 = 2000 W = 2.0 kW
In hp: 2000 / 746 = 2.68 hp
```

Principle of Conservation of Energy



KEY POINT

Energy can neither be created nor destroyed. It can only change from one form to another. The total energy of an isolated system stays constant.

For a freely falling body the mechanical energy stays constant: potential energy falls by exactly as much as kinetic energy rises.

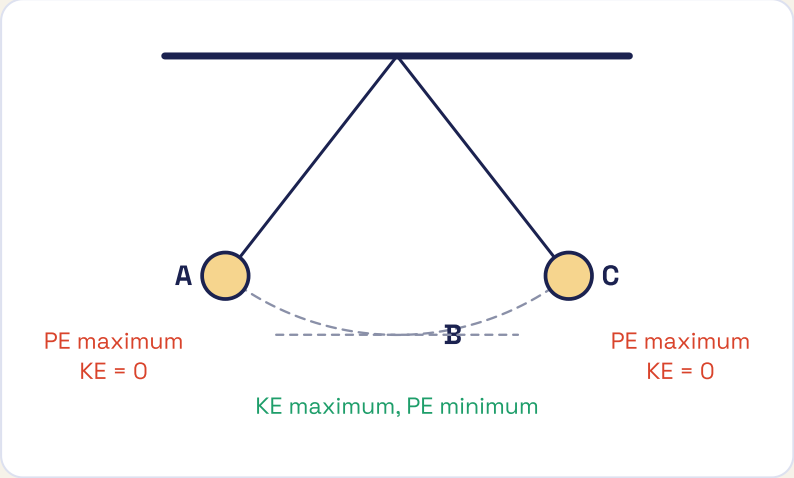
Verification for a Freely Falling Body

A body of mass m is released from rest at height H. After it has fallen a distance x, its height is (H − x) and its speed is given by $v^2 = 2\,g\,x$.

```
Potential energy  U = m g (H - x)
Kinetic energy    K = 1/2 m v^2 = 1/2 m (2 g x) = m g x
U + K = m g H - m g x + m g x = m g H   (a constant)
```

At the top x = 0, so U = mgH and K = 0. At the ground x = H, so U = 0 and K = mgH. At every point in between, the total is the same.

2 kg mass released from 20 m (g = 10)	height	speed²	U (J)	K (J)	U + K (J)
Start	20 m	0	400	0	400
After falling 5 m	15 m	100	300	100	400
After falling 10 m	10 m	200	200	200	400
Just before ground	0 m	400	0	400	400



Simple pendulum: energy swaps between potential and kinetic, and their sum stays constant (if air resistance is ignored).

Simple pendulum (qualitative). At the extreme positions the bob is momentarily at rest and highest, so all its energy is potential. At the lowest point the speed is greatest and all its energy is kinetic. In reality air resistance and friction at the pivot turn some mechanical energy into heat, so the swings slowly die out, yet the total energy is still conserved.

Example 6. A 0.20 kg ball is dropped from a height of 5.0 m. Find its speed just before it hits the ground.

```
Loss of PE = gain of KE:  m g h = 1/2 m v^2
v = sqrt(2 g h) = sqrt(2 x 10 x 5.0) = sqrt(100) = 10 m/s
Check: K = 0.5 x 0.20 x 100 = 10 J = m g h = 0.20 x 10 x 5.0 = 10 J
```



REAL LIFE

A hydroelectric station turns the potential energy of dam water into kinetic energy, then into the rotational energy of turbines, and finally into electrical energy.



FUN FACT

A cricket ball hit at 30 m/s carries about 100 times more kinetic energy than the same ball tossed gently at 3 m/s, because K goes as speed squared.



HOT QUESTION

Two bodies of masses m and 4m have the same kinetic energy. What is the ratio of their speeds, and which stops sooner under the same braking force?

Mind Map Recap

QUANTITY	FORMULA	SI UNIT
Work	$W = F\ s\ \cos\theta$; $W = mgh$	joule (J)
Potential energy	$U = mgh$	J
Kinetic energy	$K = \frac{1}{2}\ m\ v^2$	J
Power	$P = W/t = F\ v$	watt (W)
Conservation	$U + K$ = constant (free fall)	J

Unit links: $1\ J = 10^7$ erg; 1 cal = 4.2 J; $1\ kW\ h = 3.6 \times 10^6$ J; $1\ eV = 1.6 \times 10^{-19}$ J; 1 hp = 746 W.

Practice Questions

#	QUESTION	ANSWER
1	Define work. When is the work done by a force zero?	Work = force × displacement in the direction of the force. It is zero when displacement is zero or force is perpendicular to displacement.
2	A coolie carries a 20 kg load along a level road for 100 m. Is any work done against gravity? Explain.	No. The force (weight) is perpendicular to the displacement, θ = 90°, cos 90° = 0.
3	A boy of mass 45 kg climbs a 4 m high wall. Find the work done. (g = 10)	W = mgh = 45 × 10 × 4 = 1800 J.
4	Derive an expression for kinetic energy of a body of mass m moving at speed v.	W = F s = m a × $v^2/2a$ = ½ m v^2.
5	A 2.0 kg stone is thrown vertically up with 20 m s^{-1}. Find the maximum height by energy conservation. (g = 10)	½ m v^2 = mgh, h = $v^2/2g$ = 400/20 = 20 m.
6	Convert 5 kW h to joule.	5 × 3.6 × 10^6 = 1.8 × 10^7 J.
7	A motor of 3 hp runs for 10 minutes. How much work does it do?	P = 3 × 746 = 2238 W; W = 2238 × 600 = 1.34 × 10^6 J.
8	State the energy conversions in (a) a dynamo (b) a microphone (c) a solar cell.	(a) Mechanical to electrical (b) Sound to electrical (c) Light to electrical.
9	A body of mass 3 kg moves at 4 m s^{-1}. By how much does its K change if its speed becomes 8 m s^{-1}?	K_1 = 24 J, K_2 = 96 J, increase = 72 J.
10	Show that U + K is constant for a body falling freely.	U = mg(H − x), K = mgx. Sum = mgH. A constant.
11	Why does the pendulum finally stop though energy is conserved?	Mechanical energy is changed into heat by air resistance and friction at the support. The total energy is still conserved.
12	A 60 W lamp is on for 5 hours. Energy used in kW h and in joule?	0.06 kW × 5 h = 0.30 kW h = 0.30 × 3.6 × 10^6 = 1.08 × 10^6 J.

03

CHAPTER · PHYSICS

Machines

A crowbar lifts a car's corner. A pulley hoists a bucket from a well. Machines do not create energy, they let a small effort handle a big load, or change its direction. Learn the three numbers that describe them: MA, VR and efficiency.



IN THIS CHAPTER

- Load, effort, MA, VR
- Efficiency
- Levers: three classes
- Pulley systems
- Block and tackle

What a Machine Does

A **machine** is a device that makes work easier by changing the size or the direction of a force. It does **not** save work. What you gain in force you lose in distance.

- **Force multiplier:** a small effort moves a large load (crowbar, pulley block, spanner).
- **Speed multiplier:** a big movement of the effort gives a larger movement of the load (a cricket bat, tongs).
- **Change of direction:** pulling down to lift up (a fixed pulley on a flagpole).

Terms to Know

TERM	MEANING	FORMULA
Load (L)	The force the machine overcomes. It is the useful work we want done	—
Effort (E)	The force we apply to the machine	—
Mechanical advantage (MA)	How many times the machine multiplies the effort. No unit	MA = L / E
Velocity ratio (VR)	Distance moved by effort divided by distance moved by load in the same time. No unit	VR = d_E / d_L
Work input (W_i)	Work done by the effort	E × d_E
Work output (W_o)	Work done on the load	L × d_L
Efficiency (η)	Fraction of the input work that becomes useful output	η = W_o / W_i (× 100 for %)

Relation Between Efficiency, MA and VR

```
Efficiency  eta = Work output / Work input = (L x d_L) / (E x d_E)
             = (L / E) x (d_L / d_E)
             = MA x (1 / VR)

    eta = MA / VR        (as a percentage:  eta = (MA / VR) x 100)
```



KEY POINT

For all real machines, η < 1 (below 100%) and MA < VR. Some input work is always wasted against friction and in lifting the moving parts of the machine. An **ideal** machine would have η = 1 and MA = VR.

SIDEKICK TIP

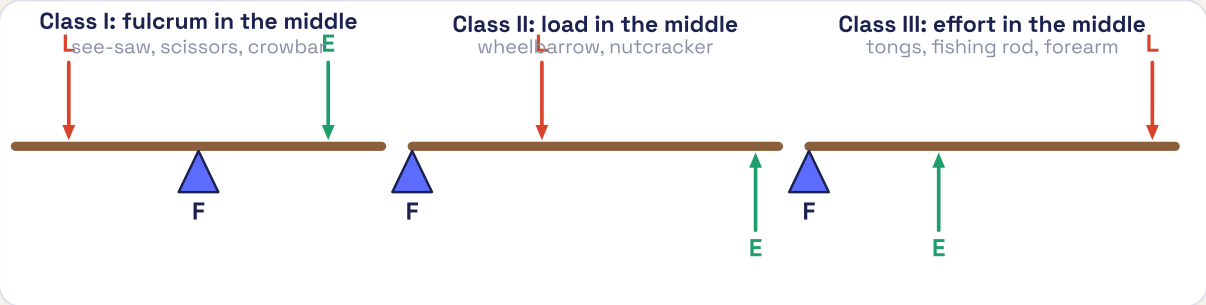
VR depends only on the **geometry** of the machine (distances, number of strands), so it is a fixed number. MA depends on how much friction there is. Lubricating a machine increases MA and efficiency, but VR stays the same.

Levers

A **lever** is a rigid bar that turns about a fixed point called the **fulcrum** (F). The principle comes from moments: in balance, the load moment equals the effort moment.

$$\text{Load} \times \text{load arm} = \text{Effort} \times \text{effort arm} \Rightarrow \mathbf{MA = L/E = \text{effort arm} / \text{load arm}} \text{ (ideal lever)}$$

The **load arm** is the perpendicular distance from the fulcrum to the line of action of the load, and the **effort arm** is the same for the effort. For an ideal lever, VR = effort arm / load arm too. So MA = VR for a lever with no friction and a weightless beam.



The three classes of lever. F = fulcrum, L = load, E = effort. Only the relative order along the beam changes.

CLASS	ARRANGEMENT ALONG THE BAR	EFFORT ARM VS LOAD ARM	MA	EXAMPLES	HUMAN BODY
I	Fulcrum between load and effort	Either can be longer	Greater than, equal to or less than 1	See-saw, crowbar, scissors, pliers, claw hammer	Head nodding on the neck joint (the skull pivots on the spine)
II	Load between fulcrum and effort	Effort arm is always longer	Always greater than 1	Wheelbarrow, nutcracker, bottle opener	Standing on tiptoe (ball of foot is the fulcrum, body weight the load, calf muscle the effort)
III	Effort between fulcrum and load	Effort arm is always shorter	Always less than 1	Tongs, fishing rod, forceps, broom	Forearm lifting a weight (elbow is the fulcrum, biceps the effort)

SIDEKICK TIP

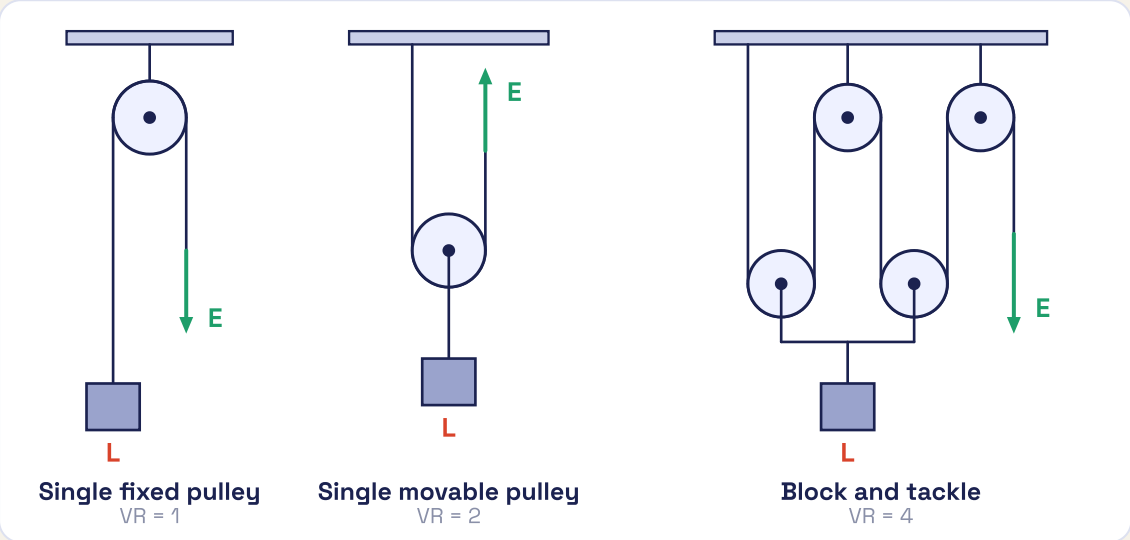
Memory hook: the **middle** item names the class in the order F-L-E. Class I: F in middle. Class II: L in middle. Class III: E in middle.

REAL LIFE

Class III levers look wasteful (MA < 1), but they give **speed and range**. The tip of a fishing rod moves much farther than your hand, and tongs let you pick up a hot coal with delicate control.

Pulley Systems

A pulley is a grooved wheel around which a rope runs. A **fixed** pulley is attached to a support. A **movable** pulley moves up and down with the load.



Pulley systems. VR equals the number of rope strands that support the load (not counting the strand the effort pulls downward over the last fixed pulley).

SYSTEM	VR	MA (IDEAL)	SPECIAL FEATURE
Single fixed pulley	1	1	Changes direction of the effort only. A downward pull raises the load
Single movable pulley (effort upward)	2	2	Load is shared by two strands. Effort moves twice as far as the load
Block and tackle with n supporting strands	n	n	Can lift very heavy loads with a small effort. Needs a long length of rope

- **VR of any pulley system = number of strands of rope supporting the movable block (including the load).** If the load rises 1 m, each of those strands must shorten by 1 m, so the effort moves n m.
- In a real system the effort has to be larger than L/n, because of friction and the weight of the movable pulley. So MA < VR.
- A single movable pulley can be combined with a fixed pulley to change the effort's direction to downward without altering the VR of 2.

Worked Examples

Example 1 (lever). A lever has a load arm of 0.20 m and an effort arm of 0.60 m. A load of 300 N is raised by a real effort of 120 N. Find VR, MA and efficiency.

```
VR = effort arm / load arm = 0.60 / 0.20 = 3
MA = L / E = 300 / 120 = 2.5
Efficiency = MA / VR = 2.5 / 3 = 0.833 = 83.3 %
Ideal effort would have been 300 / 3 = 100 N
```

Example 2 (block and tackle). A pulley system with 4 supporting strands raises a load of 800 N through 2.0 m with an effort of 250 N. Find MA, VR, efficiency, and the work input and output.

```
VR = 4,  so effort moves d_E = 4 x 2.0 = 8.0 m
MA = 800 / 250 = 3.2
Work input  = E x d_E = 250 x 8.0 = 2000 J
Work output = L x d_L = 800 x 2.0 = 1600 J
Efficiency  = 1600 / 2000 = 0.80 = 80 %   (check: MA/VR = 3.2/4 = 0.80)
```

Example 3 (single movable pulley). A load of 500 N is lifted by a single movable pulley using a force of 280 N. Find the efficiency.

```
VR = 2,   MA = 500 / 280 = 1.786
Efficiency = MA / VR = 1.786 / 2 = 0.893 = 89.3 %
```

Example 4 (class II lever). A wheelbarrow carries 600 N of sand, placed 0.40 m from the wheel axle. The handles are 1.2 m from the axle. Find the effort needed (ideal).

```
Taking moments about the axle (fulcrum):  E x 1.2 = 600 x 0.40
E = 240 / 1.2 = 200 N      MA = 600 / 200 = 3
```

Example 5 (class III lever). A pair of tongs holds a 2 N coal 20 cm from the hinge. Fingers press 8 cm from the hinge. Find the effort.

```
E x 8 = 2 x 20   so   E = 40 / 8 = 5 N      MA = 2 / 5 = 0.4  (less than 1)
```



BOARD ALERT

Always check: a MA bigger than VR, or an efficiency over 100%, means an error. Real machines cannot do that.



HOT QUESTION

A pulley system has VR = 5 and efficiency 80%. What effort lifts a 400 N load? And how far does the effort end move when the load rises 3 m?

Mind Map Recap

QUANTITY	FORMULA
MA	L / E
VR	d_E / d_L (lever: effort arm / load arm; pulley: number of supporting strands)
Efficiency	$\eta = W_o / W_i = MA / VR$ (× 100 for %)
Ideal machine	$\eta = 1$ and $MA = VR$
Real machine	$\eta < 1$ and $MA < VR$
Lever balance	L × load arm = E × effort arm

Practice Questions

#	QUESTION	ANSWER
1	Define mechanical advantage and velocity ratio.	MA = load/effort. VR = distance moved by effort / distance moved by load.
2	Derive the relation between efficiency, MA and VR.	$\eta = (L\ d_L)/(E\ d_E) = (L/E)(d_L/d_E) = MA/VR$.
3	Why is MA of a real machine always less than VR?	Friction and the weight of moving parts mean extra effort is needed, so the actual effort is greater than the ideal, which lowers MA.
4	State the class of lever for (a) scissors (b) nutcracker (c) tongs (d) wheelbarrow.	(a) I (b) II (c) III (d) II.
5	A crowbar 1.5 m long has its fulcrum 0.3 m from the load. What is its VR? What effort lifts a 600 N load (ideal)?	Effort arm 1.2 m, load arm 0.3 m. VR = 4. E = 600/4 = 150 N.
6	A pulley system with VR = 3 lifts 450 N with an effort of 200 N. Find MA and efficiency.	MA = 2.25; η = 2.25/3 = 75%.
7	In a block and tackle of VR 5 the load is raised by 0.6 m. How far does the effort move?	0.6 × 5 = 3.0 m.
8	Why is a single fixed pulley still useful if its MA is 1?	It changes the direction of the effort, so we can pull downward and use our body weight.
9	An effort of 50 N moves 4 m and raises a load of 120 N by 1.5 m. Find efficiency.	Wi = 200 J, Wo = 180 J, η = 90%.
10	Name one lever in the human body for each class.	I: skull on the neck. II: foot on tiptoe. III: forearm lifting a weight.
11	How can efficiency of a pulley system be increased?	Lubricate the pulleys, use lighter movable pulleys and light rope.
12	Classify a fishing rod and explain why MA < 1 is acceptable.	Class III. It trades force for speed and range of motion at the tip.

04

CHAPTER · PHYSICS

Refraction and Total Internal Reflection

A straw looks bent in a glass of water and a swimming pool looks shallower than it is. Light changes direction when it moves between media. This chapter builds the laws behind that bending, and then shows how light can be trapped inside glass.



IN THIS CHAPTER

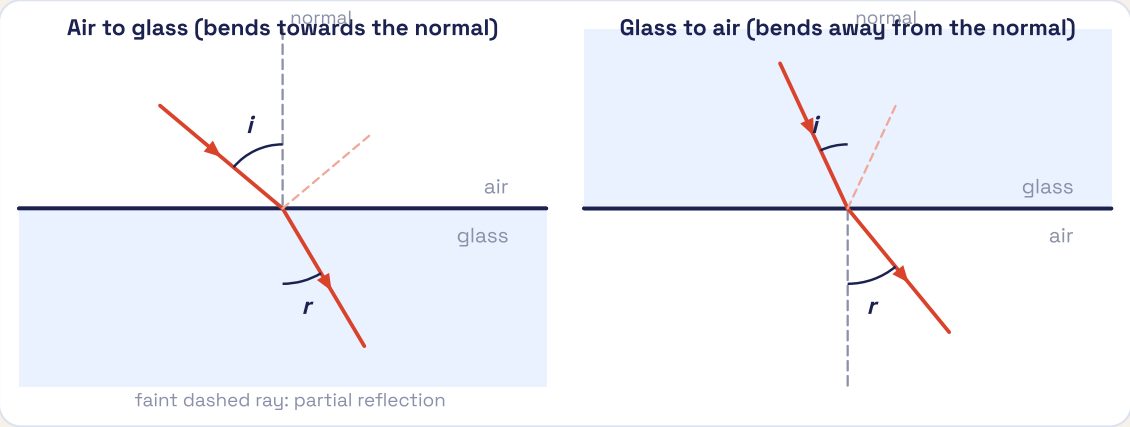
- Laws of refraction
- Refractive index
- Glass slab and prism
- Real and apparent depth
- Critical angle and TIR
- Right-angled prisms

Light Meets a New Medium

When a ray of light reaches the boundary between two transparent media, a small part of it bounces back (**partial reflection**) and the rest enters the second medium with a change in direction. That change of direction is called **refraction**.

KEY POINT

Refraction of light is the change in the direction of a light ray when it passes from one transparent medium to another, caused by a change in the speed of light.



At a boundary part of the light reflects (dashed, faint) and the rest refracts. Here μ = 1.5.

- **Rarer to denser** (air to glass): the ray bends **towards** the normal, so r < i.
- **Denser to rarer** (glass to air): the ray bends **away from** the normal, so r > i.
- The angle of incidence i and the angle of refraction r are both measured from the **normal**, the line perpendicular to the surface, and never from the surface itself.

Laws of Refraction

1. First law

The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.

2. Second law (Snell's law)

For light of a given colour passing between two given media, sin i / sin r is a constant. This constant is the refractive index of the second medium with respect to the first.

$$\sin i / \sin r = \mu \text{ (a constant for the two media)}$$

Refractive Index

KEY POINT

Refractive index (μ) of a medium is the ratio of the speed of light in vacuum to its speed in that medium: **μ = c / v**. It has no unit.

$$\mu = c / v \quad \text{where } c = 3 \times 10^8 \text{ m s}^{-1} \text{ (vacuum, almost the same in air)}$$

MEDIUM	REFRACTIVE INDEX M	SPEED OF LIGHT (M S^{-1})
Vacuum / air	1.00	3.00×10^8
Water	1.33	2.26×10^8
Glass (crown)	1.50	2.00×10^8
Diamond	2.42	1.24×10^8

Light slows down in a denser medium. A bigger μ means a slower speed and more bending. A medium that bends light more is called **optically denser**, which is not the same as having higher mass density.

What Changes on Refraction?

QUANTITY	AIR TO GLASS	WHY
Speed (v)	Decreases	Light travels slower in glass
Wavelength (λ)	Decreases	v = f λ, and f stays constant, so λ falls in step with v
Frequency (f)	**Unchanged**	The source fixes the frequency; it does not depend on the medium
Direction	Changes (unless i = 0)	Different speeds on the two sides of the boundary

$$v = f\lambda \quad \Rightarrow \quad \mu = c/v = \lambda_{vacuum} / \lambda_{medium}$$



SIDEKICK TIP

Memory line: **frequency is the survivor**. It never changes at a boundary. Speed and wavelength change together.

When Does Light Pass Without Bending?

- When it hits the surface **normally** (i = 0), so r = 0.
- When the two media have the **same refractive index** (a glass rod looks invisible in a liquid of equal index).

Verifying Snell's Law

1. Place the slab
Fix a sheet of paper on a board and trace the outline of a rectangular glass slab.

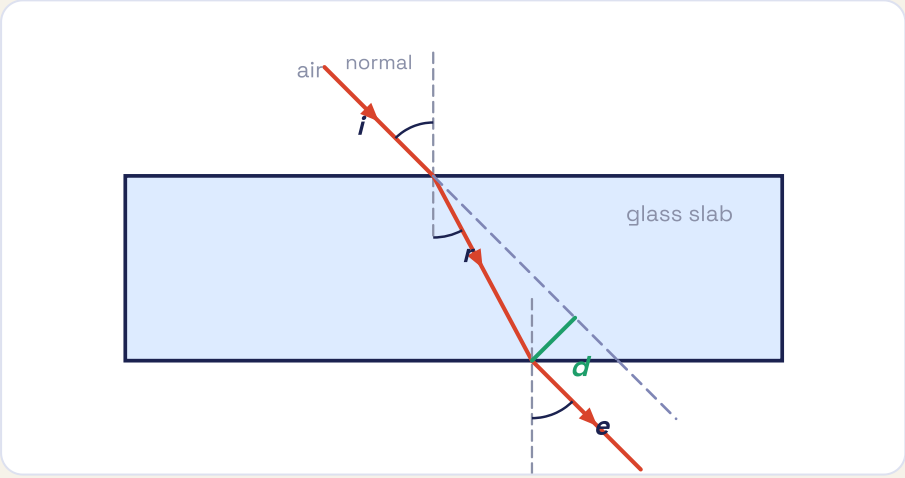
2. Pins on one side
Draw a normal and an incident line at angle i. Push two pins P_1 and P_2 on this line.

3. Pins on the other side
Looking through the slab from the opposite face, place pins P_3 and P_4 so that all four pins appear in one line.

4. Join and measure
Remove the slab, join P_3P_4 to meet the outline, join the point of entry to the point of exit, then measure i and r.

5. Plot
Repeat for 5 angles, plot sin i against sin r. A straight line through the origin confirms Snell's law and its slope is μ.

Refraction Through a Rectangular Glass Block

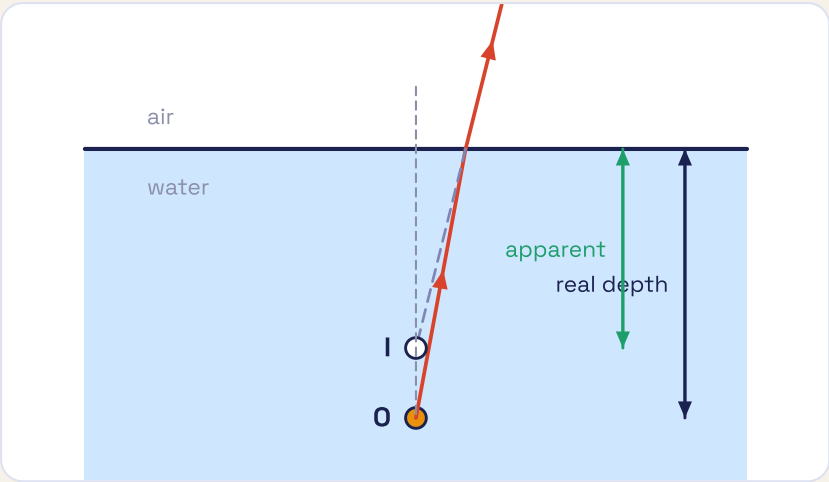


Rectangular glass slab: the emergent ray is parallel to the incident ray (e = i) but shifted sideways by the lateral displacement d.

- The ray bends towards the normal on entering, and by the same amount away from it on leaving. Since the two faces are parallel, the **emergent ray is parallel to the incident ray** (e = i).
- The ray is not deviated overall but is displaced sideways. This shift is the **lateral displacement**, the perpendicular distance between the incident ray (extended) and the emergent ray.
- Lateral displacement increases with the **thickness** of the slab, the **angle of incidence** and the **refractive index**. It is zero for normal incidence.

Multiple images. Look at a candle flame through a thick glass plate or a thick mirror at an angle and you see several images, with one very bright. The front surface and the back (silvered) surface both reflect, and the light also bounces back and forth inside the glass, giving a series of fainter images.

Real and Apparent Depth



An object in water appears raised. The refracted rays bend away from the normal and seem to come from the shallower point I.

KEY POINT

Refractive index of water = real depth / apparent depth (for viewing from directly above).

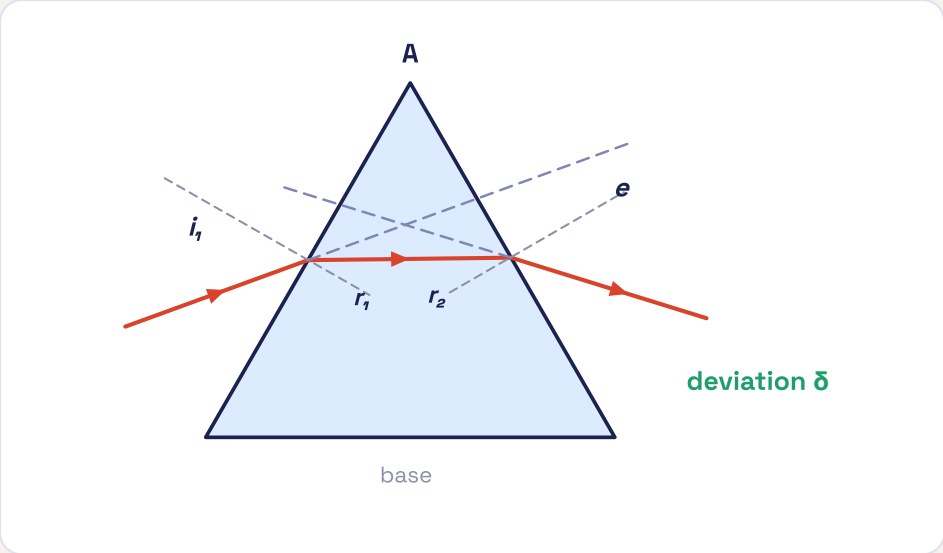
$$\mu = \text{real depth} / \text{apparent depth} \quad \Rightarrow \quad \text{apparent depth} = \text{real depth} / \mu$$

- A swimming pool, a pond or a glass of water looks shallower than it really is.
- A coin at the bottom of a bowl seems to rise when the bowl is filled with water.
- A stick dipped in water at a slant seems bent upward at the surface, because the part underwater seems raised.
- Through a slab of thickness t, an object appears raised by t (1 − 1/μ).

Example 1. A pool is 2.4 m deep. How deep does it appear from above? (μ of water = 1.33)

```
apparent depth = real depth / mu = 2.4 / 1.33 = 1.80 m
```

Refraction Through a Triangular Prism



Refraction through a prism: the ray bends towards the base twice. δ is the angle between the incident ray and the emergent ray. Relations: $r_1 + r_2 = A$ and $\delta = i_1 + e - A$.

A prism has two refracting faces inclined at the **angle of the prism A**. The ray bends towards the base on both faces, so the emergent ray is bent from its original direction. The angle between the incident and emergent rays is the **angle of deviation δ**.

$$r_1 + r_2 = A \quad \text{and} \quad i_1 + e = A + \delta$$

In the diagram, with μ = 1.5 and i_1 = 50° on a 60° prism, r_1 = 30.7°, r_2 = 29.3°, e = 47.2° and δ = 37.2°. (To compute: sin r_1 = sin 50°/1.5, r_2 = 60° − r_1, sin e = 1.5 sin r_2.)

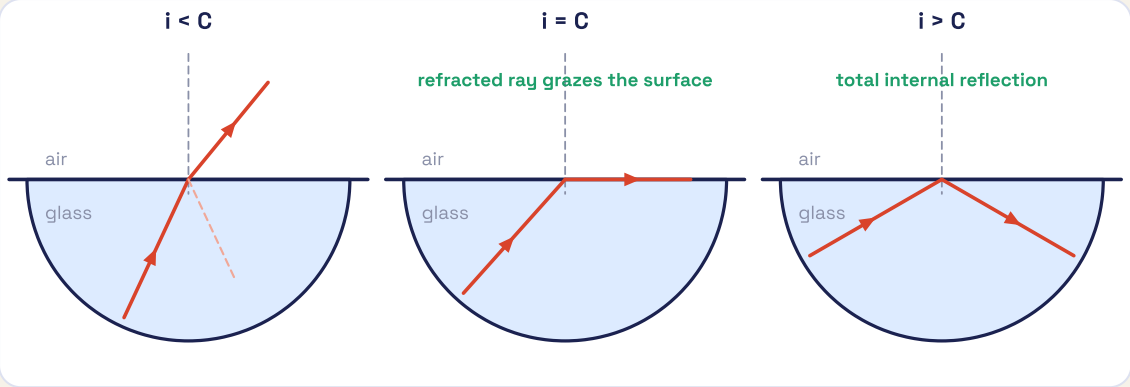
- A ray that enters **normally** (i_1 = 0) goes straight inside to the second face, where the angle of incidence equals A. It then either refracts out with deviation or is totally reflected, depending on A and the critical angle.
- In the prism experiment you trace rays at i_1 = 40°, 50°, 60° and 70°, measure i_1, i_2 (= e), A and δ, and check that $i_1 + e = A + \delta$ and $r_1 + r_2 = A$.

Total Internal Reflection (TIR)

Send light from glass or water towards air. As the angle of incidence i grows, the refracted ray bends further from the normal. At one special angle of incidence the refracted ray skims along the surface with r = 90°. Beyond that angle, no light leaves at all.

KEY POINT

Critical angle (C) is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90°.



Glass (μ = 1.5) to air through a semicircular block. The critical angle here is C = 41.8°. For i > C all the light stays in the glass.

$$\mu = \sin 90° / \sin C \quad \Rightarrow \quad \mu = 1 / \sin C$$

MEDIUM (TO AIR)	M	CRITICAL ANGLE C
Water	1.33	48.8°
Crown glass	1.50	41.8°
Diamond	2.42	24.4°

KEY POINT

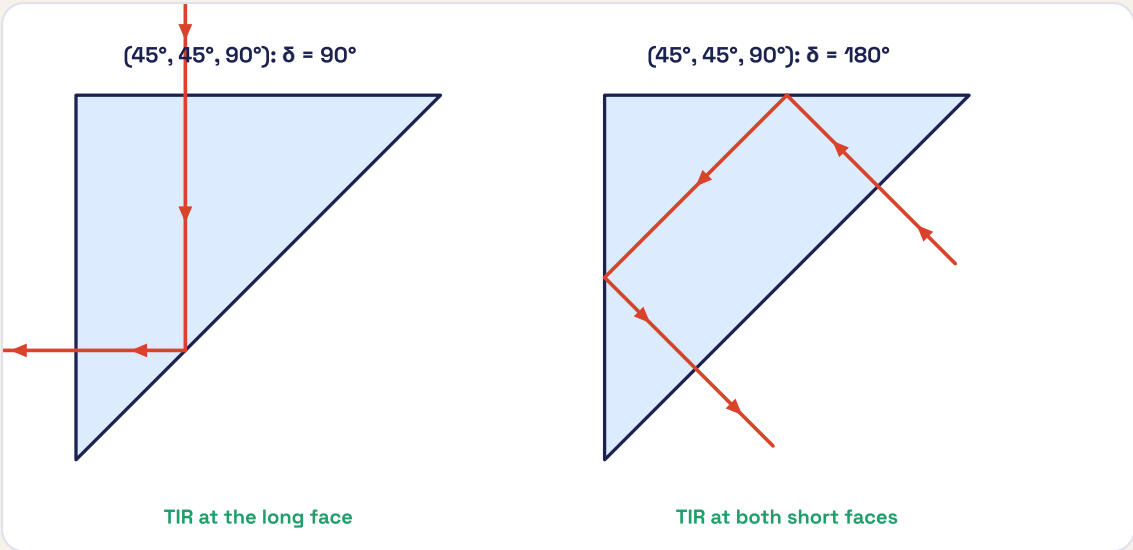
Conditions for TIR: (1) light travels from a **denser to a rarer** medium; (2) the angle of incidence is **greater than the critical angle**.

Diamond's tiny critical angle (24.4°) means light inside is trapped and reflected several times before exiting, which is why it sparkles so much.

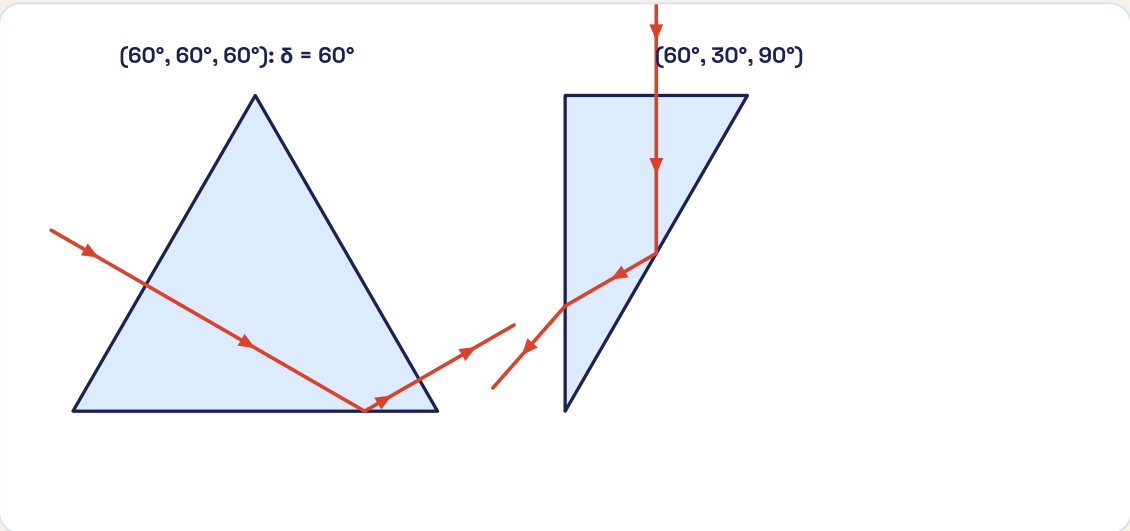
Total Internal Reflection in Prisms

In a prism the second face is where the ray can be trapped. For a ray that enters a face **normally**, the angle of incidence at the next face equals the angle between the two faces. With glass (C = 41.8°):

PRISM ANGLES	RAY ENTERS NORMALLY ON	ANGLE OF INCIDENCE AT NEXT FACE	RESULT
45°, 45°, 90°	A short face	45° > C	TIR at the long face, ray leaves the other short face. **δ = 90°**
45°, 45°, 90°	The long face	45° > C twice	TIR at both short faces, ray returns parallel to the incident path. **δ = 180°**
60°, 60°, 60°	One face (lower half)	60° > C	TIR at the base, then leaves the third face normally. δ = 60°
60°, 30°, 90°	The shorter face	60° > C	TIR at the long face, then leaves the other short face by refraction



Right-angled prisms with a ray entering normally. Light is turned through 90° (left) or sent straight back, δ = 180° (right). The incidence angle inside is 45°, larger than C = 41.8°.



Left: in an equilateral prism a normally incident ray meets the base at 60°, suffers TIR and leaves normally through the third face. Right: in a 30-60-90 prism a ray entering the shorter face hits the hypotenuse at 60°, is totally reflected and then leaves the other face.

TIR Compared with a Plane Mirror

	TOTAL INTERNAL REFLECTION	REFLECTION AT A SILVERED MIRROR
Reflected fraction	About 100%, no loss	Roughly 90 to 95%, some light is absorbed
Image	Very bright	Slightly dimmer
Surface	Needs no coating, does not tarnish	Silvering wears and tarnishes
When it happens	Only if i > C in a denser medium	Always, at any angle

Applications of TIR

- **Right-angled prism** in periscopes and prismatic binoculars turns the light through 90° or 180° with no silvering.
- **Optical fibres** carry light along bent glass threads by repeated TIR. They are used for internet cables and in the **endoscope** that lets a doctor look inside the body.

- **Reflectors** on bicycles and road signs, made of tiny prisms that send light straight back (180°).
- **Sparkle of diamond** because its critical angle is small.
- **Mirage** in deserts: hot air near the ground is rarer, and light from the sky is totally reflected upward.

REAL LIFE

Fibre-optic cables send nearly all of our internet data under the ocean. The pulse of light inside never leaks out as long as it keeps striking the glass-air boundary at angles bigger than C.

Worked Examples

Example 2. A ray of light strikes a glass slab (μ = 1.5) at 45°. Find the angle of refraction.

```
sin r = sin i / mu = sin 45 / 1.5 = 0.7071 / 1.5 = 0.4714
r = 28.1 degrees
```

Example 3. Light of wavelength 6.0×10^{-7} m in vacuum enters glass of μ = 1.5. Find its frequency, speed and wavelength in glass.

```
f = c / lambda = 3.0e8 / 6.0e-7 = 5.0e14 Hz (unchanged in glass)
v = c / mu = 3.0e8 / 1.5 = 2.0e8 m/s
lambda_glass = v / f = 2.0e8 / 5.0e14 = 4.0e-7 m
```

Example 4. The refractive index of diamond is 2.4. Find the speed of light in diamond and the critical angle.

```
v = c / mu = 3.0e8 / 2.4 = 1.25e8 m/s
sin C = 1 / mu = 1 / 2.4 = 0.4167   so  C = 24.6 degrees
```

Example 5. A coin lies under a 6 cm thick glass slab (μ = 1.5). By how much does it appear to be raised when viewed from above?

```
shift = t (1 - 1/mu) = 6 x (1 - 1/1.5) = 6 x 0.333 = 2.0 cm
```

BOARD ALERT

Write refractive index as **μ = sin i / sin r** for air to medium. When going from medium to air the ratio is inverted: sin i / sin r = 1/μ.

HOT QUESTION

A ray goes from glass (μ = 1.5) into water (μ = 1.33). Will total internal reflection occur at an angle of incidence of 60°? Compute the critical angle for this pair first.

Mind Map Recap

IDEA	FORMULA OR FACT
Snell's law	$\sin i / \sin r = \mu$ (constant)
Refractive index	$\mu = c / v = \lambda_{vac} / \lambda_{med}$
Unchanged on refraction	Frequency
Real and apparent depth	μ = real depth / apparent depth
Prism	$r_1 + r_2 = A$; $i_1 + e = A + \delta$
Critical angle	$\mu = 1 / \sin C$
TIR conditions	Denser to rarer, $i > C$
Right-angled prism	$\delta = 90°$ or $180°$

Practice Questions

#	QUESTION	ANSWER
1	State the laws of refraction of light.	(1) Incident ray, refracted ray and normal lie in one plane. (2) sin i / sin r is constant for a given pair of media and colour.
2	Light goes from air into water at i = 50°. Find r (μ = 1.33).	sin r = 0.766/1.33 = 0.576; r = 35.2°.
3	Why does a pencil in a glass of water look bent?	Rays from the part in water bend away from the normal as they leave the water, so that part appears raised.
4	Speed of light in a liquid is 2.25×10^8 m s^{-1}. Find the refractive index.	μ = 3/2.25 = 1.33.
5	A tank is 1.2 m deep and filled with water (μ = 4/3). How deep does it appear?	1.2 × 3/4 = 0.90 m.
6	What is the critical angle for a medium of μ = 1.6?	sin C = 1/1.6 = 0.625, C = 38.7°.
7	State two conditions for TIR.	Ray goes from a denser to a rarer medium; angle of incidence exceeds the critical angle.
8	Why does a diamond sparkle more than glass?	Its critical angle (24.4°) is much smaller, so more light is totally reflected inside.
9	A ray strikes a face of a right-angled glass prism normally and emerges after TIR from another face at 90° to its path. Name the angles of the prism.	45°, 45°, 90°, with the ray entering a short face.
10	Draw a ray diagram for refraction through a rectangular slab and mark the lateral displacement.	As in the chapter: parallel emergent ray shifted by d.
11	Compare the frequency, wavelength and speed of red light in air and in glass.	Frequency same; wavelength and speed smaller in glass by factor 1.5 (for μ = 1.5).
12	Name two uses of optical fibres and explain the principle.	Communication and endoscopy; they use repeated total internal reflection.

05

CHAPTER · PHYSICS

Lenses

Cameras, spectacles, microscopes and your own eyes all depend on lenses. A lens bends light in a controlled way to form images. You will learn to draw the rays, describe the image and use the lens formula.



IN THIS CHAPTER

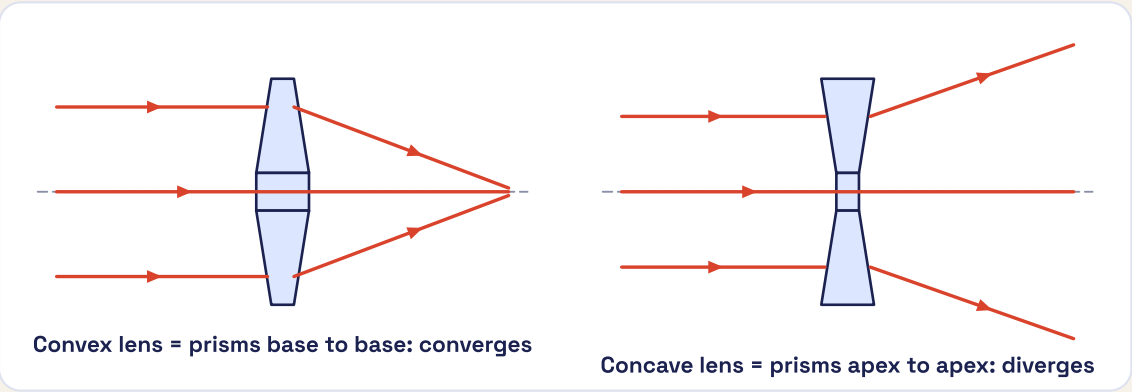
- Convex and concave lenses
- Principal rays
- Image characteristics
- Lens formula
- Power of a lens
- Magnifying glass

Types of Lens

A **lens** is a piece of transparent material (usually glass) bounded by two curved surfaces, or one curved and one plane surface.

	CONVEX (CONVERGING) LENS	CONCAVE (DIVERGING) LENS
Shape	Thicker at the centre than at the edges	Thinner at the centre than at the edges
Effect on a parallel beam	Brings the rays together to a real point	Spreads the rays so they seem to come from a point
Principal focus	Real	Virtual
Image	Real or virtual	Always virtual
Other names	Bi-convex, plano-convex, converging	Bi-concave, plano-concave, diverging

Lens as a Set of Prisms



A lens behaves like a stack of prisms. Each prism bends light towards its base. The centre acts like a thin parallel slab and passes light undeviated.

Technical Terms

TERM	MEANING
Optical centre (O)	The central point of the lens. A ray passing through it goes straight without bending
Principal axis	The line through the optical centre joining the centres of curvature of the two surfaces
Centre of curvature (C)	Centre of each spherical surface of the lens. A lens has two, written 2F on each side for thin lenses
Radius of curvature (R)	Radius of the spherical surface. R = 2f
Aperture	The diameter of the lens surface that takes part in refraction
Principal focus (F)	The point where rays parallel to the principal axis meet (convex) or seem to come from (concave) after refraction. A lens has two, F_1 and F_2, on either side
Focal length (f)	The distance from the optical centre to the principal focus
Focal plane	The plane through the principal focus perpendicular to the principal axis

Rules for Drawing Rays

To locate the image of one point of an object (usually its tip), you only need two of the three rays below. The image point is where the refracted rays meet (real) or appear to meet (virtual).

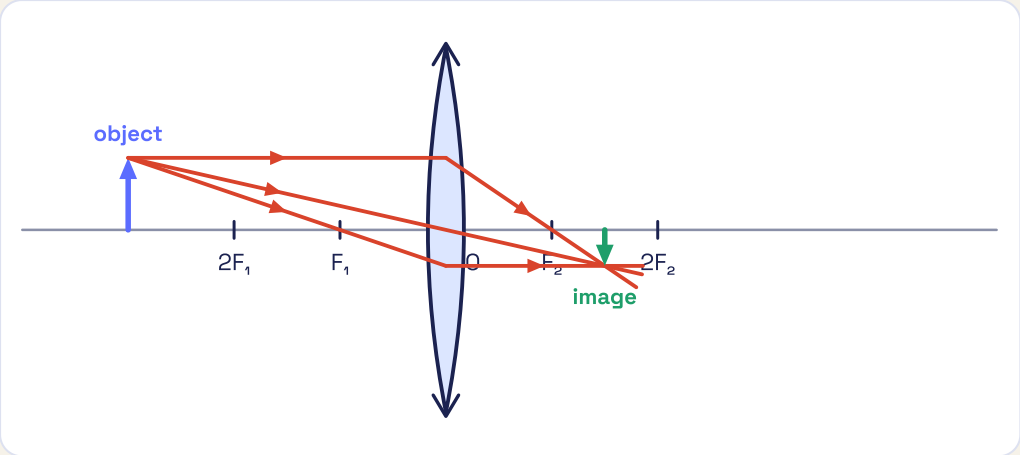
RAY	CONVEX LENS	CONCAVE LENS
1. Parallel to the principal axis	Refracts through the second focus F_2	Refracts so that it appears to come from F_1 (on the object side)
2. Through the optical centre	Goes straight, undeviated	Goes straight, undeviated
3. Directed through the first focus F_1 (convex) or towards F_2 (concave)	Emerges parallel to the principal axis	Emerges parallel to the principal axis



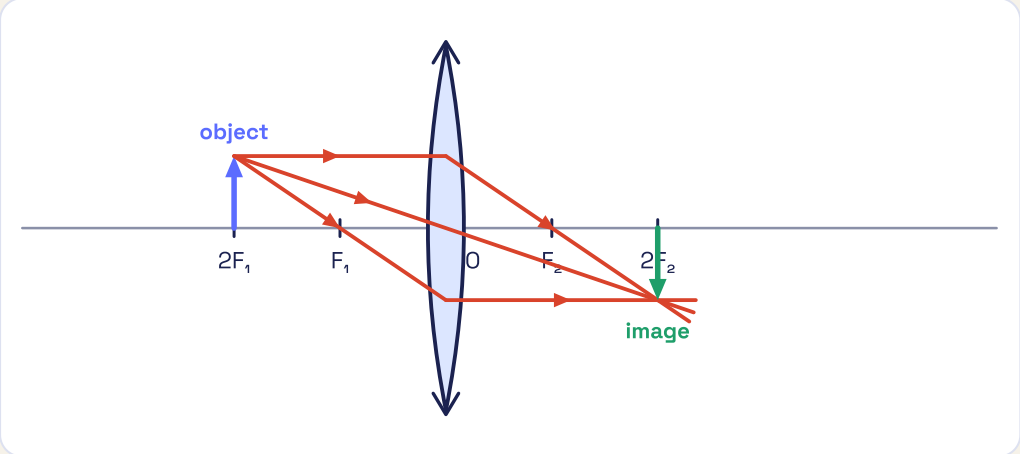
SIDEKICK TIP

Solid lines for real rays. Dotted lines for imaginary extensions that you trace back to a virtual image. Add arrowheads to every ray.

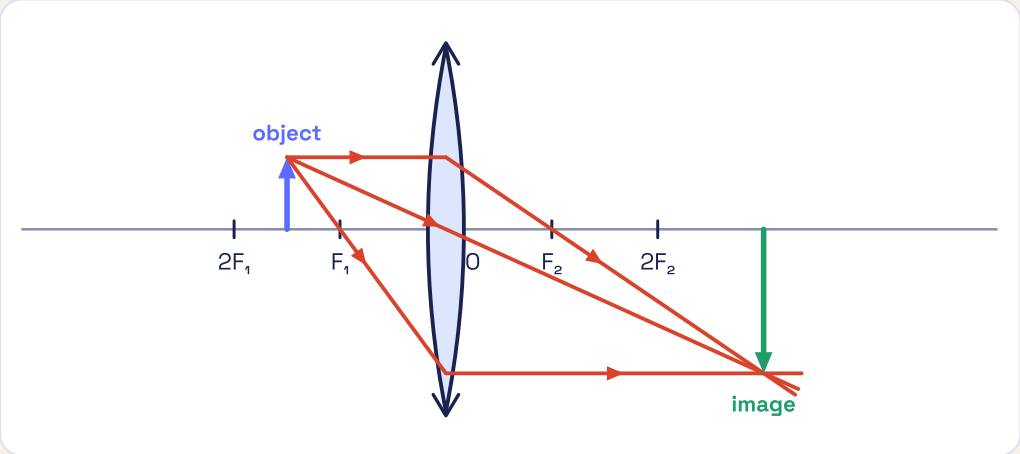
Image Formation by a Convex Lens



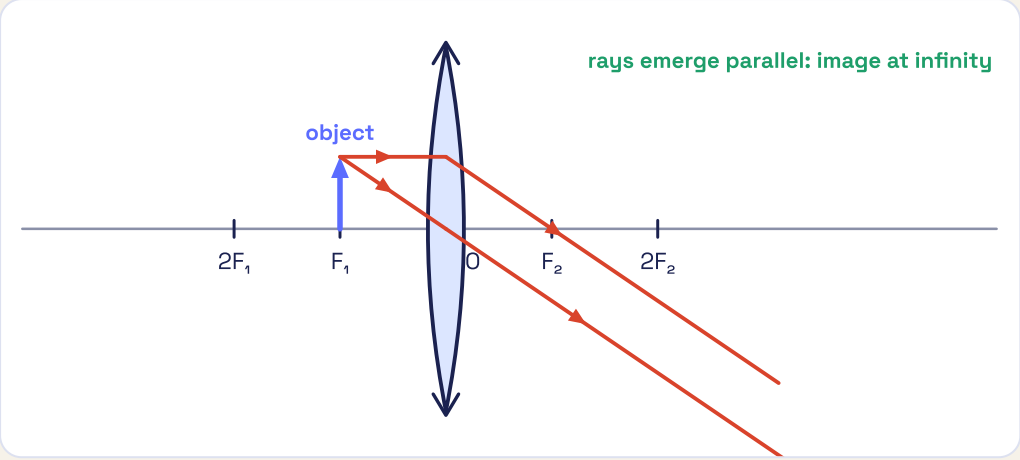
Object beyond $2F_1$: image between F_2 and $2F_2$, real, inverted, diminished.



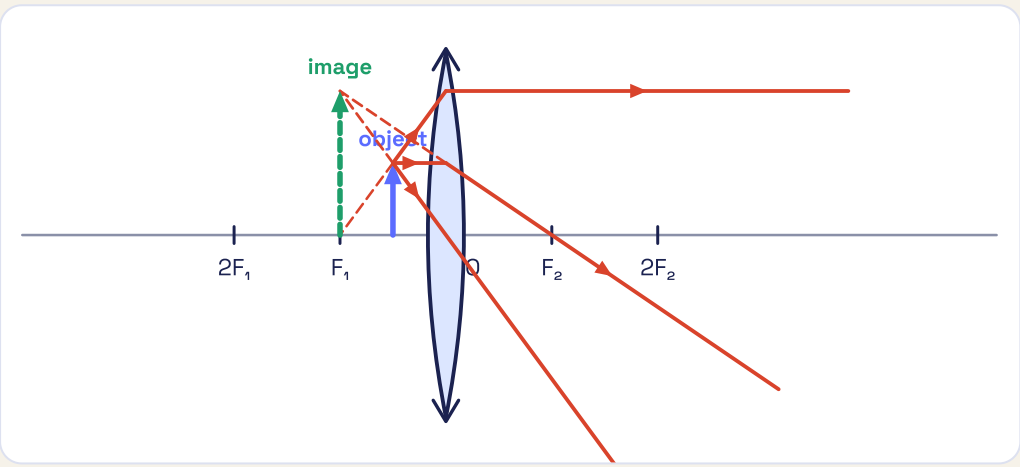
Object at $2F_1$: image at $2F_2$, real, inverted, same size.



Object between F_1 and $2F_1$: image beyond $2F_2$, real, inverted, magnified.



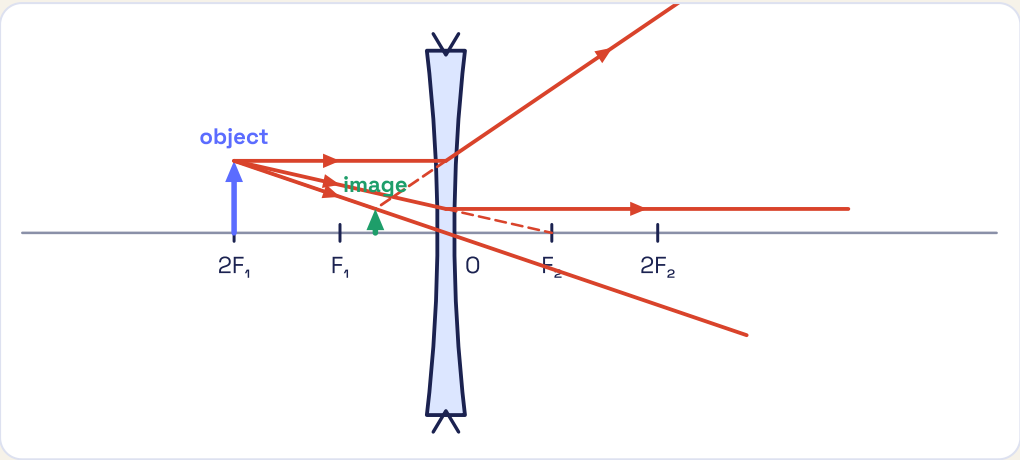
Object at F_1: refracted rays emerge parallel, the image is at infinity, real, inverted, highly magnified.



Object between F_1 and O: image on the same side as the object, virtual, erect, magnified. This is the magnifying glass.

OBJECT POSITION	IMAGE POSITION	NATURE	SIZE
At infinity	At F_2	Real, inverted	Highly diminished (a point)
Beyond $2F_1$	Between F_2 and $2F_2$	Real, inverted	Diminished
At $2F_1$	At $2F_2$	Real, inverted	Same size
Between F_1 and $2F_1$	Beyond $2F_2$	Real, inverted	Magnified
At F_1	At infinity	Real, inverted	Highly magnified
Between F_1 and O	Same side as the object, farther than the object	Virtual, erect	Magnified

Image Formation by a Concave Lens



A concave lens gives a virtual, erect and diminished image between F_1 and O for every position of the object.

OBJECT POSITION	IMAGE POSITION	NATURE	SIZE
At infinity	At F_1	Virtual, erect	Highly diminished
Anywhere between infinity and O	Between F_1 and O	Virtual, erect	Diminished

KEY POINT

Real image: formed where refracted rays actually meet, can be caught on a screen, and is inverted.
Virtual image: formed where rays only appear to meet, cannot be caught on a screen, and is erect.

Sign Convention and Lens Formula

Use the **New Cartesian convention**: measure all distances from the optical centre O. Distances in the direction of the incident light (usually to the right) are **positive**, and opposite to it are **negative**. Heights above the axis are positive and below are negative.

- Object distance u is always **negative** for a real object (it lies to the left).
- Focal length f is **positive** for a convex lens and **negative** for a concave lens.
- Image distance v is **positive** for a real image (right of the lens) and **negative** for a virtual image.

$$\mathbf{1/v - 1/u = 1/f} \qquad \text{Magnification} \quad \mathbf{m = h'/h = v/u}$$

- m is **negative** for a real, inverted image and **positive** for a virtual, erect one.
- $|m| > 1$ means magnified, $|m| < 1$ means diminished.

(The derivation of the lens formula is not required by the syllabus. Direct numericals using it are.)

Power of a Lens



KEY POINT

Power of a lens (P) is its ability to bend light, defined as the reciprocal of its focal length in metres. A shorter focal length means more bending, and so a higher power.

$$P = 1 / f \text{ (f in metres)} \quad \text{SI unit: dioptre (D)} \quad 1\ \text{D} = 1\ \text{m}^{-1}$$

P is positive for a convex lens and negative for a concave lens. An optician's lens of +2 D is convex with f = 0.5 m, and one of −4 D is concave with f = −0.25 m.

Worked Examples

Example 1. An object is placed 15 cm from a convex lens of focal length 10 cm. Find the image distance and the magnification.

```
u = -15 cm, f = +10 cm
1/v = 1/f + 1/u = 1/10 - 1/15 = (3 - 2)/30 = 1/30
v = +30 cm   (real image, 30 cm on the far side)
m = v / u = 30 / (-15) = -2   (inverted, twice the size)
```

Example 2. A concave lens has f = −20 cm. Find the image formed of an object 30 cm from it.

```
u = -30 cm, f = -20 cm
1/v = 1/f + 1/u = -1/20 - 1/30 = -(3 + 2)/60 = -1/12
v = -12 cm   (virtual, on the object side)
m = v / u = (-12) / (-30) = +0.4   (erect, diminished)
```

Example 3. A convex lens of focal length 10 cm is used as a magnifying glass with the object 6 cm away.

```
u = -6, f = +10:  1/v = 1/10 - 1/6 = (3 - 5)/30 = -1/15
v = -15 cm  (virtual, same side as the object)
m = (-15)/(-6) = +2.5   (erect, 2.5 times bigger)
```

Example 4. Find the power of (a) a convex lens of f = 25 cm, (b) a concave lens of f = 50 cm.

```
(a) P = 1/f = 1/0.25 = +4 D        (b) P = 1/(-0.50) = -2 D
```

Example 5. How far from a convex lens of focal length 12 cm must an object be placed so the real image is three times its size?

```
m = -3 so v = -3u. With u = -x, v = 3x.
1/v - 1/u = 1/(3x) + 1/x = 4/(3x) = 1/12
x = 16 cm (object), v = +48 cm (image)
```

Magnifying Glass (Simple Microscope)

A magnifying glass is a convex lens of short focal length. Placing the object **between the lens and its first focus** gives a **virtual, erect, enlarged** image on the same side as the object. You look through the lens and see the image behind the object. For the syllabus, location of the image and the magnification are found from the ray diagram only.

Applications of Lenses

DEVICE	LENS USED	WHY
Camera	Convex lens forms a real, diminished, inverted image on the film or sensor	Object lies beyond 2F
Projector	Convex lens gives a real, magnified image on a screen	Slide is between F and 2F, placed upside down
Simple microscope / magnifying glass	One short-focus convex lens	Virtual magnified image
Spectacles	Convex lens for long-sightedness, concave for short-sightedness	Corrects where the eye's image falls
Telescope and compound microscope	Two lens combination	Magnify distant or tiny objects
Human eye	The eye lens (convex) forms a real inverted image on the retina	The brain turns it upright



REAL LIFE

A water drop on a leaf acts as a tiny convex lens and magnifies the veins. Sunlight focused by a convex lens can ignite paper at the focus, which is why a glass ball left in the sun is a fire risk.



BOARD ALERT

Do **not** use scale drawing for ray diagrams in the exam: the syllabus asks for neat, labelled but approximate diagrams. But always mark F, 2F, O, arrows and the three characteristics.



HOT QUESTION

A convex lens forms a real image the same size as the object at 40 cm from the lens. What is its focal length and power? What happens to the image if the object is moved to 30 cm?

Mind Map Recap

IDEA	FORMULA OR FACT
Lens formula	1/v − 1/u = 1/f
Magnification	m = v/u = h′/h
Power	P = 1/f (metre) in dioptre
Convex	f +, converging, real or virtual image
Concave	f −, diverging, always virtual, erect, diminished
Radius of curvature	R = 2f
Magnifying glass	Object within F, virtual erect magnified image

Practice Questions

#	QUESTION	ANSWER
1	Define principal focus of a convex lens.	The point on the principal axis where rays parallel to the axis meet after refraction.
2	An object is 20 cm from a convex lens of f = 15 cm. Find v and m.	1/v = 1/15 − 1/20 = 1/60, v = +60 cm, m = −3.
3	A concave lens of f = −15 cm forms an image of an object 30 cm away. Find v.	1/v = −1/15 − 1/30 = −1/10, v = −10 cm (virtual).
4	Find the power of a convex lens of focal length 40 cm.	P = 1/0.40 = +2.5 D.
5	A lens has power −4 D. Find f and state its type.	f = −0.25 m = −25 cm; concave.
6	An object 4 cm high gives a real image with m = −2. Find the image height and state its nature.	Image 8 cm tall, inverted.
7	List three characteristics of the image formed by a concave lens.	Virtual, erect, diminished.
8	Where must an object be placed for a convex lens to act as a magnifying glass?	Between the optical centre and the first principal focus.
9	A convex lens of f = 5 cm is used as a magnifier with the object at 4 cm. Find v and m.	1/v = 1/5 − 1/4 = −1/20, v = −20 cm, m = +5.
10	Explain how a lens works as a set of prisms.	A convex lens is like prisms base to base and bends rays toward the axis; a concave lens is like prisms apex to apex and spreads rays.
11	State the difference between a real and a virtual image.	Real: rays actually meet, can be caught on a screen, inverted. Virtual: rays only appear to meet, cannot be caught on a screen, erect.
12	What is the nature and position of the image when an object is at 2F of a convex lens?	Real, inverted, same size, at 2F on the other side.

06

CHAPTER · PHYSICS

Spectrum

A glass prism turns white sunlight into a rainbow. Beyond the colours your eyes can see, the same family of waves includes radio, X-rays and gamma rays. This chapter explains dispersion, the electromagnetic spectrum, and why the sky is blue.



IN THIS CHAPTER

- Deviation and colour
- Dispersion
- Electromagnetic spectrum
- Infrared and ultraviolet
- Scattering of light

Deviation by a Prism and Colour

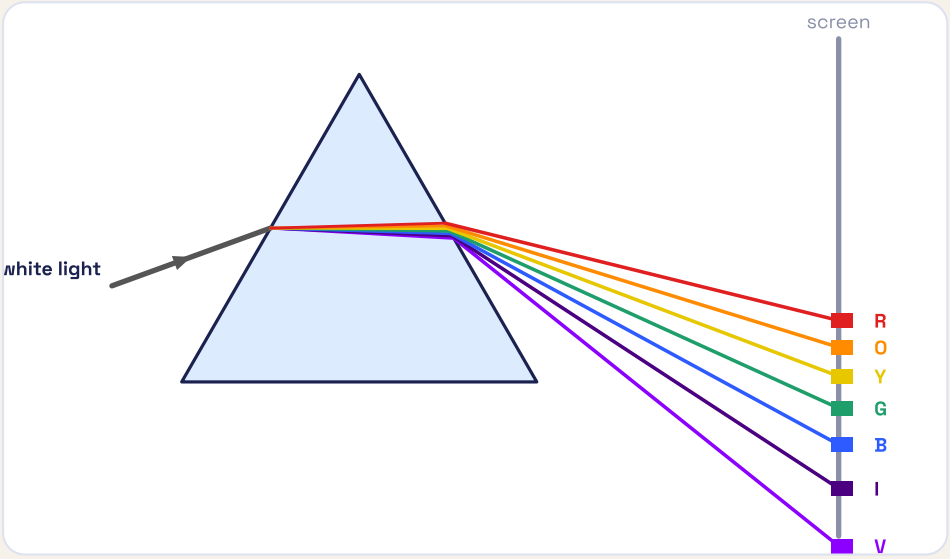
In the last chapter, a prism bent a ray towards its base. The size of that bend, the **deviation δ**, depends on the angle of the prism, the angle of incidence, the refractive index of the glass, and the **colour (wavelength)** of the light.

Light of a shorter wavelength is slowed more by glass. Its refractive index is higher, so it is bent more. **Violet** has the shortest wavelength of visible light and is deviated the most. **Red** has the longest and is deviated the least.

Dispersion and the Visible Spectrum

KEY POINT

Dispersion is the splitting of white light into its component colours on passing through a prism. The band of colours obtained is called the **spectrum**.



A prism splits white light. Violet bends most and red bends least. (The colour gap is exaggerated for clarity.)

- From most bent to least bent the colours run **V I B G Y O R** (violet, indigo, blue, green, yellow, orange, red).
- In glass, red travels fastest and violet slowest. In vacuum, all colours travel at the same speed c.
- White light is a mixture. A second inverted prism placed after the first can recombine the colours into white light (Newton's experiment).
- A single colour, such as red from a laser, is not dispersed by a prism. It only deviates.

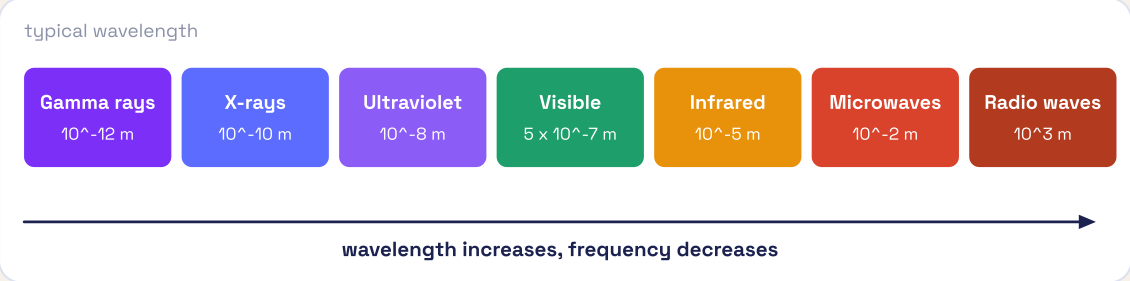
SIDEKICK TIP

The mnemonic **VIBGYOR** reads from the most-bent colour to the least-bent colour. Red is at the top of a rainbow and violet is at the bottom.

COLOUR	WAVELENGTH (APPROX.)	FREQUENCY (APPROX.)	DEVIATION IN A PRISM
Violet	400 nm	7.5×10^{14} Hz	Largest
Green	550 nm	5.5×10^{14} Hz	Medium
Red	700 nm	4.3×10^{14} Hz	Smallest

The Electromagnetic Spectrum

Visible light is one small part of a much larger family of waves, the **electromagnetic waves**. All of them are produced by oscillating electric and magnetic fields and differ only in wavelength and frequency.



The electromagnetic spectrum in order of increasing wavelength. Visible light is a thin slice from about 400 nm (violet) to 700 nm (red).

KEY POINT

In order of **increasing wavelength**: gamma rays, X-rays, ultraviolet, visible light, infrared, microwaves, radio waves.

SIDEKICK TIP

Memory line for the order of increasing wavelength (G, X, U, V, I, M, R): **G**ood **X**-ray **U**ses **V**ery **I**nteresting **M**edical **R**ays. Frequency and energy run the opposite way.

Properties Common to All Electromagnetic Waves

- They are **transverse waves**: electric and magnetic fields oscillate at right angles to the direction of travel.
- They need **no material medium**, and travel through vacuum.
- In vacuum they all travel at the **same speed, $c = 3 \times 10^8$ m s^{-1}**.
- They carry **energy** but have no charge, so electric and magnetic fields do not deflect them.
- They obey **$v = f\lambda$**, and can be reflected, refracted, and diffracted.

Infrared Radiation

- Lies just beyond red, wavelength roughly 700 nm to 1 mm. Emitted by every warm object. We feel it as heat.
- **Uses:** TV and AC remote controls, night vision cameras, heating and cooking (infrared grills), physiotherapy lamps, photography of haze, and detecting fever or heat loss in buildings.
- Absorbed strongly by water and carbon dioxide, which is the basis of greenhouse warming.

Ultraviolet Radiation

- Lies just beyond violet, wavelength roughly 10 nm to 400 nm. The Sun is the main source, along with mercury lamps.
- **Uses:** sterilising water and medical tools, detecting forged currency notes and fingerprints (fluorescence), vitamin D production in the skin.
- **Danger:** overexposure causes sunburn, eye damage and skin cancer. The ozone layer absorbs most of the harmful UV.

RADIATION	TYPICAL SOURCE	TYPICAL USE
Radio waves	Antenna circuits	Broadcast, mobile and TV signals
Microwaves	Magnetron, klystron	Ovens, radar, satellite links
Infrared	Hot objects	Remote controls, thermal imaging
Visible	Sun, lamps	Seeing, photosynthesis
Ultraviolet	Sun, mercury lamps	Sterilisation, forgery detection
X-rays	X-ray tube	Medical imaging, security scans
Gamma rays	Radioactive nuclei	Cancer treatment, sterilising food

Scattering of Light

When light passes through air, the gas molecules and tiny particles absorb a bit of it and re-emit it in all directions. This is called **scattering**. Short wavelengths scatter much more than long ones. A rough rule (for tiny particles) is that the scattered intensity is proportional to $1/\lambda^4$.

- **Blue sky:** blue and violet light are scattered by air molecules in all directions, and reach our eyes from every part of the sky. Violet is scattered even more, but the Sun emits less of it, and our eyes are less sensitive to it, so we see blue.
- **Red Sun at sunrise and sunset:** the Sun is low and its light passes through a thicker layer of air. Most of the blue is scattered out of the beam, leaving red and orange to reach us.
- **Danger signals are red** because red is scattered the least and can be seen from far away even in fog.
- **Black sky for astronauts:** there is no air in space to scatter the light.



FUN FACT

Blue light of 450 nm scatters about $(700/450)^4 \approx 5.9$ times more than red light of 700 nm. That is the reason the sky is blue and sunsets glow red.

Worked Examples

Example 1. A radio station broadcasts at 100 MHz. Find the wavelength.

```
lambda = c / f = 3.0e8 / 1.0e8 = 3.0 m
```

Example 2. X-rays of wavelength 1.0×10^{-10} m. Find their frequency.

```
f = c / lambda = 3.0e8 / 1.0e-10 = 3.0e18 Hz
```

Example 3. Red light of wavelength 7.0×10^{-7} m enters glass of $\mu = 1.5$. Find its frequency and wavelength in glass.

```
f = 3.0e8 / 7.0e-7 = 4.29e14 Hz (does not change)
lambda_glass = 7.0e-7 / 1.5 = 4.67e-7 m
```



BOARD ALERT

Do not write that red light has the greatest **refractive index**. Red has the **lowest** refractive index and the **highest** speed in glass. Violet has the highest refractive index.



HOT QUESTION

Why is the sky black on the Moon, and why does the Sun look red when it is near the horizon? Use the idea of scattering in both answers.

Mind Map Recap

IDEA	KEY FACTS
Dispersion	White light → VIBGYOR through a prism. Violet bends most
Speed in glass	Red fastest, violet slowest. $\mu_{violet} > \mu_{red}$
EM spectrum (increasing λ)	Gamma, X-ray, UV, visible, IR, microwave, radio
Common properties	Transverse, no medium, speed c in vacuum, carry energy, uncharged
Scattering	Shorter λ scatters more. Blue sky, red sunset
Relation	$c = f \lambda$

Practice Questions

#	QUESTION	ANSWER
1	What is dispersion? Which colour deviates most?	Splitting of white light into colours by a prism. Violet.
2	Why does red light deviate least in a prism?	It has the longest wavelength and the smallest refractive index in glass.
3	Name the EM waves, in order of increasing wavelength.	Gamma, X-rays, ultraviolet, visible, infrared, microwaves, radio waves.
4	State four common properties of EM waves.	Transverse; need no medium; same speed in vacuum; carry energy; not deflected by fields.
5	Give two uses each of infrared and ultraviolet radiation.	IR: remote controls, thermal imaging. UV: sterilising, detecting forged notes.
6	Why does the sky look blue?	Shorter wavelength blue light is scattered more by air molecules in all directions.
7	Why does the Sun look red at sunset?	Light travels through more air, blue is scattered out, red reaches the eye.
8	Find the frequency of 600 nm light in vacuum.	$f = 3\times10^{8}/6\times10^{-7} = 5.0\times10^{14}$ Hz.
9	An FM station has frequency 90 MHz. Find its wavelength.	$3\times10^{8}/9\times10^{7} = 3.33$ m.
10	Why are danger signals red?	Red is scattered least, so it is visible from the greatest distance.
11	Which has the greater speed in glass, red or violet?	Red.
12	How can the colours of the spectrum be recombined to white light?	Pass them through a second inverted prism (or a lens).

07

CHAPTER · PHYSICS

Sound

Shout in a quiet valley and your own voice comes back. Bats and dolphins find their way using such echoes. This chapter covers how sound reflects, how vibrations behave, and the three features that let you tell any two sounds apart.



IN THIS CHAPTER

- Echoes and SONAR
- Natural, damped, forced vibrations
- Resonance
- Loudness, pitch, quality
- Decibel and noise pollution

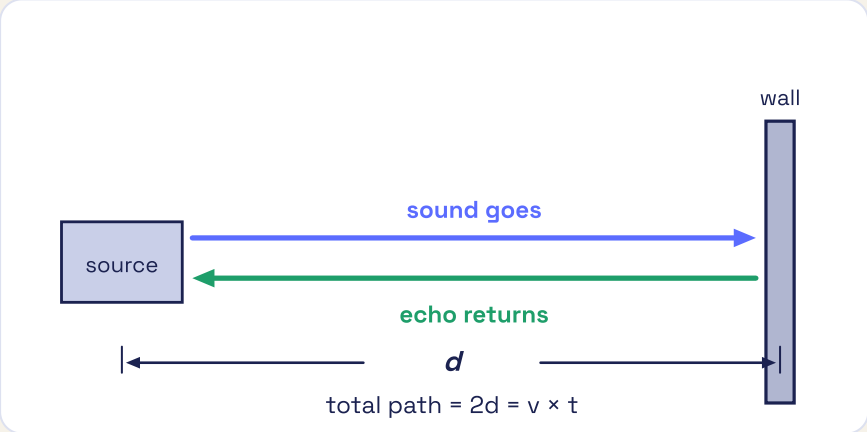
What Sound Is

Sound is produced by a **vibrating body**. The vibrations pass through a material medium such as air, water or steel as a **longitudinal wave**: the particles of the medium move back and forth along the direction in which the wave travels, forming alternate **compressions** (high pressure) and **rarefactions** (low pressure). Sound cannot travel through a vacuum.

- **Frequency (f):** number of vibrations per second, in hertz (Hz).
- **Wave speed:** $v = f\lambda$, where λ is the wavelength.
- Speed of sound in air at room temperature is about **340 m s^{-1}** (use this unless told otherwise), in water about 1500 m s^{-1} and in steel about 5000 m s^{-1}. Sound is fastest in solids and slowest in gases.
- The audible range for humans is **20 Hz to 20 000 Hz**. Below 20 Hz is infrasonic and above 20 kHz is ultrasonic.

Reflection of Sound and Echoes

Sound reflects from hard, large surfaces such as walls, cliffs and buildings in the same way light reflects from a mirror: the angle of incidence equals the angle of reflection. A reflected sound heard separately after the original sound is called an **echo**.



An echo is sound that returns after reflecting from a distant surface. The sound covers the distance d twice.

KEY POINT

Echo is the repetition of sound heard after reflection from a distant obstacle, as a distinct sound separate from the original.

1. Time gap

The ear keeps a sound for about 0.1 s (persistence of hearing). The echo must arrive at least 0.1 s after the original, or the two blend together.

2. Distance

In 0.1 s sound travels 340 × 0.1 = 34 m. That is the to-and-fro path, so the reflector must be at least 17 m away.

3. Surface

The surface must be large and hard (a wall, cliff, hillside), not a soft or small obstacle that absorbs sound.

4. Loudness

The sound must be strong enough to be heard after the return trip.

$$d = v\,t / 2 \quad \text{or} \quad v = 2d / t \quad (t = \text{time between sending the sound and hearing the echo})$$

Uses of Echoes

- **Bats** send out ultrasonic squeaks and listen for echoes to find insects and avoid obstacles in the dark. **Dolphins** use clicks in the same way.
- **SONAR** (Sound Navigation And Ranging): a ship sends ultrasonic pulses down, and the time for the echo gives the depth of the sea, the position of a shoal of fish, submarines or a wreck.
- **Fishermen** use echo sounders to locate shoals.
- **Medical field:** ultrasound scans echo from organs and a foetus to give images without radiation.
- Echo-based range finders measure distances inside buildings.



REAL LIFE

Auditoriums have curtains, carpets and rough walls to absorb sound, so that echoes do not blur the speaker's words. Concert halls use curved ceilings to direct sound evenly to the audience.

Example 1. A man claps near a cliff and hears the echo after 0.60 s. How far is the cliff? (Speed of sound = 340 m s^{-1})

```
d = v t / 2 = 340 x 0.60 / 2 = 102 m
```

Example 2. A SONAR on a ship sends a pulse and receives the echo from the sea bed after 0.80 s. If sound travels at 1500 m s^{-1} in water, find the depth.

```
d = v t / 2 = 1500 x 0.80 / 2 = 600 m
```

Example 3. A gun is fired at a distance of 1.0 km from a hill. After how long is the echo heard?

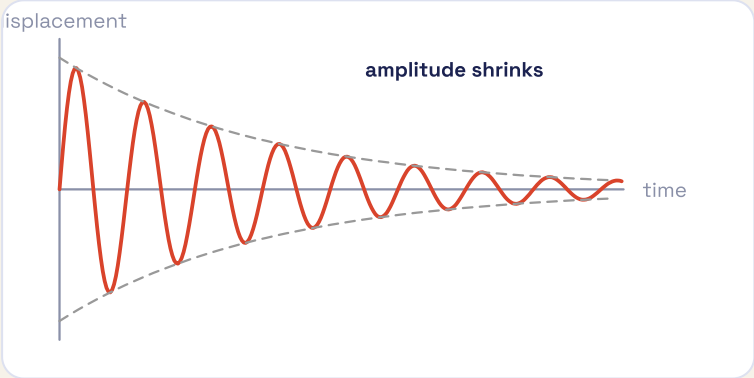
```
t = 2 d / v = 2 x 1000 / 340 = 5.88 s
```

Example 4. An echo is heard 0.50 s after a shout when the wall is 85 m away. Find the speed of sound.

```
v = 2 d / t = 2 x 85 / 0.50 = 340 m/s
```

Natural, Damped and Forced Vibrations

TYPE	WHAT HAPPENS	FREQUENCY	AMPLITUDE	EXAMPLE
Natural (free) vibration	The body is disturbed once and then left alone	Its own natural frequency, fixed by its size, shape, material and tension	Ideally constant (in practice it dies slowly)	A plucked guitar string, a tuning fork, a swing pushed once
Damped vibration	Friction and air resistance take energy away	Natural frequency, nearly unchanged	**Decreases** with time until the motion stops	A pendulum that gradually stops, a struck bell fading out
Forced vibration	A periodic external force keeps driving the body	**Frequency of the driver**, not its own	Set by how close the driver is to the natural frequency	A tuning fork pressed on a table makes the table vibrate and the sound louder



Damped vibration: friction and air resistance take energy away, so the amplitude decays with time until the body stops.

Resonance

KEY POINT

Resonance is a special case of forced vibration in which the frequency of the external periodic force equals the natural frequency of the body. The body then vibrates with a very large amplitude.

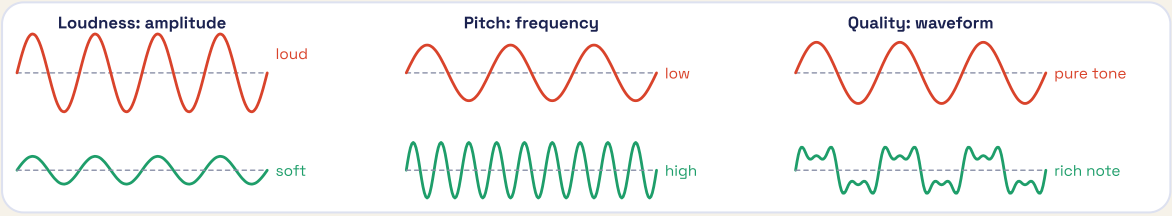
- A child on a swing goes high when pushed at the swing's own rhythm, in time with each swing.
- Soldiers **break step** when crossing a bridge so that their marching does not match the bridge's natural frequency.
- A radio is **tuned** by matching its circuit's natural frequency to the station's, so that this one station resonates.
- Musical instruments (guitar, violin) have hollow bodies that resonate with the strings and make the sound louder.
- A resonating column of air in a tube above a tuning fork gives a loud sound, used to measure the speed of sound.

BOARD ALERT

Resonance is **not** always good. Resonant vibrations can break glass or damage buildings and machinery. In resonance the amplitude is large, but the energy still comes from the driver, so no energy is created.

Characteristics of Sound

CHARACTERISTIC	OBJECTIVE (PHYSICAL) BASIS	SUBJECTIVE (HEARD) FEATURE	DEPENDS ON
Loudness	Amplitude and intensity	How loud it seems to the ear	Amplitude of vibration, distance from the source, the sensitivity of the ear
Pitch	Frequency	How shrill or flat a note sounds	Frequency only: high f gives a high pitch
Quality (timbre)	Waveform, which comes from the mix of harmonics	Lets you tell a violin from a flute playing the same note	The shape of the wave



Same note, different property: a taller wave is louder, a closer-packed wave has higher pitch, and a different wave shape means a different quality (timbre).

Loudness, Intensity and the Decibel

Intensity is the energy passing per second through unit area at right angles to the wave (unit W m^{-2}). It is a physical, measurable quantity. **Loudness** is the effect of that intensity on the ear. It is subjective and depends on the listener, though a larger amplitude and a larger intensity both make a sound louder.

The **sound level** is measured in **decibels (dB)** (the syllabus asks for the unit only).

SOUND	APPROXIMATE LEVEL
Threshold of hearing	0 dB
Whisper	20 to 30 dB
Normal conversation	60 dB
Heavy traffic	80 to 90 dB
Loud music, drilling	100 to 110 dB
Jet engine nearby (pain)	120 dB and above

Pitch, Frequency, Quality

- A thin string, a short string and a tight string all vibrate fast and give a **high pitch**.
- A female voice has a higher pitch than a male voice because the vocal cords vibrate at a higher frequency.
- Two instruments playing the same note at the same loudness sound different because each adds different harmonics, which gives a different wave shape. This is their **quality**.

Noise Pollution

Unwanted, harsh sound is **noise**. Excessive noise is a form of pollution.

SOURCES	HARMFUL EFFECTS	CONTROL
Traffic, aircraft, loudspeakers, firecrackers, construction machinery, factories	Hearing loss, headaches, high blood pressure, sleeplessness, stress, poor concentration	Silencers on engines, no honking zones, planting trees as sound absorbers, soundproofing, ear protectors, laws on loudspeakers



FUN FACT

A blue whale's call can reach about 180 dB, and travel hundreds of kilometres underwater because water carries sound so well.



HOT QUESTION

A 512 Hz tuning fork resonates with a column of air closed at one end. Taking the speed of sound as 340 m s^{-1}, estimate the shortest column length (use λ/4). How does this change if a 1024 Hz fork is used?

Mind Map Recap

IDEA	FORMULA OR FACT
Echo	d = v t / 2; reflecting surface at least 17 m away (v = 340 m s^{-1})
Wave relation	v = f λ
SONAR	Ultrasonic pulses; depth = v t / 2
Natural vs damped vs forced	Fixed f; amplitude decays; driver's f
Resonance	Driver f = natural f, maximum amplitude
Loudness / pitch / quality	Amplitude / frequency / waveform
Unit of sound level	decibel (dB)

Practice Questions

#	QUESTION	ANSWER
1	State two conditions for an echo.	Reflecting surface at least 17 m away and large and hard; time gap of at least 0.1 s.
2	A boy hears the echo of a clap from a wall after 0.4 s. How far is the wall? ($v = 340$ m s^{-1})	$d = 340 \times 0.4 / 2 = 68$ m.
3	A ship's SONAR gets the echo after 1.2 s. The speed of sound in sea water is 1500 m s^{-1}. Find the depth.	$1500 \times 1.2 / 2 = 900$ m.
4	What does SONAR stand for? Give two uses.	Sound Navigation And Ranging. Depth measurement, locating fish shoals or submarines.
5	Differentiate natural and forced vibrations.	Natural: vibrate at the body's own frequency after a single disturbance. Forced: vibrate at the frequency of an external periodic force.
6	What is resonance? Give one use and one harmful effect.	Forced vibration at the natural frequency, giving maximum amplitude. Use: tuning radio. Harm: shattering glass or damaging a bridge.
7	Why do soldiers break step on a bridge?	To avoid resonance, which could make the bridge vibrate with a dangerously large amplitude.
8	Which characteristic of sound depends on frequency? On waveform? On amplitude?	Pitch; quality; loudness.
9	A tuning fork gives louder sound when its stem is pressed on a table. Why?	Forced vibration of the larger table surface moves more air, so the sound is louder (energy lasts a shorter time).
10	Name the unit of sound level and give two harmful effects of noise.	Decibel (dB). Hearing loss, stress, sleeplessness.
11	A bat emits an ultrasonic pulse and hears the echo from an insect after 0.02 s. How far is the insect?	$d = 340 \times 0.02 / 2 = 3.4$ m.
12	Why can't sound travel in the vacuum of space?	It needs a material medium of particles to pass on the vibrations.

08

CHAPTER · PHYSICS

Current Electricity

Every gadget you own runs on moving charge. This chapter defines current, voltage and resistance, states Ohm's law, and shows you how to find the combined resistance of any simple network.



IN THIS CHAPTER

- Charge, current, pd, resistance
- Ohm's law and its graph
- Resistivity
- emf and internal resistance
- Series and parallel
- Simple networks

The Four Basic Quantities

QUANTITY	DEFINITION	FORMULA	SI UNIT
Charge (Q)	Property of matter that feels electric force. Carried by electrons (each has 1.6×10^{-19} C)	$Q = I\,t$	coulomb (C)
Current (I)	Rate of flow of charge through a conductor	$I = Q / t$	ampere (A) = 1 C s^{-1}
Potential difference (V)	Work done per unit charge in moving it between two points	$V = W / Q$	volt (V) = 1 J C^{-1}
Resistance (R)	Opposition a conductor offers to the flow of current	$R = V / I$	ohm (Ω) = 1 V A^{-1}

- Conventional current flows from the positive terminal to the negative terminal outside the cell, opposite to the direction in which electrons actually move.
- An **ammeter** is connected in **series** and has very low resistance. A **voltmeter** is connected in **parallel** across the component and has very high resistance.

KEY POINT

1 volt is the potential difference between two points if 1 joule of work is done to move 1 coulomb of charge between them.

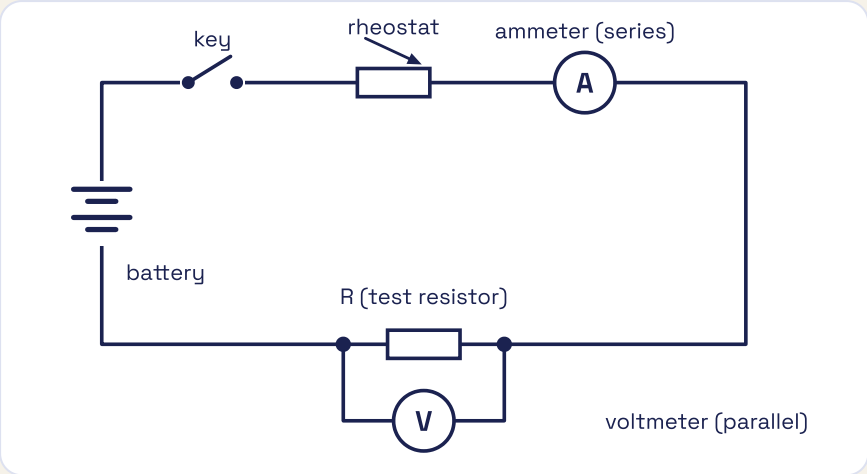
Ohm's Law

KEY POINT

Ohm's law: The current flowing through a conductor is directly proportional to the potential difference across its ends, provided the temperature and other physical conditions stay the same.

$$V = I R \quad (V \propto I \text{ at constant temperature; } R \text{ is the constant of proportionality})$$

Verifying Ohm's Law



Circuit to verify Ohm's law. The ammeter is in series; the voltmeter is connected across (in parallel with) the resistor. The rheostat varies the current.

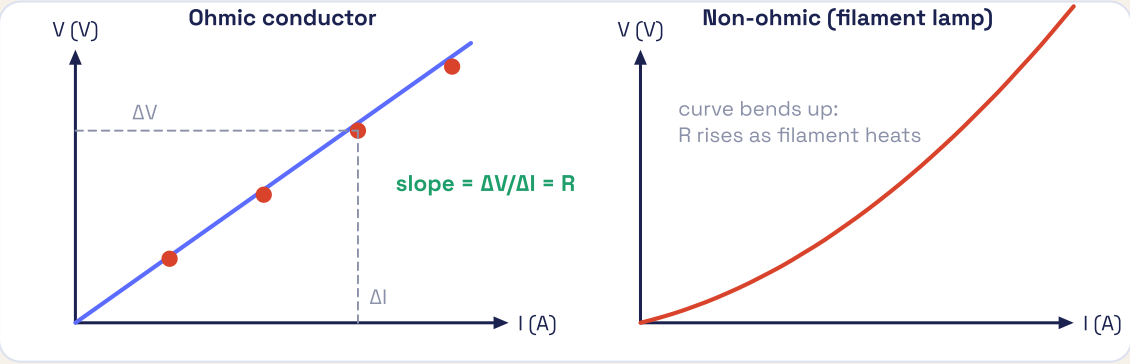
1. Connect
Join the cell, key, rheostat, ammeter and test resistor in series, with the voltmeter across the resistor.

2. Take readings
Close the key. Move the rheostat to get a series of currents and note the matching voltage each time.

3. Tabulate
Write I (A) and V (V) in two columns. Compute V/I for each row.

4. Plot
Draw a graph of V (y-axis) against I (x-axis). A straight line through the origin shows $V \propto I$. Its slope = $\Delta V/\Delta I$ = R.

I (A)	0.4	0.8	1.2	1.6
V (V)	2.0	4.0	6.0	8.0
V/I (Ω)	5.0	5.0	5.0	5.0



V against I. For an ohmic conductor the graph is a straight line through the origin, and its slope gives the resistance. A filament lamp gives a curve because its resistance rises with temperature.

- **Ohmic conductors** (metals at constant temperature) obey Ohm's law and give a straight-line V–I graph.
- **Non-ohmic conductors** (filament lamp, diode, thermistor) do not. Their V–I graph is curved.
- Do not leave the key closed for long: the current heats the wire and changes R, which spoils the result.

Factors Affecting Resistance

FACTOR	EFFECT ON R
Length l	R ∝ l. Doubling the length doubles R
Cross-section area A	R ∝ 1/A. A thicker wire has less R. Halving the diameter makes A one-quarter, so R becomes four times
Material	Different materials resist differently (copper low, nichrome high)
Temperature	R of a metal rises as it gets hotter

$$R = \varrho l / A \quad \varrho = \text{specific resistance (resistivity) of the material}$$



KEY POINT

Specific resistance (resistivity) of a material is the resistance of a conductor of that material of unit length and unit area of cross-section. SI unit: **ohm metre (Ω m)**. It depends on the material and its temperature, but not on the shape or size.

MATERIAL	P (Ω M), APPROX.	USE
Silver	1.6×10^{-8}	Best conductor, costly
Copper	1.7×10^{-8}	Connecting wires, cables
Nichrome	1.1×10^{-6}	Heating elements
Glass, rubber	10^{10} to 10^{14}	Insulators

Superconductors. Some materials drop to **zero resistance** below a certain critical temperature (mercury at about 4 K). A current set up in a superconducting loop keeps flowing with no loss. Uses: very strong electromagnets in MRI scanners and maglev trains, lossless power cables.

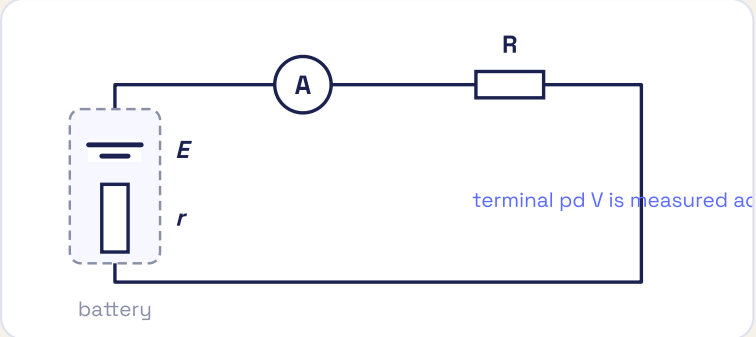
emf and Internal Resistance



KEY POINT

Electromotive force (emf, E) of a cell is the work done by the cell in moving one coulomb of charge around the complete circuit. It is the pd across the cell terminals when **no current** is drawn. Unit: volt.

A cell is not perfect. The chemicals and electrodes inside it resist the flow of current. This is the cell's **internal resistance (r)**. When a current I flows, some voltage, I r, is lost inside the cell.

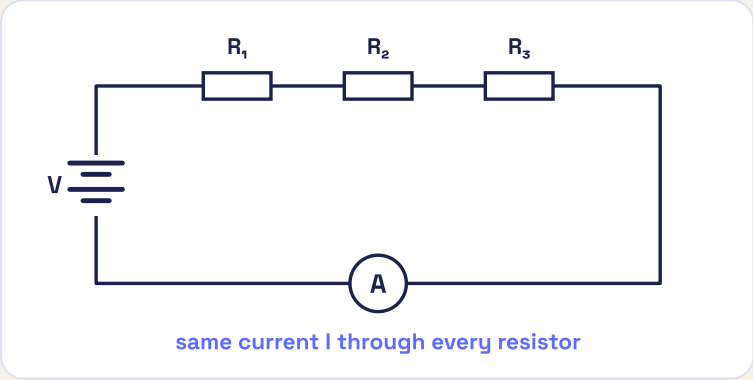


A real cell: it has an emf E and an internal resistance r in series. The current is I = E/(R + r), and the terminal pd is V = E − I r.

$$\mathbf{E = V + I\,r} \quad \Rightarrow \quad \mathbf{I = E / (R + r)} \quad \text{terminal pd} \quad V = E - I\,r$$

- emf is larger than the terminal pd whenever current flows, and equals it only on open circuit.
- Terminal pd drops as more current is drawn. That is why a torch bulb dims as the cell gets old (r grows).

Combination of Resistors: Series



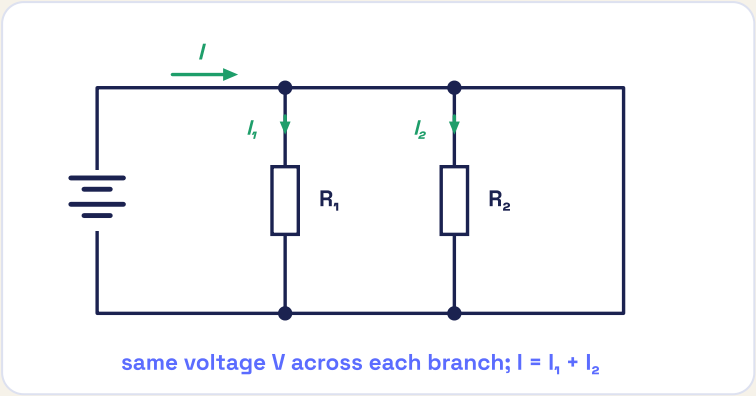
Resistors in series: one path, so the same current flows through all, and the supply voltage divides across them.

Derivation. Three resistors R_1, R_2, R_3 carry the same current I. The total voltage is the sum of the individual pds:

```
V = V1 + V2 + V3 = I R1 + I R2 + I R3 = I (R1 + R2 + R3)
But V = I R_s  so
    R_s = R1 + R2 + R3
```

- The equivalent resistance is **larger** than the largest individual resistor.
- The voltage divides in proportion to the resistances: $V_1 : V_2 = R_1 : R_2$.
- If one resistor fails (breaks), the whole circuit stops, like old festival lights.

Combination of Resistors: Parallel



Resistors in parallel: each branch is connected across the same two points, so the voltage is common and the current splits.

Derivation. Each branch has the same voltage V. The total current is the sum of the branch currents:

```
I = I1 + I2 + I3 = V/R1 + V/R2 + V/R3 = V (1/R1 + 1/R2 + 1/R3)
But I = V / R_p  so
    1/R_p = 1/R1 + 1/R2 + 1/R3
For two resistors:  R_p = R1 R2 / (R1 + R2)
```

- The equivalent resistance is **smaller** than the smallest individual resistor.
- Current divides inversely to the resistances: the smaller R gets the larger current.
- If one branch fails, the others keep working, which is why household appliances are wired in parallel.

	SERIES	PARALLEL
Current	Same through each	Divides, $I = I_1 + I_2 + \dots$
Voltage	Divides, $V = V_1 + V_2 + \dots$	Same across each
Equivalent resistance	$R_1 + R_2 + \dots$	$1/R_1 + 1/R_2 + \dots$ (smaller than any)
If one component breaks	All stop	Others continue

Worked Examples

Example 1. A current of 2.0 A flows for 3 minutes. How much charge passes? If the pd is 12 V, how much energy is transferred?

```
Q = I t = 2.0 x 180 = 360 C
W = Q V = 360 x 12 = 4320 J
```

Example 2. A resistor draws 0.40 A when connected to 6.0 V. Find its resistance. Find the current at 15 V.

```
R = V / I = 6.0 / 0.40 = 15 ohm
I = 15 / 15 = 1.0 A
```

Example 3. A copper wire is 10 m long with area of cross-section 1.0 mm^2. Find its resistance. ($\rho = 1.7 \times 10^{-8}$ Ω m)

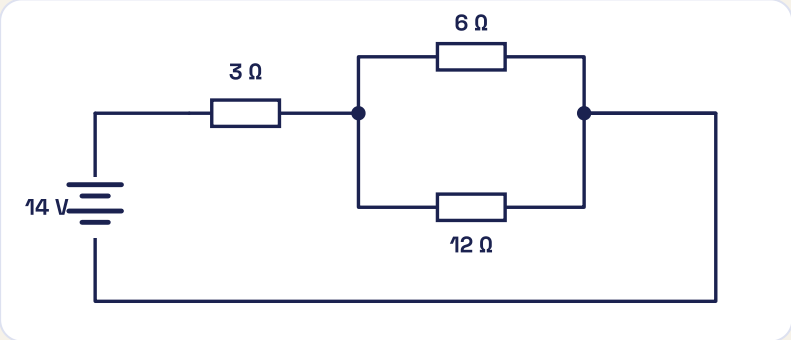
```
A = 1.0 mm^2 = 1.0e-6 m^2
R = rho l / A = 1.7e-8 x 10 / 1.0e-6 = 0.17 ohm
```

Example 4. Three resistors 4 Ω, 6 Ω and 10 Ω are in series with a 20 V supply. Find the current and the pd across each.

```
R_s = 4 + 6 + 10 = 20 ohm
I = V / R = 20 / 20 = 1.0 A
V1 = 4 V,  V2 = 6 V,  V3 = 10 V   (sum = 20 V)
```

Example 5. Resistors of 6 Ω and 3 Ω are connected in parallel across 12 V. Find the equivalent resistance and the current in each branch and in the supply.

```
R_p = (6 x 3) / (6 + 3) = 18 / 9 = 2 ohm
I1 = 12 / 6 = 2 A,   I2 = 12 / 3 = 4 A
Total I = 2 + 4 = 6 A   (check: 12 / 2 = 6 A)
```



A simple network: a 3 Ω resistor in series with a parallel pair (6 Ω and 12 Ω) across a 14 V battery.

Example 6. For the network in the figure (3 Ω in series with 6 Ω and 12 Ω in parallel, 14 V battery), find the total current and the current in each parallel resistor.

```
Parallel part: R_p = (6 x 12)/(6 + 12) = 72/18 = 4 ohm
Total R = 3 + 4 = 7 ohm
I = 14 / 7 = 2.0 A
Pd across the parallel pair = I x R_p = 2.0 x 4 = 8.0 V
I(6 ohm) = 8/6 = 1.33 A,  I(12 ohm) = 8/12 = 0.67 A  (sum = 2.0 A)
```

Example 7. A cell of emf 1.5 V and internal resistance 0.5 Ω is connected to a 2.5 Ω resistor. Find the current and the terminal pd.

```
I = E / (R + r) = 1.5 / (2.5 + 0.5) = 0.50 A
V = E - I r = 1.5 - 0.50 x 0.5 = 1.25 V   (also I R = 0.5 x 2.5 = 1.25 V)
```

SIDEKICK TIP

For parallel combinations, check the answer: R_p must be smaller than the smallest resistor. For series, R_s must be larger than the largest.

REAL LIFE

Fairy lights were once wired in series, so one dead bulb put out the whole string. Modern strings use parallel or shunted bulbs.

HOT QUESTION

Two identical resistors have an equivalent resistance of 4 Ω in series. What is their equivalent resistance in parallel? Which carries more current in a parallel combination of 2 Ω and 8 Ω?

Mind Map Recap

IDEA	FORMULA
Current	$I = Q / t$ (A)
Potential difference	$V = W / Q$ (V)
Ohm's law	$V = I R$ (R in Ω)
Resistivity	$R = \rho l / A$; unit Ω m
emf and internal resistance	$E = V + I r$; $I = E/(R + r)$
Series	$R_s = R_1 + R_2 + \ldots$
Parallel	$1/R_p = 1/R_1 + 1/R_2 + \ldots$

Practice Questions

#	QUESTION	ANSWER
1	Define potential difference and state its SI unit.	Work done per unit charge in moving it between two points. Volt (V).
2	A heater draws 5 A for 2 minutes. Find the charge that flows.	$Q = 5 \times 120 = 600$ C.
3	State Ohm's law. What does the slope of a V–I graph give?	$I \propto V$ at constant temperature. The slope gives resistance.
4	A wire has R = 8 Ω. What is its resistance if its length is doubled and its area is also doubled?	R unchanged, 8 Ω ($R \propto l/A$).
5	Find R of a nichrome wire 2.0 m long, A = 0.5 mm^2, ρ = 1.1 × 10^{-6} Ω m.	$R = 1.1\times10^{-6} \times 2.0 / 0.5\times10^{-6} = 4.4$ Ω.
6	Three resistors 2 Ω, 3 Ω and 5 Ω are in series with a 5 V battery. Find the current.	R = 10 Ω, I = 0.5 A.
7	Find the equivalent resistance of 12 Ω and 4 Ω in parallel.	(12 × 4)/16 = 3 Ω.
8	A 9 V battery is connected to 6 Ω and 3 Ω in parallel. Find the total current.	R_p = 2 Ω, I = 4.5 A.
9	A cell of emf 2 V and internal resistance 1 Ω drives a current through 3 Ω. Find the terminal pd.	I = 2/4 = 0.5 A, V = 2 − 0.5 = 1.5 V.
10	Why is a voltmeter given a large resistance and an ammeter a small one?	The voltmeter must draw almost no current; the ammeter must not change the current it measures.
11	What is a superconductor? Give one use.	A material with zero resistance below its critical temperature. Used in MRI magnets.
12	Why are household appliances connected in parallel?	Each gets the full supply voltage and can be switched independently.

09

CHAPTER · PHYSICS

Electrical Power and Energy

Your electricity bill counts energy, not power. This chapter links work done by charges to the formulas for energy and power, and shows how to find the cost of running appliances.



IN THIS CHAPTER

- Electrical energy
- Electrical power
- Heating effect
- Commercial unit kW h
- Appliance ratings and bills

Electrical Energy

When charge Q moves through a potential difference V, the work done is W = Q V. That is the **electrical energy** transferred. Since Q = I t, we get several useful forms.

$$W = Q\,V = V\,I\,t$$

Combine with Ohm's law (V = I R) to eliminate one variable:

$$W = V\,I\,t = I^2\,R\,t = (V^2 / R)\,t$$

DEVICE	ENERGY CONVERSION
Heater, geyser, iron	Electrical → heat
Lamp	Electrical → light (and heat)
Motor, fan	Electrical → mechanical (and heat)
Loudspeaker	Electrical → sound
Battery being charged	Electrical → chemical



SIDEKICK TIP

Which form to use? Known I and R: $I^2R\,t$. Known V and R: $(V^2/R)\,t$. Known V and I: $V\,I\,t$. Series circuits keep I fixed, so use I^2R; parallel keep V fixed, so use V^2/R.

Electrical Power



KEY POINT

Electrical power is the rate at which electrical energy is transferred or used.

$$P = W / t = V\,I = I^2\,R = V^2 / R$$

- **SI unit:** watt (W) = 1 J s^{-1} = 1 volt × 1 ampere.
- 1 kilowatt (kW) = 1000 W; 1 MW = 10^6 W.
- A bulb marked **60 W, 220 V** uses 60 J every second when connected to 220 V. This is its **power rating**.
- Rated power is valid only at the rated voltage. At lower voltage the bulb glows dimmer, and its power is lower ($P \propto V^2$ if R is constant).

Commercial Unit of Energy

The joule is far too small for billing. Electricity boards use the **kilowatt hour (kW h)**, also called a **unit**.

KEY POINT

1 kW h is the energy used when an appliance of power 1 kW works for 1 hour. 1 kW h = 1000 W × 3600 s = **3.6×10^6 J**.

$$\text{Energy (kW h)} = \text{Power (kW)} \times \text{Time (h)} \qquad \text{Cost} = \text{Energy} \times \text{rate per unit}$$

APPLIANCE	TYPICAL POWER	CURRENT AT 220 V
LED bulb	9 W	0.04 A
Ceiling fan	75 W	0.34 A
Television	100 W	0.45 A
Refrigerator	200 W	0.91 A
Electric iron	1000 W	4.5 A
Microwave oven	1200 W	5.5 A
Geyser	2000 W	9.1 A
Air conditioner	1500 W	6.8 A

Heating Effect of Current

When a current flows through a resistor, collisions of electrons with atoms turn electrical energy into heat. The heat produced is $H = I^2 R t$ (joule). This is the principle of heaters, irons, toasters and fuses. Heating elements are made of **nichrome**, which has high resistivity and a high melting point and does not oxidise easily.

Worked Examples

Example 1. A heater is rated 2 kW, 220 V. Find the current drawn and the resistance of its element.

```
I = P / V = 2000 / 220 = 9.09 A
R = V^2 / P = 220^2 / 2000 = 48400 / 2000 = 24.2 ohm
```

Example 2. A bulb is marked 60 W, 220 V. Find the current and the resistance.

```
I = 60 / 220 = 0.273 A
R = 220^2 / 60 = 806.7 ohm
```

Example 3. A house uses four 40 W bulbs for 6 h daily. Find the energy used in a month of 30 days and the bill at Rs 6.50 per unit.

```
Power = 4 x 40 = 160 W = 0.16 kW
Energy per day = 0.16 x 6 = 0.96 kW h
Energy in 30 days = 0.96 x 30 = 28.8 kW h = 28.8 units
Cost = 28.8 x 6.50 = Rs 187.20
```

Example 4. A 5 Ω resistor carries a current of 3 A for 2 minutes. Find the heat produced.

```
H = I^2 R t = 3^2 x 5 x 120 = 9 x 5 x 120 = 5400 J
```

Example 5. Two identical bulbs marked 100 W, 220 V are joined in series across a 220 V supply. Find the power used by each.

```
Resistance of one bulb = 220^2 / 100 = 484 ohm
Series total = 968 ohm
Total power = V^2 / R = 220^2 / 968 = 50 W
Each bulb uses 25 W   (so they are much dimmer than rated)
```

Example 6. A geyser of 3 kW works for 1.5 h. Find the energy in kW h and in joule.

```
E = 3 x 1.5 = 4.5 kW h = 4.5 x 3.6e6 = 1.62e7 J
```



BOARD ALERT

Do not mix units. Convert watt to kilowatt and minutes to hours before you multiply for kW h. Writing 'units' for kW h is acceptable, but joule and kW h cannot be added without conversion.



REAL LIFE

Switching off standby devices, using LED bulbs and star-rated appliances reduces the electrical energy you buy. A 9 W LED gives about the same light as a 60 W filament bulb.



HOT QUESTION

A 1000 W heater and a 100 W bulb are both designed for 220 V. Which has the larger resistance? If both are connected in series across 220 V, which one gets hotter, and why?

Mind Map Recap

QUANTITY	FORMULA	UNIT
Energy	$W = QV = VIt = I^2Rt = (V^2/R)t$	J
Power	$P = W/t = VI = I^2R = V^2/R$	W
Commercial energy	$E = P\ (kW) \times t\ (h)$	kW h
Conversion	$1\ kW\ h = 3.6 \times 10^6\ J$	
Cost	Units × rate per unit	

Practice Questions

#	QUESTION	ANSWER
1	Derive $W = I^2Rt$ from $W = QV$.	$Q = It$ and $V = IR$, so $W = (It)(IR) = I^2Rt$.
2	A bulb is marked 100 W, 220 V. Find the resistance.	$R = 220^2/100 = 484\ \Omega$.
3	Find the energy in kW h used by a 1500 W iron in 40 minutes.	$1.5 \times (40/60) = 1.0$ kW h.
4	Define kW h and convert 2 kW h to joule.	Energy used by a 1 kW device in 1 h. $2 \times 3.6\times10^6 = 7.2\times10^6$ J.
5	A fan of 80 W runs 10 h a day. Cost for 30 days at Rs 7 per unit?	$0.08 \times 10 \times 30 = 24$ kW h; $24 \times 7 =$ Rs 168.
6	What current does a 2.2 kW geyser draw at 220 V?	10 A.
7	The heating element of a toaster is 55 Ω and carries 4 A. Find its power.	$P = I^2R = 16 \times 55 = 880$ W.
8	Why is nichrome used in heaters?	High resistivity, high melting point, does not oxidise easily.
9	A 40 W bulb and a 100 W bulb (both 220 V) are in parallel across 220 V. Which glows brighter?	The 100 W bulb, since power is larger at the same voltage.
10	Name the energy conversion in a motor and in a loudspeaker.	Electrical to mechanical; electrical to sound.
11	An electric kettle of 1.8 kW heats water for 5 minutes. Find the heat produced in joule.	$1800 \times 300 = 5.4 \times 10^5$ J.
12	Why does a bulb get dimmer if the supply voltage falls?	$P = V^2/R$ falls, so less energy per second as light.

10

CHAPTER · PHYSICS

Household Circuits

How does electricity travel from a power station to the socket on your wall, and what stops a fault from turning into a fire or a shock? This chapter traces the main circuit, explains fuses and earthing, and covers wiring and safety.



IN THIS CHAPTER

- Power distribution
- Main circuit
- Ring system
- Switches and two-way wiring
- Fuse and MCB
- Earthing and 3-pin plug
- Safety

Power Distribution: Grid to Gadget

1. Generation
Power stations generate AC at around 11 to 25 kV.

2. Step-up
Transformers raise the voltage to hundreds of kilovolts so the current is small and the I^2R heating loss in the long transmission lines is low.

3. Step-down
Substations near towns reduce it to about 11 kV, and a pole transformer reduces it again to **220 V (domestic supply in India is about 230 V)**.

4. Delivery
Cables bring the supply to the meter and then the distribution board of the house.

- Household supply is **alternating current (AC)** at **220 V and 50 Hz**: the direction reverses 100 times each second.
- Why high voltage for transmission? Power P = V I stays the same, so a higher V means a smaller I. The loss in the lines (I^2R) falls sharply.

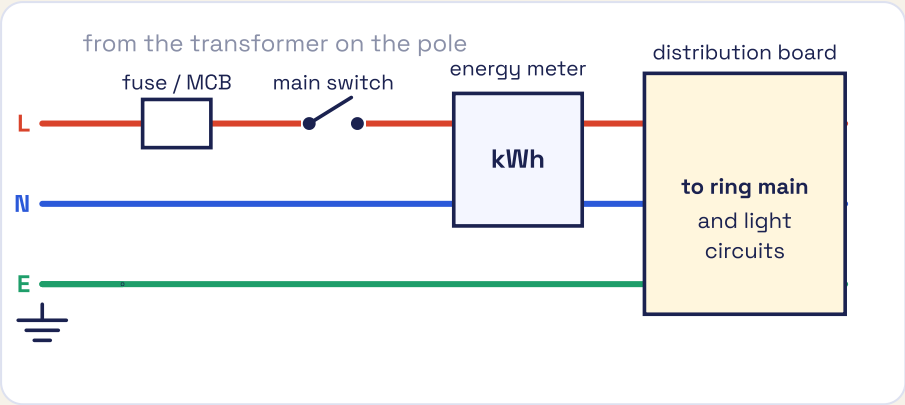
The Three Wires

WIRE	COLOUR CODE (NEW / OLD)	POTENTIAL	JOB
Live (phase)	Red (older) or brown	220 V above earth, alternating	Carries the current to the appliance. All switches and fuses go in this wire
Neutral	Black or blue	About 0 V	Returns the current to the supply
Earth	Green or green with yellow	0 V	Safety path to the ground. Carries no current in normal use

BOARD ALERT

Colour codes differ between older and newer wiring standards. Know the standard you were taught: **red = live, black = neutral, green = earth** is the ICSE convention, while brown, blue and green-yellow are the modern international ones.

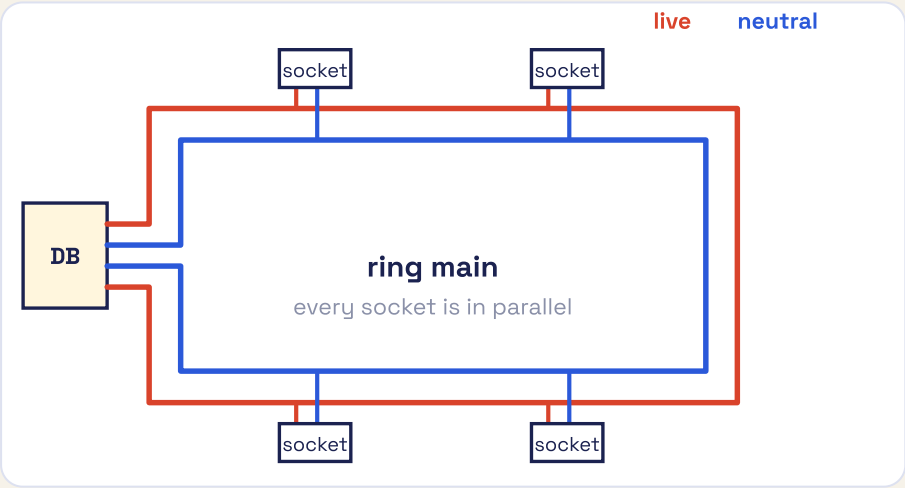
The Main Circuit



Main circuit: the live wire passes through the fuse, main switch and meter to the distribution board. The neutral goes through the meter. The earth wire runs separately to a metal plate buried in the ground.

- **Fuse or MCB** on the live wire cuts off the supply for the entire house if the current exceeds a safe limit.
- **Main switch** disconnects the entire house at once, for repairs or in emergencies. Having it on the live wire means the house wiring is dead when it is off.
- **Energy meter** records the electrical energy used, in kW h.
- **Distribution board** divides the supply into separate circuits: lights and fans (5 A, thin wires) and heavy power sockets (15 A, thick wires).

House Wiring: The Ring System



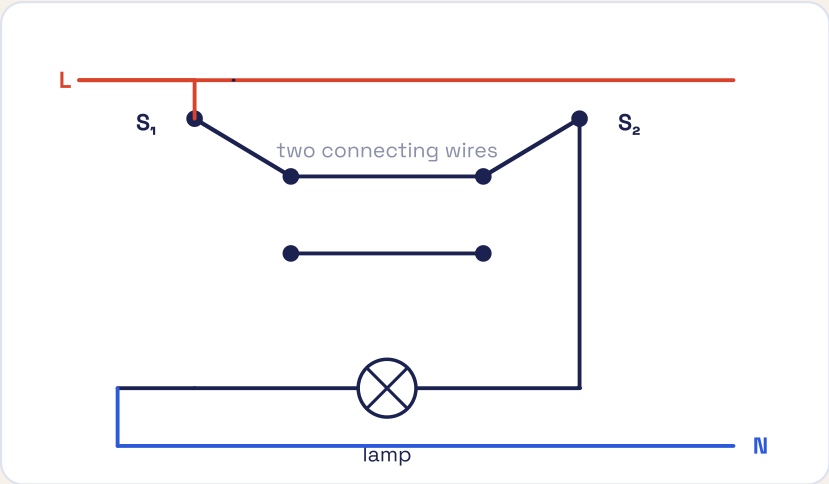
Ring main: the live (red) and neutral (blue) wires form a closed loop that starts and ends at the distribution board. Each socket is connected across the two wires, so all appliances get the full 220 V and current can reach each one by two routes.

- All sockets and appliances are connected in **parallel** across live and neutral, so each gets the full voltage and can be switched on or off independently.
- The wire loops back to the board. Current reaches each socket from both directions, so a thinner and cheaper wire can serve many sockets.
- Separate circuits for lights (5 A) and for heavy appliances such as geysers and air conditioners (15 A) prevent one overloaded line from affecting all.

Switches

A switch must be connected to the **live wire** only. When it is off, the appliance is disconnected from the live wire and so is at zero potential and safe to touch. If it were on the neutral wire, the appliance would still be live even when switched off.

A **two-way switch** lets you control one lamp from two places, for example at the top and bottom of a staircase, or at both ends of a corridor.



Staircase (two-way) wiring: each of the two switches can complete or break the circuit independently. The lamp is on when both switches connect the same one of the two connecting wires.

Fuse and MCB

KEY POINT

A fuse is a safety device with a thin wire of low melting point and high resistance (tin-lead alloy). It is connected in series with the live wire. If the current exceeds its rating, the wire heats up and melts, breaking the circuit.

- **Rating:** the maximum current it can carry. Choose a fuse slightly above the normal current of the appliance. A 1.5 kW kettle at 220 V draws 6.8 A, so a 10 A fuse is right.
- **Causes of excessive current:** a short circuit (live touches neutral) or an overload (too many appliances on one line).
- An **MCB (miniature circuit breaker)** does the same job without melting. It trips a switch when the current exceeds the limit and can be reset. It reacts faster and does not need replacing.

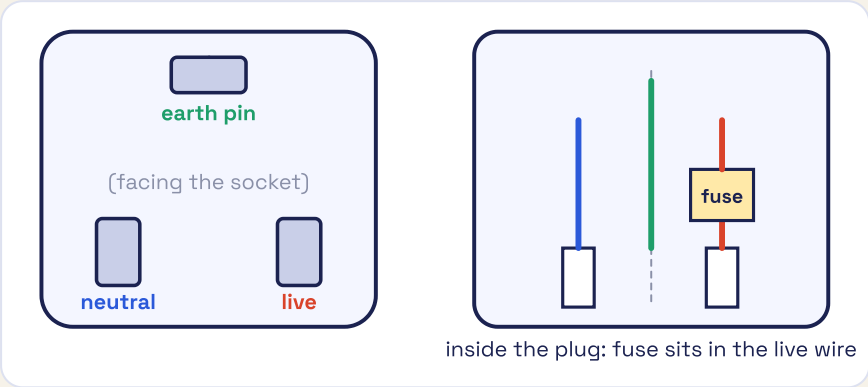
Earthing

The metal body of an appliance such as a refrigerator, iron or geyser is joined by the **earth wire** to a copper plate buried deep in the ground. If a live wire accidentally touches the metal body, a large current flows through the earth wire to the ground (the circuit's resistance is tiny). It then blows the fuse, cutting off the supply, so the body never stays at a dangerous potential and you do not get a shock.

KEY POINT

Earthing is the connection of the metal body of an appliance to the earth through a low resistance wire, as a safety measure against electric shock.

Three-Pin Plug and Socket



A three-pin plug. The earth pin (green wire) is longer than the other two so that it makes contact first and also opens the shutters of the socket. The live wire (red) is connected through the fuse.

- The **earth pin is the longest and thickest** and sits at the top. It is joined to the metal body.
- The other two pins are **live** and **neutral**. Conventionally, facing the socket with the earth pin at the top, the live pin is on the right and the neutral on the left.
- The **fuse** is placed in the live wire inside the plug.
- Heavy appliances use thick 15 A plugs; light ones use 5 A plugs.

Safety Precautions

- Never touch switches or appliances with wet hands. Water conducts.
- Switch off and unplug before repairing or cleaning any appliance.
- Use fuses of the correct rating. Never replace a fuse with a thick copper wire.
- Use proper insulating (rubber or plastic) coverings and wire of the right thickness for the current.
- Do not overload sockets with multiple plugs and extension boards.
- Ensure all metal-bodied appliances are earthed. Keep wiring away from water and damp.
- Keep metal objects (knives, keys) away from live sockets, and use switch covers if small children are around.
- Do not fly kites near overhead power lines, and keep away from fallen wires.

Worked Examples

Example 1. Choose a fuse for a geyser of 3 kW at 220 V from the ratings 5 A, 10 A, 15 A.

```
I = P / V = 3000 / 220 = 13.6 A
The fuse must exceed the normal current but be close to it: 15 A
```

Example 2. How many 100 W, 220 V bulbs can be run safely on a 5 A circuit?

```
Maximum power = V x I = 220 x 5 = 1100 W
Number of bulbs = 1100 / 100 = 11
```

Example 3. A TV (100 W), a fan (75 W) and four bulbs of 40 W work together on a 220 V line. Find the total current.

```
Total power = 100 + 75 + 4 x 40 = 335 W
I = 335 / 220 = 1.52 A
```



REAL LIFE

A tripping MCB is a signal, not a nuisance. Look for a faulty appliance or too many items on the line, and do not tape the MCB switch down.



HOT QUESTION

A fuse of 5 A is used in a circuit where an iron of rating 1.5 kW and a geyser of 3 kW (both 220 V) are plugged into the same socket. What happens, and why is the correct solution not to use a thicker fuse wire?

Mind Map Recap

IDEA	KEY FACTS
Supply	AC, 220 V, 50 Hz. Live, neutral, earth
Fuse	Thin low melting point wire, in the live wire, rating a little above normal current
Switch	Always on the live wire
Earthing	Metal body connected to ground to stop shocks and blow the fuse on a fault
Ring system	All sockets in parallel on a loop of live and neutral
3-pin plug	Earth pin top and longest; fuse in live wire
Safety	Dry hands, correct fuse, earthed appliances, no overloading

Practice Questions

#	QUESTION	ANSWER
1	Name the three wires of the household main circuit and their colour codes.	Live (red), neutral (black), earth (green).
2	Why is the switch connected in the live wire?	When off, it disconnects the appliance from the live wire, so the appliance is not at high potential.
3	What is a fuse? State two properties of fuse wire.	A safety device that melts when the current is too large. Low melting point and high resistance.
4	Find the correct fuse for a 2 kW heater at 220 V (5 A, 10 A, 15 A).	I = 9.1 A, so 10 A.
5	What is earthing and why is it needed?	Connecting the metal body to the ground, to give leaked current an easy path and prevent shocks.
6	Why is the earth pin longer than the other two?	It connects before live and neutral and opens the socket shutters.
7	State two advantages of the ring system.	Appliances work independently at full voltage; thinner wire suffices; fewer wires.
8	Give two functions of an MCB over a fuse.	Faster trip and can be reset without replacement.
9	Why is power transmitted at a high voltage?	The current is smaller, so I^2R heating loss is reduced.
10	State four safety precautions in household electricity.	Dry hands, correct fuse, earthed appliances, avoid overloading.
11	A circuit has a 10 A fuse. Can a 3 kW geyser and 1 kW heater run together at 220 V?	Total 4 kW, I = 18.2 A > 10 A, so the fuse blows.
12	What is the purpose of a two-way switch?	To control a lamp from two different places.

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CHAPTER · PHYSICS

Electromagnetism

Electricity and magnetism are two faces of the same thing. A current makes a magnetic field, and a changing magnetic field makes a current. Motors, generators, transformers, doorbells and cranes all use this link.



IN THIS CHAPTER

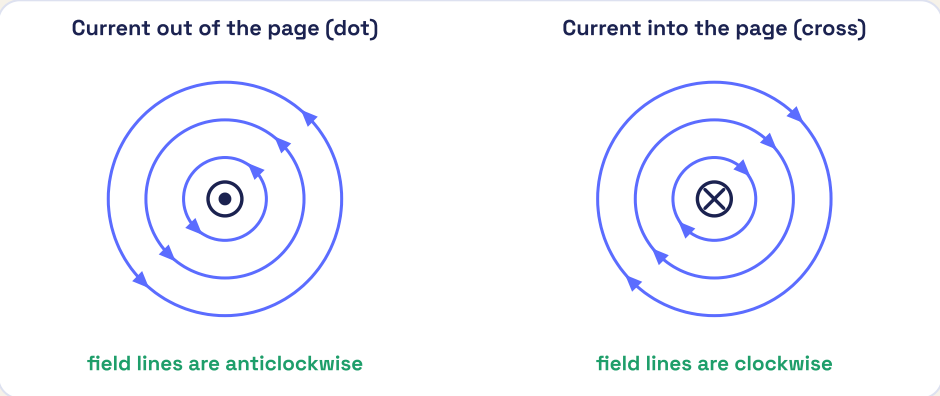
- Oersted's experiment
- Right-hand thumb rule
- Electromagnets
- Fleming's left-hand rule and DC motor
- Electromagnetic induction
- AC generator
- Transformer

Magnetic Effect of a Current

In 1820, Hans Christian Oersted noticed that a compass needle placed near a wire swung away from north when the current was switched on, and returned when it was switched off. A current-carrying conductor produces a **magnetic field** around it.

- A magnetic field (B) is the region around a magnet or current where a magnetic force can be felt.
- Field lines are drawn from N to S outside a magnet. They never cross, and are closer where the field is stronger.
- Reversing the current reverses the direction of the field.

Field Due to a Straight Wire



Magnetic field of a straight current-carrying wire: concentric circles, stronger near the wire. Right-hand thumb rule: thumb along the current, curled fingers show the field direction.

KEY POINT

Right-hand thumb rule: Hold the conductor in the right hand with the thumb pointing along the current. The curled fingers show the direction of the magnetic field lines around it.

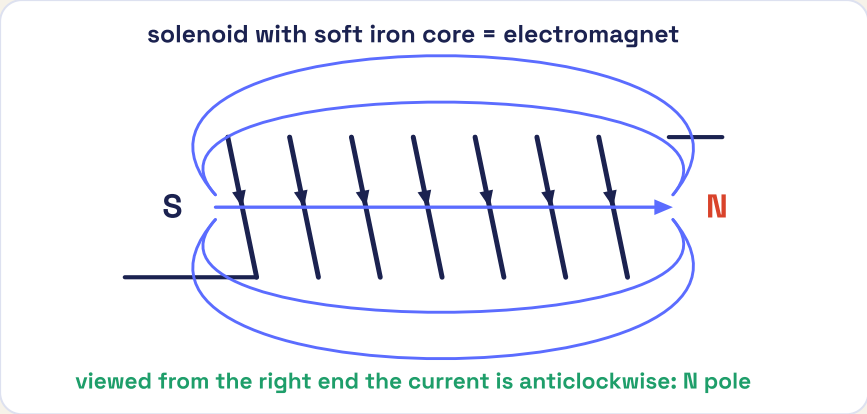
- The field lines are **concentric circles** in a plane perpendicular to the wire.
- Field strength **increases** with current and **decreases** with distance from the wire.

Field Due to a Current in a Loop

Bend the wire into a circular loop. Near the wire the field lines are almost circles, but at the centre they become almost straight, perpendicular to the plane of the loop, and all the little fields from each part of the loop point the same way, so they add up. Applying the right-hand thumb rule to every small part shows that the field lines go through the loop in one direction.

- Looking at the face of the loop, if the current is **anticlockwise** that face behaves like a **north pole**; if **clockwise**, like a **south pole**.
- A coil of many turns (N turns) has a field N times stronger.

Solenoid and Electromagnet



Field of a solenoid: like a bar magnet. Inside the coil the lines are nearly parallel and strong. The polarity follows the right-hand grip rule: curl the fingers along the current, the thumb points to the N pole.

A **solenoid** is a long coil of insulated wire. Its field is like that of a bar magnet. When a **soft iron core** is placed inside, the field becomes much stronger and the combination is an **electromagnet**.

- **Strength depends on:** the number of turns, the current, and the presence and kind of core.
- Soft iron is used because it magnetises strongly and loses its magnetism as soon as the current stops. Steel would stay magnetised.

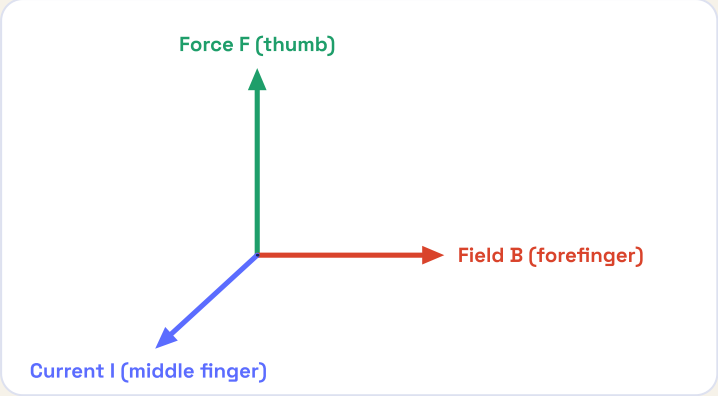
	ELECTROMAGNET	PERMANENT MAGNET
Magnetism	Temporary, only while the current flows	Always present
Strength	Can be changed by changing current or turns	Fixed
Polarity	Can be reversed by reversing the current	Fixed
Core material	Soft iron	Steel or alloys (alnico)
Uses	Cranes for scrap iron, electric bells, relays, loudspeakers, MRI	Compass, fridge magnets, small motors

REAL LIFE

A scrap-yard crane carries tonnes of iron with a huge electromagnet. When the driver switches off the current, the iron falls off at once.

Force on a Current in a Magnetic Field: Fleming's Left-Hand Rule

A conductor carrying a current in a magnetic field feels a force at right angles to both the current and the field. The force is larger for a larger current, a stronger field and a longer conductor. It is zero when the conductor is parallel to the field.



Fleming's left-hand rule (motors): the forefinger, middle finger and thumb of the left hand, held mutually at right angles, point along the field, the current and the force (motion).

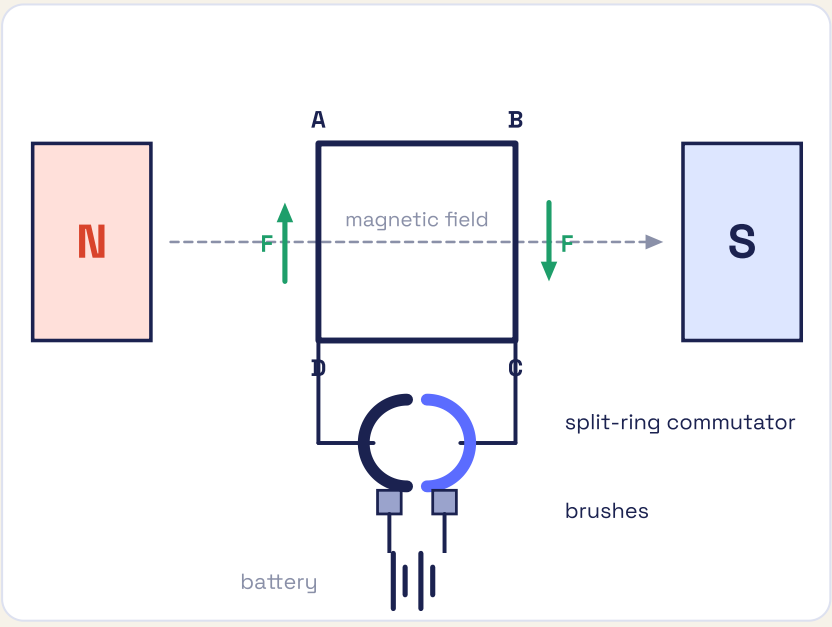
KEY POINT

Fleming's left-hand rule: Stretch the forefinger, the middle finger and the thumb of the left hand so that they are mutually perpendicular. If the forefinger points along the magnetic field and the middle finger along the current, then the **thumb** points in the direction of the **force** (motion) on the conductor.

SIDEKICK TIP

Motor = left hand (current and field in, motion out). **Generator = right hand** (motion and field in, current out). Both rules use the same finger order: forefinger = field, middle finger = current, thumb = motion.

The DC Electric Motor



DC motor: current in the coil ABCD (armature) in the field of the magnet. By Fleming's left-hand rule the forces on the two sides are opposite, so the coil rotates. The split ring reverses the current every half turn so the rotation continues the same way. Energy change: electrical to mechanical.

PART	FUNCTION
Armature (coil)	A rectangular coil of insulated wire on a soft iron core, free to rotate
Field magnet	Provides the magnetic field (permanent magnet or electromagnet)
Split-ring commutator	Two halves of a ring that reverse the current in the coil every half turn
Brushes	Carbon contacts that press on the commutator and join it to the battery

Energy transfer: electrical energy → mechanical (kinetic) energy. Motors are used in fans, mixers, washing machines, trains and electric vehicles. (A detailed working is not required.)

Electromagnetic Induction



KEY POINT

Electromagnetic induction is the production of an emf (and so a current in a closed circuit) in a coil when the magnetic flux through it changes, for example by moving a magnet in or out of the coil.

1. Move a magnet in
Push the N pole of a bar magnet into a coil connected to a sensitive galvanometer. The needle deflects.

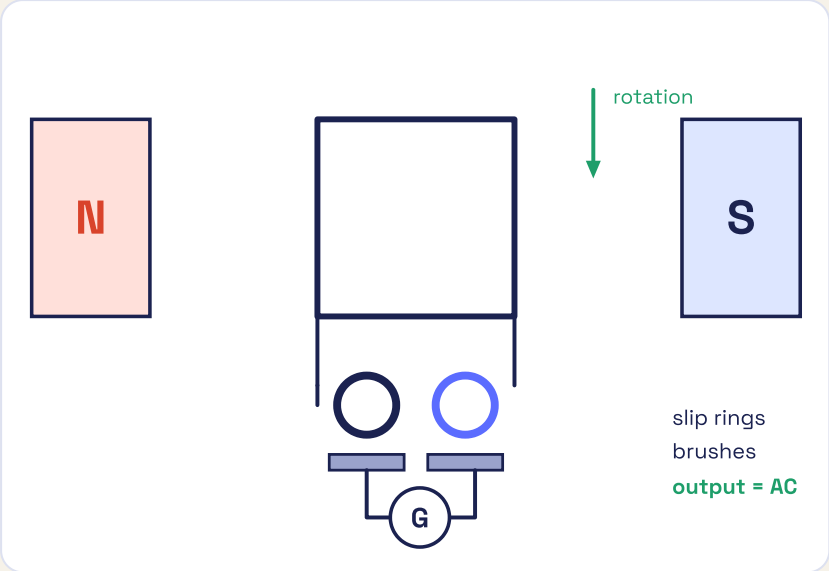
2. Hold it still
The needle returns to zero. A steady magnet in the coil induces nothing.

3. Pull it out
The needle deflects the other way.

4. Faster means more
A faster movement, more turns or a stronger magnet gives a larger deflection.

- Current is induced only **while there is relative motion** between the magnet and the coil.
- **Fleming's right-hand rule:** stretch the thumb, forefinger and middle finger of the right hand mutually at right angles. If the thumb is along the motion of the conductor and the forefinger along the field, the **middle finger** gives the direction of the **induced current**.

The AC Generator (Dynamo)



AC generator: a coil rotated in a magnetic field cuts the field lines, so an emf is induced (Fleming's right-hand rule gives its direction). Each end of the coil is joined to its own slip ring, so the output reverses every half turn: alternating current. Energy change: mechanical to electrical.

PART	FUNCTION
Armature coil	Many turns of insulated wire that rotate
Strong field magnet	Gives the magnetic field
Slip rings	Two separate rings, each joined to one end of the coil
Brushes	Carbon contacts that lead the current to the external circuit

- **Energy transfer:** mechanical energy → electrical energy.
- Induced emf changes its direction twice per revolution, so the current is **alternating**. In household supply it changes direction 100 times each second, and has **frequency 50 Hz** (50 cycles per second).
- Power stations turn the coil with steam, water or wind turbines.

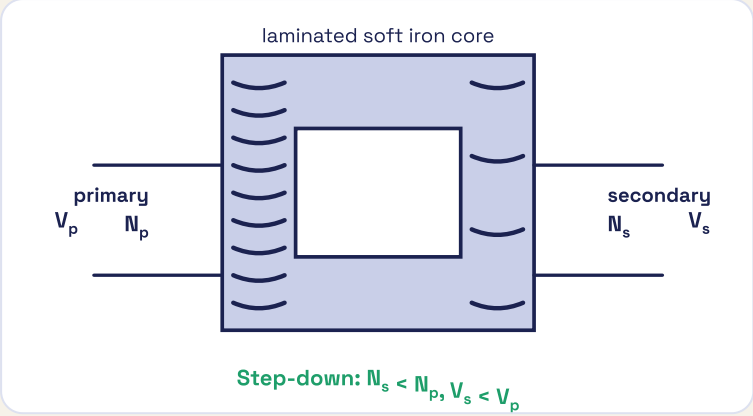
Advantages of AC over DC

- AC voltage can be easily stepped up or down with transformers, but DC cannot.
- High-voltage AC transmits power over long distances with low loss.
- AC generators are cheaper and simpler than DC generators.
- (DC has its own uses: batteries, electronics and electroplating, and it is needed by cell phones.)

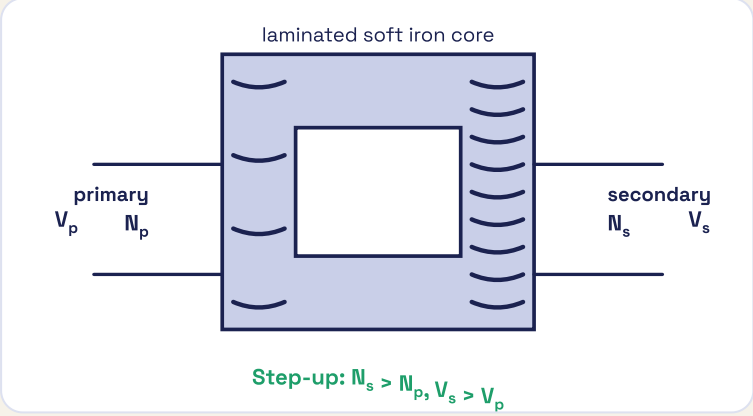
The Transformer

KEY POINT

A **transformer** is a device that changes the voltage of an alternating supply. It works on the principle of **mutual induction**: a changing current in the primary induces an emf in the secondary.



A transformer has two coils wound on one laminated soft iron core. AC in the primary sets up a changing magnetic field in the core, which induces an alternating emf in the secondary.



A transformer has two coils wound on one laminated soft iron core. AC in the primary sets up a changing magnetic field in the core, which induces an alternating emf in the secondary.

	STEP-UP TRANSFORMER	STEP-DOWN TRANSFORMER
Turns	$N_s > N_p$	$N_s < N_p$
Voltage	$V_s > V_p$	$V_s < V_p$
Current	$I_s < I_p$	$I_s > I_p$
Secondary wire	Thinner (carries smaller current)	Thicker (carries larger current)
Use	Power station to transmission lines	Pole transformer, phone charger, adapters

$$\mathbf{V_s / V_p = N_s / N_p} \quad \text{and for an ideal transformer} \quad \mathbf{V_p I_p = V_s I_s}$$

- A transformer works **only with AC**. With DC, the field is steady, so nothing is induced in the secondary.
- The core is **laminated** (made of insulated sheets) to reduce eddy-current heating. Energy is also lost as heat in the windings and by flux leakage, so real transformers are 90 to 99% efficient.
- A transformer cannot create energy: if the voltage goes up, the current goes down.

Worked Examples

Example 1. A step-down transformer has 1000 turns in the primary and is connected to a 220 V supply. It gives 11 V. Find the number of turns in the secondary.

```
Ns / Np = Vs / Vp  so  Ns = 1000 x 11 / 220 = 50 turns
```

Example 2. The above transformer supplies a current of 2.0 A at 11 V. Find the primary current (ideal case) and the power.

```
Output power = Vs Is = 11 x 2.0 = 22 W
Ideal: Vp Ip = 22  so  Ip = 22 / 220 = 0.10 A
If efficiency is 90 %: input = 22 / 0.90 = 24.4 W, Ip = 24.4 / 220 = 0.111 A
```

Example 3. A step-up transformer has 100 turns in the primary and 1000 in the secondary. The primary voltage is 220 V and the primary current is 5.0 A. Find the secondary voltage and the secondary current (ideal).

```
Vs = Vp x Ns / Np = 220 x 1000 / 100 = 2200 V
Is = Vp Ip / Vs = 220 x 5.0 / 2200 = 0.50 A
```



BOARD ALERT

Never write that the transformer increases power. A step-up transformer raises the voltage and lowers the current by the same factor.



HOT QUESTION

Why does a transformer not work on DC, and what happens if a DC battery is connected to the primary for a long time? Why is the primary of a step-down transformer made of thinner wire than the secondary?

Mind Map Recap

IDEA	KEY FACTS
Oersted	Current produces a magnetic field
Thumb rule	Thumb along the current, fingers curl along the field
Electromagnet	Solenoid + soft iron core. Strength depends on turns, current, core
Fleming's left hand	Motor. Forefinger field, middle finger current, thumb force
Fleming's right hand	Generator. Thumb motion, forefinger field, middle finger induced current
DC motor	Electrical to mechanical; split ring commutator
AC generator	Mechanical to electrical; slip rings; 50 Hz
Transformer	$V_s/V_p = N_s/N_p$; $V_pI_p = V_sI_s$; AC only

Practice Questions

#	QUESTION	ANSWER
1	What did Oersted observe?	A compass needle deflects near a current-carrying wire: current produces a magnetic field.
2	State the right-hand thumb rule.	Thumb along the current, curled fingers show the direction of the field lines.
3	Name three ways of strengthening an electromagnet.	More turns, larger current, soft iron core.
4	State Fleming's left-hand rule and one use.	Forefinger = field, middle finger = current, thumb = force. Used to find the direction of rotation of a motor.
5	What is the function of the split-ring commutator?	Reverses the current in the coil every half turn so the torque keeps the same sense.
6	What is electromagnetic induction?	Production of emf in a coil when the magnetic flux linked with it changes.
7	State two differences between a DC motor and an AC generator.	Motor: electrical to mechanical, split ring. Generator: mechanical to electrical, slip rings.
8	Why is the core of a transformer laminated?	To reduce eddy-current heating loss.
9	A transformer has 500 turns in the primary and 50 in the secondary. A 220 V supply is connected. Find the secondary voltage.	22 V (step-down).
10	A 12 V, 30 W lamp is run from a step-down transformer fed from 240 V. Find the primary current (ideal).	P = 30 W; I_p = 30/240 = 0.125 A.
11	State two advantages of AC over DC.	Easily stepped up or down; less loss in transmission.
12	Why can a transformer not work on DC?	A steady current produces a steady field, so no emf is induced in the secondary.

12

CHAPTER · PHYSICS

Heat

Why does a coastal city stay cooler than a desert at noon? Why does ice cool a drink better than cold water? Heat transfer, specific heat capacity and latent heat answer these, and let you calculate temperature changes of mixtures exactly.



IN THIS CHAPTER

- Heat and temperature
- Heat capacity
- Specific heat capacity
- Method of mixtures
- Change of state
- Latent heat of fusion

Heat and Temperature

	HEAT	TEMPERATURE
Meaning	Energy that flows from a hotter to a colder body	Degree of hotness; decides the direction of heat flow
Nature	Form of energy	A measure, not energy
SI unit	joule (J). Also calorie: 1 cal = 4.2 J	kelvin (K). Also degree Celsius (°C)
Instrument	Calorimeter	Thermometer

$$T\ (\text{kelvin}) = t\ (^\circ\text{C}) + 273 \quad \text{so a change of 1 K is the same as a change of 1 } ^\circ\text{C}$$

- **1 calorie** is the heat needed to raise the temperature of 1 g of water through 1 °C. 1 cal = 4.2 J.
- Heat flows from the body at the higher temperature to the one at the lower, until both are at the same temperature (thermal equilibrium).

Heat Capacity

KEY POINT

Heat capacity (thermal capacity, C′) of a body is the heat needed to raise its temperature by 1 K (or 1 °C). SI unit: **joule per kelvin (J K^{-1})**.

$$C' = Q / \Delta T \quad (Q = \text{heat absorbed}, \Delta T = \text{rise in temperature})$$

A big pan has more heat capacity than a small one of the same material. It needs more heat for the same rise in temperature.

Specific Heat Capacity

KEY POINT

Specific heat capacity (c) of a substance is the heat needed to raise the temperature of **1 kg** of the substance by 1 K (or 1 °C). SI unit: **J kg^{-1} K^{-1}**.

$$c = Q / (m\,\Delta T) \quad \Rightarrow \quad Q = m\,c\,\Delta T$$

$$\text{Relation between the two:} \quad C' = m\,c$$

SUBSTANCE	C (J KG^{-1} K^{-1})	C (CAL G^{-1} °C^{-1})
Water	4200	1.0
Ice	2100	0.5
Copper	400 (approx. 390)	0.095
Aluminium	900	0.21
Iron	450	0.11

Water has a very **high** specific heat capacity. It takes a lot of heat to warm it up, and it gives out a lot when it cools.

Consequences of the High Specific Heat of Water

- **Climate near the sea:** water warms and cools slowly, so coastal regions have mild summers and winters, while deserts have hot days and cold nights.
- **Land and sea breezes:** by day land heats faster than the sea, so cool air blows from the sea to land. At night the direction reverses.
- **Coolant:** water in car radiators and nuclear plants carries away large amounts of heat with small rise in temperature.
- **Hot water bottles** and water-filled heating systems give out heat for a long time.
- **Our body** is mostly water, so body temperature stays nearly steady.
- The sand on a beach is burning hot at noon but the sea is cool, because sand has a lower specific heat.

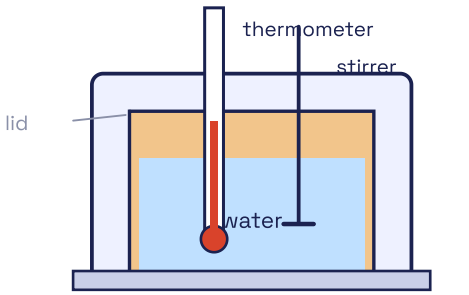
Method of Mixtures



KEY POINT

Principle of method of mixtures: when two bodies at different temperatures are mixed and no heat is lost to the surroundings, the heat **lost** by the hotter body is equal to the heat **gained** by the colder body, until a common temperature is reached.

Heat lost by hot body = Heat gained by cold body i.e. $m_1 c_1 (T_1 - T) = m_2 c_2 (T - T_2)$



A calorimeter: a thin metal (copper) vessel with a stirrer and thermometer, standing inside an insulating jacket so that very little heat is lost to the surroundings.

In experiments the mixing is done inside a **calorimeter**. Copper is used for the calorimeter because it conducts heat well and has a low specific heat. Heat loss is cut by an insulating jacket, a lid, and by stirring so the temperature is uniform.

1. Weigh
Find the mass of the empty calorimeter with the stirrer, then with some water (m_w).

2. Initial reading
Record the temperature of the water, T_2.

3. Add hot metal
Heat the metal block in boiling water to T_1 and drop it quickly into the calorimeter. Close the lid.

4. Final reading
Stir gently and note the highest steady temperature T.

5. Calculate
Equate heat lost by the metal to heat gained by water and calorimeter, then solve for c.

Worked Examples: Heat Capacity and Mixtures

Example 1. How much heat is needed to raise the temperature of 2.0 kg of water from 20 °C to 70 °C? (c = 4200 J kg^{-1} K^{-1})

```
Q = m c dT = 2.0 x 4200 x (70 - 20) = 2.0 x 4200 x 50 = 4.2e5 J
```

Example 2. A copper block of mass 0.50 kg (c = 400 J kg^{-1} K^{-1}) is heated through 30 K. Find its heat capacity and the heat absorbed.

```
C' = m c = 0.50 x 400 = 200 J/K
Q = C' dT = 200 x 30 = 6000 J
```

Example 3. 200 g of water at 80 °C is mixed with 300 g of water at 20 °C. Find the final temperature.

```
m1 (80 - T) = m2 (T - 20)  (same c, cancels)
0.2 (80 - T) = 0.3 (T - 20)
16 - 0.2 T = 0.3 T - 6
22 = 0.5 T   so   T = 44 degrees C
```

Example 4. A 0.40 kg block of an unknown metal at 100 °C is dropped into 0.30 kg of water at 25 °C in an insulated calorimeter (ignore the calorimeter). The final temperature is 30 °C. Find the specific heat capacity of the metal.

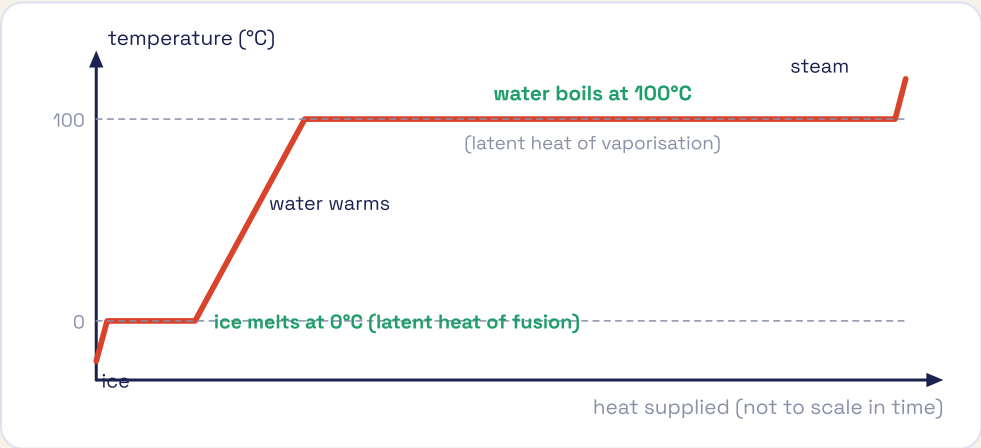
```
Heat gained by water = 0.30 x 4200 x (30 - 25) = 6300 J
Heat lost by metal = 0.40 x c x (100 - 30) = 28 c
28 c = 6300   so   c = 225 J/(kg K)
```

Example 5. A copper calorimeter of mass 100 g (c = 400) holds 200 g of water at 20 °C. 50 g of hot water at 80 °C is poured in. Find the final temperature.

```
Heat capacity of calorimeter + cold water = 0.1 x 400 + 0.2 x 4200 = 40 + 840 = 880 J/K
Heat lost by hot water = 0.05 x 4200 x (80 - T) = 210 (80 - T)
880 (T - 20) = 210 (80 - T)
880 T - 17600 = 16800 - 210 T   so   1090 T = 34400   T = 31.6 degrees C
```

Change of State and Latent Heat

Matter changes between solid, liquid and gas. During a change of state, the temperature **does not change**, even though heat keeps flowing in or out. This hidden heat is called **latent heat**. It goes into loosening the bonds between molecules (increasing their potential energy), not into making them move faster.



Heating curve of ice, water and steam at constant heating rate. While the state changes the temperature stays fixed: the heat supplied goes into breaking bonds (latent heat).

KEY POINT

Specific latent heat of fusion (L) of a solid is the quantity of heat needed to change **1 kg** of it from solid to liquid **at its melting point without any change in temperature**. SI unit: **J kg^{-1}**.

$$\mathbf{Q = m\,L} \quad \text{For ice: } L = 3.36 \times 10^{5}\ \text{J kg}^{-1}\ (336\ \text{J g}^{-1}, \text{ or about } 80\ \text{cal g}^{-1})$$

The same heat is given out when water freezes. The syllabus asks for calculations involving fusion only. Vaporisation is also a latent heat effect: 1 kg of water at 100 °C needs about 2.26×10^6 J to turn into steam.

Phenomena Involving Latent Heat of Fusion

- **Ice cools a drink better than the same mass of cold water at 0 °C** because melting absorbs 336 kJ for every kilogram.
- **Freezing** of lakes in winter releases heat to the surroundings, so the water below cools slowly and aquatic life survives.
- Farmers sprinkle water on crops before a frost: as it freezes it releases heat and protects the plants.
- Snow melting needs a lot of heat, so snow takes a long time to melt on a warm day.
- Fish and ice-cream are packed in ice because it absorbs heat while it melts at a constant 0 °C.

Example 6. How much heat is required to melt 0.50 kg of ice at 0 °C? ($L = 3.36 \times 10^5$ J kg^{-1})

```
Q = m L = 0.50 x 3.36e5 = 1.68e5 J
```

Example 7. 100 g of ice at 0 °C is added to 300 g of water at 40 °C in an insulated container. Find the final temperature.

```
Heat lost by water cooling to 0 degrees C = 0.30 x 4200 x 40 = 50 400 J
Heat needed to melt all the ice = 0.10 x 3.36e5 = 33 600 J
Heat left over = 50 400 - 33 600 = 16 800 J, which warms 0.40 kg of water:
dT = 16 800 / (0.40 x 4200) = 10 degrees C   so the final temperature = 10 degrees C
```

Example 8. A 1.0 kW heater melts 2.0 kg of ice at 0 °C. How long does it take if all the heat goes into the ice?

```
Q = 2.0 x 3.36e5 = 6.72e5 J
t = Q / P = 6.72e5 / 1000 = 672 s = 11.2 minutes
```



BOARD ALERT

In mixture problems, first check whether all the ice melts. Find heat available from the water cooling to 0 °C and compare it with m L before solving for the final temperature.



REAL LIFE

A pressure cooker, a thermos flask and a refrigerator all depend on heat transfer ideas from this chapter: heat must be kept in, kept out, or carried away.



HOT QUESTION

Equal masses of water and copper, both at 100 °C, are placed on separate large blocks of ice. Which melts more ice, and why? What does this say about the ability of each to store heat?

Mind Map Recap

IDEA	FORMULA / FACT
Heat capacity	$C' = Q / \Delta T$ (J K^{-1})
Specific heat capacity	$c = Q / (m \Delta T)$; unit J kg^{-1} K^{-1}; $C' = m c$
Heat needed	$Q = m c \Delta T$
Mixtures	Heat lost = heat gained
Latent heat of fusion	$Q = m L$; $L(ice) = 3.36 \times 10^5$ J kg^{-1}
Values	c(water) = 4200, c(ice) = 2100, c(copper) = 400 J kg^{-1} K^{-1}
Conversions	1 cal = 4.2 J; T(K) = t(°C) + 273

Practice Questions

#	QUESTION	ANSWER
1	Define specific heat capacity and give its SI unit.	Heat required to raise the temperature of 1 kg of a substance by 1 K. J kg^{-1} K^{-1}.
2	How are heat capacity and specific heat capacity related?	C′ = m c.
3	How much heat raises 5 kg of water by 4 °C?	5 × 4200 × 4 = 84 000 J.
4	A 0.2 kg iron piece at 100 °C is dropped into 0.5 kg water at 20 °C. If c(iron) = 450, find the final temperature (ignore the vessel).	0.2 × 450 (100 − T) = 0.5 × 4200 (T − 20); 9000 − 90T = 2100T − 42000; T = 51000/2190 = 23.3 °C.
5	State two consequences of high specific heat of water.	Mild climate near the sea; good coolant; sea breeze.
6	What is latent heat? Why does temperature stay constant during melting?	Heat absorbed or released in a change of state at constant temperature. The heat goes into overcoming the forces between molecules.
7	Heat needed to melt 3 kg of ice at 0 °C?	3 × 3.36×10^{5} = 1.008 × 10^{6} J.
8	Convert 27 °C to kelvin.	300 K.
9	A heater of 500 W works for 5 minutes on a liquid of mass 1 kg. If the temperature rises by 30 °C, find c (all heat absorbed).	Q = 500 × 300 = 1.5×10^{5} J; c = $1.5\times10^{5}/30$ = 5000 J kg^{-1} K^{-1}.
10	Why is ice at 0 °C more effective than water at 0 °C for cooling?	Melting ice absorbs latent heat on top of the heating of the meltwater.
11	State the principle of method of mixtures.	Heat lost by the hot body equals heat gained by the cold body, with no loss to the surroundings.
12	Why is a calorimeter made of copper and kept inside a jacket?	Copper conducts quickly and has a low heat capacity; the jacket reduces heat loss.

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CHAPTER · PHYSICS

Radioactivity

Some atoms are unstable and shoot out particles or energy without being asked. This chapter looks at the nucleus, the three kinds of radiation, how the nucleus changes, uses and dangers, and the two ways of getting energy from nuclei: fission and fusion.



IN THIS CHAPTER

- Nucleus, Z and A
- Alpha, beta, gamma
- Changes in the nucleus
- Uses and safety
- Background radiation and X-rays
- Fission and fusion

The Nucleus

An atom has a tiny, dense, positively charged **nucleus** at its centre, made of **protons** and **neutrons** (together called **nucleons**), with electrons orbiting around it.

PARTICLE	CHARGE	RELATIVE MASS
Proton	+1	1
Neutron	0	1
Electron	−1	about 1/1840

- **Atomic number (Z)** = number of protons (and of electrons in a neutral atom).
- **Mass number (A)** = number of protons + number of neutrons.
- Number of neutrons = A − Z.
- An element X is written ${}^{A}_{Z}X$. For example, ${}^{14}_{6}C$ has 6 protons and 8 neutrons.
- **Isotopes** are atoms of the same element (same Z) with different mass numbers (different neutron numbers), such as ${}^{12}C$ and ${}^{14}C$.

Radioactivity



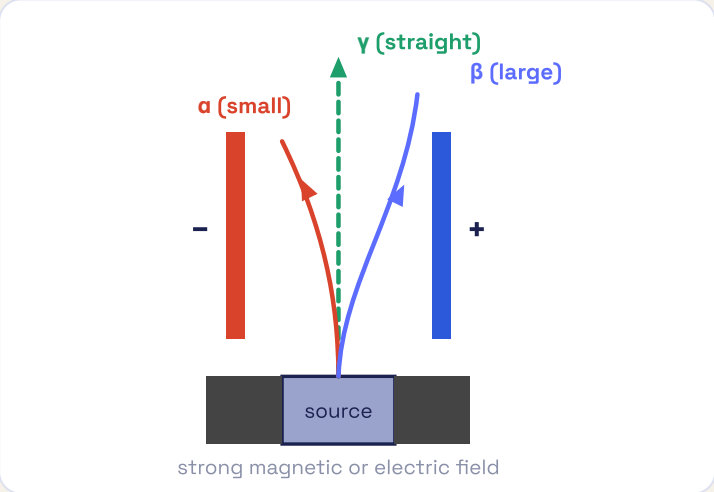
KEY POINT

Radioactivity is the spontaneous disintegration of an unstable nucleus, with the emission of alpha particles, beta particles or gamma rays, and the formation of a more stable nucleus.

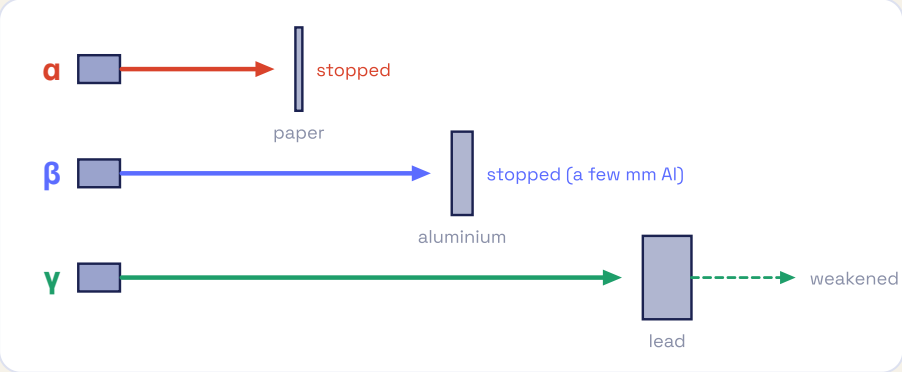
- Discovered by Henri Becquerel in 1896 in uranium salts. Marie and Pierre Curie then found polonium and radium.
- **Spontaneous:** it happens by itself, with no outside trigger.
- It is not affected by temperature, pressure, or chemical combination. It is a property of the nucleus, not of the electrons.
- Heavy nuclei (Z above 82), such as uranium, radium and thorium, are naturally radioactive. Some isotopes of light elements (${}^{14}C$) are also radioactive.

Alpha, Beta and Gamma Radiation

PROPERTY	ALPHA (A)	BETA (B)	GAMMA (Γ)
Nature	Helium nucleus, $^{4}_{2}He$ (2 protons + 2 neutrons)	High-speed electron, $^{0}_{-1}e$	Electromagnetic wave of very short wavelength
Charge	+2	−1	0
Mass	4 u (heavy)	Very small	None
Speed	About 10% of c	Up to 99% of c	Speed of light c
Ionising power	Very strong	Moderate	Weak
Penetrating power	Weak: stopped by paper or skin	Moderate: stopped by a few mm of aluminium	Strong: needs thick lead or concrete
Deflection in electric / magnetic field	Small, towards the negative plate	Large, towards the positive plate (opposite to alpha)	None



Behaviour in an electric field: alpha (positive) bends towards the negative plate by a small amount, beta (negative, much lighter) bends towards the positive plate by a large amount, gamma (neutral) is not deflected.



Penetrating power: alpha is stopped by paper, beta by a few millimetres of aluminium, and gamma needs thick lead or concrete to be greatly reduced.

SIDEKICK TIP

Memory: **ionising power** goes α > β > γ, **penetrating power** goes γ > β > α. The heavier and slower the particle, the more it ionises and the sooner it is stopped.

Changes Within the Nucleus

EMISSION	WHAT HAPPENS IN THE NUCLEUS	CHANGE IN Z	CHANGE IN A
Alpha	The nucleus loses 2 protons and 2 neutrons	Decreases by 2	Decreases by 4
Beta	A neutron changes into a proton and an electron. The electron is emitted at once	Increases by 1	No change
Gamma	The nucleus drops from an excited state to a lower energy state	No change	No change

Examples

```
Alpha decay:  U-238 → Th-234 + He-4     (A: 238 = 234 + 4,   Z: 92 = 90 + 2)
Beta decay:   Th-234 → Pa-234 + e-        (A: 234 = 234 + 0,   Z: 90 = 91 - 1)
```

In symbols: $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$ and $^{234}_{90}Th \rightarrow {}^{234}_{91}Pa + {}^{0}_{-1}e$

- Another beta decay: $^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$. This is used in carbon dating.
- Alpha decay: $^{218}_{84}Po \rightarrow {}^{214}_{82}Pb + {}^{4}_{2}He$.
- The nucleus left after the decay is called the **daughter nucleus**. If it is still unstable it decays again, giving a **decay series**.
- Alpha or beta emission leaves the daughter in an excited state, which then emits **gamma** to settle down.

Example 1. $^{238}_{92}U$ decays by α, then β, then β. Find the final nucleus.

```
After alpha:  A = 238 - 4 = 234, Z = 92 - 2 = 90
After beta:   A = 234, Z = 91
After beta:   A = 234, Z = 92     final nucleus: U-234 (an isotope of uranium)
```

Example 2. How many alpha and beta particles are emitted when ^{238}U becomes ^{206}Pb (Z = 82)?

```
Alpha particles = (238 - 206) / 4 = 8     (Z falls by 16 to 76)
Beta particles needed to raise Z from 76 to 82 = 6
```

Uses of Radioisotopes

A **radioisotope** is a radioactive isotope of an element.

FIELD	USE	RADIATION USED
Medicine	Radiotherapy to destroy cancer cells (cobalt-60); tracers to study thyroid or blood flow (iodine-131); sterilising syringes and instruments	Gamma, beta
Archaeology / geology	Carbon-14 dating of fossils, wood and bones	Beta
Industry	Checking thickness of paper or metal foil; finding leaks in pipes using a tracer; detecting flaws in welds	Beta, gamma
Agriculture	Studying uptake of fertiliser by plants; developing new crop varieties; preserving food by irradiation	Gamma, beta
Household	Smoke detectors (americium-241)	Alpha
Power	Nuclear reactors for electricity; power sources for satellites and spacecraft	Fission heat

Harmful Effects and Safety

- Radiation **ionises** atoms in living cells, damaging DNA and proteins. Effects: skin burns, cataract, hair loss, radiation sickness, cancer (leukaemia) and genetic mutations that can pass to children.
- Large doses kill. Small doses over a long time are also dangerous.

1. Time
Keep exposure as short as possible.

2. Distance
Stay far from the source. Handle sources with long tongs.

3. Shielding
Use lead aprons and thick lead or concrete walls. Store sources in thick lead containers.

4. Monitoring
Wear film badges or dosimeters. Wash after handling and do not eat or drink in the lab.

Background Radiation

Background radiation is the low-level radiation we all receive all the time, even with no radioactive source nearby. It is detected by a Geiger–Müller counter even without a source.

- **Natural sources:** cosmic rays from space, radon gas from rocks and soil, radioactive potassium and carbon in food and our own bodies, and rocks such as granite.
- **Man-made sources:** medical X-rays, nuclear power stations, fallout from nuclear tests, and waste.
- Its average level is small (a few millisieverts a year) and is not harmful, but must be subtracted from a measured count rate in an experiment.

X-rays

- **X-rays** are electromagnetic waves of very short wavelength (about 10^{-10} m), made when fast electrons strike a heavy metal target in an X-ray tube. They come from outside the nucleus, unlike gamma rays.
- **Uses:** seeing broken bones and teeth, CT scans, security checks at airports, studying crystal structure, finding flaws in metal castings.
- **Harmful:** they ionise tissue. Lead shields protect the patient and the operator, and exposure is kept short.

Nuclear Fallout and Waste

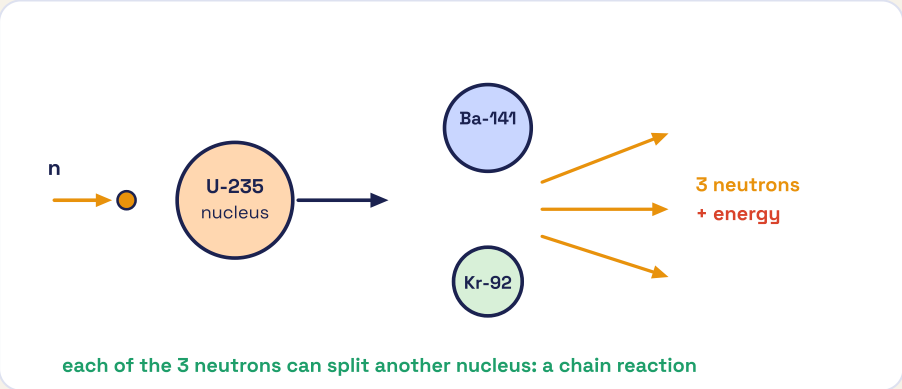
Radioactive fallout is the radioactive dust and debris that settles from the air after a nuclear explosion or reactor accident (Chernobyl in 1986, Fukushima in 2011). It contaminates soil, water and food, and can stay dangerous for years.

Nuclear waste from reactors stays radioactive for a very long time. Safe disposal means cooling spent fuel in water pools, mixing it with glass (vitrification), sealing it in steel and concrete casks, and burying it deep underground in stable rock, far from water. Strict rules and inspection of reactors are essential.

Nuclear Fission

KEY POINT

Nuclear fission is the splitting of a heavy nucleus into two lighter nuclei of comparable size, with the release of a large amount of energy and extra neutrons.



Nuclear fission of uranium-235: a slow neutron splits the nucleus into two medium-sized nuclei, releases 2 to 3 neutrons and a large amount of energy.

```
U-235 + n → Ba-141 + Kr-92 + 3 n + energy
A: 235 + 1 = 141 + 92 + 3 = 236      Z: 92 + 0 = 56 + 36 + 0 = 92
```

- The extra neutrons trigger more fissions: a **chain reaction**. Uncontrolled, it makes an atom bomb. Controlled, it runs a power reactor.
- In a reactor, **control rods** (boron or cadmium) absorb neutrons to hold the reaction steady, a **moderator** (graphite or heavy water) slows the neutrons, and a coolant carries heat away to make steam.
- The energy comes from a tiny loss of mass: $E = m c^2$. One fission releases about 200 MeV.

- **Drawbacks:** dangerous radioactive waste and risk of accidents.

Nuclear Fusion

KEY POINT

Nuclear fusion is the joining of two light nuclei to form a heavier nucleus, with the release of a large amount of energy.

```
H-2 + H-3 → He-4 + n + energy (17.6 MeV)
A: 2 + 3 = 4 + 1      Z: 1 + 1 = 2 + 0
```

- It needs **extremely high temperature** (millions of kelvin) and pressure to push the positive nuclei together against their repulsion.
- It powers the Sun and stars, where hydrogen is fused into helium.
- Per kilogram of fuel, fusion gives more energy than fission, with little long-lived waste. Controlled fusion for power is still being developed.

	FISSION	FUSION
Process	Heavy nucleus splits	Light nuclei join
Typical fuel	Uranium-235, plutonium-239	Hydrogen isotopes (deuterium, tritium)
Conditions	Slow neutron; room temperature	Very high temperature and pressure
Chain reaction	Yes	No chain, needs continuous heat
Waste	Long-lived radioactive waste	Little radioactive waste
Example	Nuclear reactor, atom bomb	The Sun, hydrogen bomb

FUN FACT

Every second the Sun turns about 600 million tonnes of hydrogen into helium, and converts about 4 million tonnes of mass into energy, using fusion. That is where almost all energy on Earth comes from.

BOARD ALERT

Beta particles do not come from the electron shells. They come from the nucleus, when a neutron changes into a proton. And in alpha or beta decay, always check that A and Z add up on both sides.

HOT QUESTION

A radioactive source emits α, β and γ. A sheet of paper is placed in front of a counter, then a sheet of aluminium, then thick lead. Describe the changes in the count rate at each step.

Mind Map Recap

IDEA	FACTS
Nucleus	Protons + neutrons; Z = protons; A = protons + neutrons
α decay	Z − 2, A − 4
β decay	Z + 1, A unchanged
γ emission	No change in Z or A
Penetration	γ > β > α
Ionisation	α > β > γ
Fission	U-235 + n → Ba-141 + Kr-92 + 3n + energy
Fusion	H-2 + H-3 → He-4 + n + energy

Practice Questions

#	QUESTION	ANSWER
1	Define radioactivity.	The spontaneous emission of radiation (alpha, beta or gamma) from the unstable nucleus of an atom.
2	Name the particle emitted when (a) A drops by 4 and Z by 2 (b) Z rises by 1 with A unchanged.	(a) Alpha particle (b) Beta particle.
3	Complete: $^{226}_{88}Ra \rightarrow ? + {}^{4}_{2}He$.	$^{222}_{86}Rn$.
4	Complete: $^{60}_{27}Co \rightarrow ? + {}^{0}_{-1}e$.	$^{60}_{28}Ni$.
5	Compare alpha and gamma in nature, charge and penetration.	Alpha: helium nucleus, +2, low penetration. Gamma: EM wave, 0, high penetration.
6	How many neutrons are in $^{235}_{92}U$?	235 − 92 = 143.
7	Give two uses of radioisotopes in medicine and one in industry.	Radiotherapy, tracer studies; thickness gauges or leak detection.
8	Why is lead used for shielding?	It has high density and absorbs gamma rays and X-rays strongly.
9	What is background radiation? Give two natural sources.	Low-level radiation always present around us. Cosmic rays and radon gas.
10	Distinguish between nuclear fission and nuclear fusion.	Fission: heavy nucleus splits, chain reaction. Fusion: light nuclei join, needs very high temperature.
11	State two safety measures in handling radioactive sources.	Lead shielding, handling with tongs, short exposure, dosimeter badges.
12	What is the function of control rods and moderator in a reactor?	Control rods absorb neutrons to control the rate; the moderator slows the neutrons.

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CHAPTER · PHYSICS

Practical Lab Companion

Twenty marks of Internal Assessment come from experiments. You must follow instructions, take readings, present them in a table and, where asked, plot a graph. This chapter gives the method, formula and common errors for each suggested experiment.



IN THIS CHAPTER

- What the examiner tests
- Measuring instruments
- Significant figures and errors
- The suggested experiments
- SI units and prefixes

What the Examiner Tests

- Follow simple instructions and take **suitable readings**.
- Present readings **systematically in a table** and, if required, **graphically**.
- Use the concepts of **least count, significant figures** and elementary error handling.
- Experiments may be based on topics outside the syllabus, but no theoretical knowledge of them is required.



SIDEKICK TIP

Always write: Aim, Apparatus, Diagram, Procedure, Observation Table with units, Calculation, Result, Precautions. A neat table is half the marks.

Measuring Instruments and Least Count

Least count is the smallest value an instrument can measure. For a ruler it is the smallest division. For vernier callipers and a screw gauge it is calculated.

INSTRUMENT	LEAST COUNT	USED FOR
Metre rule	1 mm = 0.1 cm	Lengths up to 1 m
Vernier callipers	0.1 mm = 0.01 cm (1 main scale division = 1 mm, 10 vernier divisions = 9 mm)	Diameter of a bob, thickness of a block
Screw gauge	0.01 mm (pitch 1 mm ÷ 100 circular divisions)	Thin wires, sheets
Stop watch	0.1 s or 0.01 s	Time for oscillations
Measuring cylinder	1 mL (or as marked)	Volume of liquids, volume of solids by displacement
Spring balance	As marked (0.1 N for example)	Forces and weights
Thermometer	1 °C (or 0.5 °C)	Temperature

Significant Figures and Error Handling

- All non-zero digits are significant. Zeros between digits are significant. Leading zeros are not (0.0042 has 2 sig. figs). Trailing zeros after a decimal point are (4.20 has 3).
- In a product or quotient, keep as many significant figures as the **least precise** number has.
- **Repeat readings** and take the mean. Take a reading at eye level to avoid **parallax**. Check for **zero error** in vernier callipers and screw gauge.
- **Absolute error** = least count (for a single reading). **Percentage error** = (error / measured value) × 100. For a time of 20 oscillations the error in one period is error/20, which is why we time many oscillations.

Graph Drawing

- Choose a suitable scale (about 1 cm for a convenient number) so the graph fills most of the page.
- Label axes with the quantity and unit. Mark each point with a small cross or a dot in a circle.
- Draw a **smooth best-fit line**, with equal numbers of points on both sides. Do not join the dots.
- Find the slope using two points far apart on the line, not data points. Show the slope triangle.

1. Lever: Metre Rule as a Lever

Aim. To find the mass of a metre rule and to study MA and VR when the pivot is placed at the 40 cm, 50 cm and 60 cm marks.

1. Mass of the rule
Balance the rule on a knife edge at a point away from the middle (say 30 cm) with a 50 g weight on the other side. The weight of the rule acts at 50 cm, so M × (50 − 30) = 50 × (30 − position of weight). Find M.

2. Pivot at 40 cm
Suspend a known load L near the left end (say at 10 cm). Hang weights E from the right end (90 cm) until the rule balances. Note L, E, load arm and effort arm.

3. Repeat
Do the same with the pivot at 50 cm and 60 cm.

4. Calculate
MA = L / E and VR = effort arm / load arm for each pivot position.

Sample results (rule weight 1.0 N acting at 50 cm, L = 0.60 N at 10 cm, effort at 90 cm):

PIVOT AT	LOAD ARM	EFFORT ARM	EFFORT NEEDED E	MA = L / E	VR	COMPARISON
40 cm	30 cm	50 cm	(0.60 × 30 − 1.0 × 10)/50 = 0.16 N	3.75	1.67	MA > VR
50 cm	40 cm	40 cm	0.60 N	1.00	1.00	MA = VR
60 cm	50 cm	30 cm	(0.60 × 50 + 1.0 × 10)/30 = 1.33 N	0.45	0.60	MA < VR

Explanation. At 50 cm the weight of the rule passes through the pivot and has no moment, so the rule behaves as an ideal lever. At 40 cm the weight of the rule (acting at 50 cm) is on the effort side and it **helps** the effort, so a smaller E is needed and MA exceeds VR. At 60 cm the rule's weight is on the load side and it **adds to the load**, so a larger E is needed and MA falls below VR. The **real load** is L plus the moment of the rule's weight divided by the load arm, and the **real effort** is the one corrected by the same method.



SIDEKICK TIP

Verification of the **principle of moments** uses the same apparatus with two spring balances. See the procedure in Chapter 1.

2. VR and MA of a Pulley System

1. Setup
Assemble the given pulley system (for example 2 fixed and 2 movable pulleys) with a stand, string and a pan for effort.

2. VR
Count the strands supporting the movable block. Check by raising the load by 10 cm and measuring the distance the effort moves.

3. MA
Put a known load L on the hook. Add weights to the effort pan until the load rises at a slow steady speed when tapped gently. Note E.

4. Result
MA = L / E; VR = d_E / d_L; efficiency = MA / VR × 100. Repeat for different loads and tabulate.

3. Refraction Through a Rectangular Glass Slab

Trace the outline of the slab on paper. Draw the normal and incident rays at 30°, 40°, 50° and 60°. Fix two pins on the incident ray and look from the other side to fix two pins in line with the images. Join them for the emergent ray. Measure **i, r, e** and the **lateral displacement**. Result: e = i, and sin i / sin r is constant (μ). Precautions: use sharp pins, keep them vertical, and put pins at least 5 cm apart.

4. Focal Length of a Convex Lens

- **(a) Distant object method:** focus a distant tree or window on a screen. Measure the distance between lens and sharp image = f. The object must be very far (more than 10 m away).
- **(b) Needle and plane mirror:** place the lens on a plane mirror and move a pin along the axis until the pin and its inverted image coincide with no parallax. The distance from pin to lens is f. (The rays return along their own path after reflecting from the mirror, so they were parallel between the lens and the mirror.)

5. Focal Length by u and v

Place a bright object pin at a distance u (more than f) from the lens and move a screen to get a sharp real image at distance v. Take readings for 5 values of u. For each, compute **f = uv / (u + v)** (all distances taken positive in this form). Take the mean. A graph of 1/v against 1/u is a straight line with intercept 1/f.

U (CM)	V (CM)	F = UV/(U+V) (CM)
30.0	60.0	20.0
40.0	40.0	20.0
50.0	33.3	20.0

(Sample data for a lens of 20 cm: check 1/20 = 1/30 + 1/60 = 1/40 + 1/40 = 1/50 + 1/33.3 ✓.)

6 and 7. Triangular Prism

- Place the prism on paper and draw its outline. Fix pins P_1, P_2 on the incident ray at i_1 = 40°, 50°, 60° and 70°, then locate the emergent ray with P_3, P_4.
- Measure i_1, i_2 (emergent angle), A and δ. Verify $i_1 + i_2 = A + \delta$ and $A = r_1 + r_2$ (r_1, r_2 can be found from Snell's law, or by measurement).
- **Normal incidence ($i_1 = 0$):** the ray goes in straight. For A = 60° and 45° (glass) the ray meets the second face at A, which is above the critical angle, so it is totally reflected. For A = 90° it is not possible for a ray to leave. Describe what you see and measure δ where the ray emerges.

8, 9. Specific Heat Capacity by Method of Mixtures

- **Calorimeter material:** weigh the empty calorimeter (m_c), then with cold water (m_1); note T_1. Pour in warm water (m_2 at T_2), stir, note final T. Heat lost by warm water = heat gained by cold water + calorimeter: $m_2\, c_w\, (T_2 - T) = m_1\, c_w\, (T - T_1) + m_c\, c_c\, (T - T_1)$. Solve for c_c.
- **Metal solid:** heat the metal in boiling water, transfer it quickly to cold water in the calorimeter, stir and note the highest temperature. Use heat lost by metal = heat gained by water and calorimeter.
- **Precautions:** insulating jacket, stir gently, transfer the metal quickly with minimal splashing, and read the thermometer at eye level.

10. Specific Latent Heat of Ice

Put warm water at about 40 °C (mass m_w) in a weighed calorimeter. Add small pieces of dried ice and stir until all the ice melts and the temperature falls to about 5 °C. Find the mass of ice from the final weight. Equate heat lost by water and calorimeter in cooling to the heat to melt ice plus the heat to warm the meltwater from 0 °C. Precaution: dry the ice with blotting paper before adding it, or the extra water will make the answer too small.

11. Verifying Ohm's Law

Connect the circuit as in Chapter 8. Vary the current with the rheostat. Record 6 pairs of I and V. Plot V against I. The graph is a straight line through the origin, and the slope gives R. Precautions: close the key only while taking readings, check zero error of meters, and use thick connecting wires with tight joints.

12. Model of Household Wiring

On a wooden board, set up a main switch, fuse, a bulb with its own switch, a two-pin or three-pin socket, and an MCB if available. All fixtures are joined in **parallel**, with the fuse and switches in the **live** wire only. Test: removing the fuse should turn the entire circuit dead, and each bulb works independently. Use a low voltage (say a 6 V battery) for the school model.

Other Suggested Experiments

EXPERIMENT	WHAT YOU FIND OUT
Density of a glass block	Length, breadth, thickness with a metre rule; mass with a balance; density = mass / volume, in cgs and SI units
Volume of a metal bob	Displacement method in a measuring cylinder compared with the vernier callipers radius method ($V = 4/3\ \pi\ r^3$)
Simple pendulum	Time for 20 oscillations at lengths 70 to 110 cm; plot l against T^2; $g = 4\pi^2 \times$ slope
Celsius and Fahrenheit thermometers	Plot T_F against T_C; slope compared with 9/5 and intercept 32
Reflection from a plane mirror	Angle of incidence equals angle of reflection; image distance equals object distance
Focal length of a concave mirror	Distant object method and one-needle method
Bulbs in series and in parallel	Compare brightness to understand current and resistance in each circuit
Magnetic field of the earth and a bar magnet	Plot field lines with a compass; locate neutral points
Archimedes' principle	Weight in air and in water gives upthrust; compare with weight of water displaced

Sample calculation (pendulum). A simple pendulum of length 1.00 m takes 40.2 s for 20 oscillations. $T = 40.2/20 = 2.01$ s. $g = 4\pi^2\ l\ /\ T^2 = 39.48 \times 1.00 / 4.04 =$ **$9.77\ m\ s^{-2}$**.

SI Base Units

QUANTITY	UNIT	SYMBOL
Mass	kilogram	kg
Length	metre	m
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Luminous intensity	candela	cd
Amount of substance	mole	mol

Derived units with special names: force = newton (N), pressure = pascal (Pa), energy and work = joule (J), power = watt (W), frequency = hertz (Hz), charge = coulomb (C), resistance = ohm (Ω), emf and pd = volt (V). Others: velocity m s^{-1}, acceleration m s^{-2}, density kg m^{-3}, momentum kg m s^{-1}. When a unit is named after a person, its symbol is a capital letter.

Standard Prefixes

MULTIPLE	PREFIX	SYMBOL
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f

Viva Questions

#	QUESTION	ANSWER
1	What is the least count of a vernier callipers?	0.1 mm (0.01 cm) when 10 vernier divisions equal 9 main scale divisions.
2	Why do we time 20 oscillations, not one?	Reaction time error is spread over 20 swings, so the error in one period is 20 times smaller.
3	What is parallax and how is it avoided?	Apparent shift between two objects due to eye position. Keep the eye in line, perpendicular to the scale.
4	Why must the ice be dried before use in the latent heat experiment?	Extra water on the ice adds mass that is counted as ice, making L look smaller.
5	In the lever experiment, why is MA > VR when the pivot is at 40 cm?	The rule's weight on the effort side helps the effort, so less effort is needed.
6	What is the slope of a V–I graph?	Resistance of the conductor.
7	Why is the key closed only while taking readings in the Ohm's law experiment?	To avoid heating the wire, which would change its resistance.
8	Why are all fixtures in the model house wiring connected in parallel?	So each gets full voltage and can work independently.

Done reading?
Now go break the board.



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